

ENGLISH 3

COURSE CODE: ENG203

**FOR THE STUDENTS OF BS PROGRAMMES
IN THE FACULTY OF INFORMATION
TECHNOLOGY & COMPUTER SCIENCES
(FOIT&CS)**



**English Language Centre,
Faculty of Languages & Literature,
University of Central Punjab**

Acknowledgement

This book meets the objectives of the course, English for Specific Purposes, designed for the students of the Faculty of Information Technology & Computer Science. We acknowledge the support of Dr. Fehmida Sultana, Dean FoLL; Dr. Muhammad Amjad Iqbal, Dean FOIT&CS; and, Dr. Mohsin Ashraf, the Head of Department at FOIT&CS, in this initiative. We also extend our gratitude to Ms. Tahira Munir, Lecturer at the English Language Centre, for tailoring the book to the required content of the course.

Course Outline

Course Title: English 3

Course Code: ENG203

Credit Hours: 3

Course Prerequisite: English 2

Course Description:

This course provides comprehensive training in effective communication within technical and professional contexts. It will guide the students from information technology and computer science to develop the essential skills to convey complex information clearly, concisely, and ethically. The course covers various aspects of technical communication, including document planning, style, visual design, correspondence, documentation with citation, process explanations, reports, proposals, and presentations. The resource book presents the fundamental concepts, techniques, and genres commonly used in the workplace—supported by useful examples and exercises. As this course is for undergraduate students, it offers a solid starting point for communicating effectively in a future of rapidly changing technology and workplace settings.

Course Objectives:

By the end of this course, students should be able to:

1. Understand the principles and importance of technical communication in professional settings.
2. Plan, organize, and structure technical documents effectively.
3. Apply principles of visual design to enhance document readability and navigation.
4. Craft various forms of correspondence, including letters, memoranda, and emails.
5. Learn the ethical standard methods for source documentation.
6. Create process explanations and instructions effectively.
7. Produce different types of reports and proposals, both informal and formal, following established guidelines.
8. Deliver impactful presentations using appropriate content and visual aids.

Course Learning Outcomes:

Upon successful completion of the course, students will be able to:

1. Prepare technical documents that address specific purposes and audiences.
2. Employ the standard writing process while composing any professional document.
3. Design documents with effective visual elements to enhance understanding and navigation.
4. Compose different forms of professional communication like memoranda, emails, and letters.
5. Apply IEEE citation style and standard format in their papers.
6. Create process explanations and instructions that are clear and user-friendly.
7. Produce various types of reports and proposals following established guidelines.
8. Deliver engaging and informative presentations using suitable visual aids.

Methodology:

This course is based on English for Specific Purposes. The objective behind the design and content of the course is to equip the students of information technology and computer science with a practical understanding of writing and speaking skills that are essential in their professional context. So, this course adopts a comprehensive methodology that blends theoretical understanding with hands-on application. Through engaging lectures, interactive class discussions, and practical exercises, participants delve into the intricacies of technical communication.

Specimens from the real world have been incorporated into collaborative group projects to immerse participants in authentic communication scenarios, fostering teamwork and practical skills. Peer review sessions provide constructive feedback to refine both writing and presentation abilities. Participants also tackle individual assignments that facilitate the direct application of newly acquired techniques. The culmination of the course is the final presentation, showcasing participants' learning of technical communication strategies. With access to resources and materials, participants are guided on a dynamic learning journey, ultimately evaluated through their engagement, assignments, quizzes, and a project.

Calendar of Activities

Week	Chapter	Assessment/ Activity
1	Introduction to Technical Communication - Defining Technical Communication - Determining the Purpose of Your Documents - Analyzing Your Readers - Planning Your Documents	
2	Style in Technical Writing - Overview of Style - Writing Clear Sentences - Being Concise	
3	- Being Accurate in Wording - Using the Active Voice - Using Nonsexist Language - Plain English and Simplified English	
4	Visual Design - Elements of Page Design and Navigation - Font - Guidelines for Graphics - Misuse of Graphics	Quiz 1 on Style in Technical Writing
5	Letters, Memos, and Electronic Communication - General Guidelines for Correspondence - Letters	Assignment 1 from the chapter on visual design
6	Letters, Memos, and Electronic Communication - Memoranda - E-mail - Memoranda versus E-mail	Assignment 2 on writing an email or a memo
7	IEEE Citation	
8	Guide on Writing a Research Paper in IEEE Format	Quiz 2 on IEEE citation
9	Mid Term Examination	
10	Process Explanations & Instructions - Process Explanations versus Instructions - Guidelines for Process Explanations - Guidelines for Instructions	
11	Reports - General Guidelines for Reports - Guidelines for Informal Reports - Guidelines for Formal Reports	

12	Reports Four Common Reports	Quiz 3 on writing process explanation and instructions
13	Proposals - Guidelines for Informal Proposals - Guidelines for Formal Proposals	Assignment 3 from the chapter on reports
14	Presentations - Presentations and Your Career - Guidelines for Preparation and Delivery - Guidelines for Presentation Graphics	
15	Project Presentations On a report or proposal	Project Presentation
16	Revision	
17	Final Term Examination	

Assessment Structure:

Assessment Components	Percentage
3 Assignments	10 %
3 Quizzes	10 %
Project Presentations	10 %
Class Participation	10 %
Mid Term Exam	20 %
Final Term Exam	40 %

Text Book:

Pfeiffer, W. S., & Adkins, K. A. (2014). *Technical Communication Fundamentals*. Longman Publishing Group.

Reference Books:

Alred, G. J., Brusaw, C. T., & Oliu, W. E. (2009). *Handbook of technical writing*. Bedford Books.

Gerson, S., & Gerson, S. (2014). *Technical communication: Process and product*. Pearson Group.

Web Sources:

https://owl.purdue.edu/owl/subject_specific_writing/professional_technical_writing/index.html

https://www.ieee.org/content/dam/ieee-org/ieee/web/org/conferences/style_references_manual.pdf

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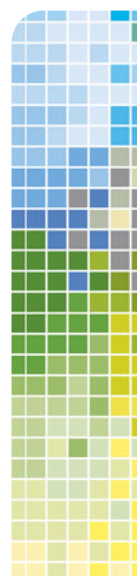
Note: The chapters are listed according to the placement of topics in the 'Calendar of Activities'.

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Chapter | 1 | Introduction to Technical Communication



>>> Chapter Outline

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Features of Technical Communication 2

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Good communication skills are essential in any career you choose. Jobs, promotions, raises, and professional prestige result from your ability to present both written and visual information effectively.

>>> Defining Technical Communication



For review, see the Writing Process section.

You may have learned how to write academic essays in previous writing courses. This book helps you transfer that basic knowledge to the kind of writing done on the job. Career writing is so practical, so well-grounded in common sense, that it will seem to proceed smoothly from your previous work. While traditional academic writing and technical communication share many common features, they also have important differences.

Features of Academic Writing

Writing you have done in school probably has had the following characteristics:

- **Purpose:** Communicating what you know about the topic, in a way that justifies a high grade
- **Your knowledge of topic:** Less than the teacher who evaluates the writing
- **Audience:** The teacher who requests the assignment and reads it from beginning to end
- **Criteria for evaluation:** Depth, logic, clarity, unity, supporting evidence, and grammar
- **Graphic elements:** Sometimes used to explain and persuade

Academic writing requires that you use words to display your learning to someone who knows more about the subject than you do. Because this person's job is to evaluate your work, you have what might be called a *captive audience*. In an academic setting, the purpose is to demonstrate your command of information to someone more knowledgeable about the subject than you are.

Features of Technical Communication

The rules for writing shift when you begin your career. *Technical communication* is a generic term for all written and oral communication done on the job—whether in business, industry, or in other settings. It is particularly identified with documents in technology, engineering, science, the health professions, and other fields with specialized vocabularies. The terms *technical writing*, *professional writing*, *business writing*, and *occupational writing* also refer to writing done in your career.

Besides writing projects, your career will also bring you speaking responsibilities, such as formal speeches at conferences and informal presentations at meetings. Thus, the term *technical communication* can encompass the full range of writing and speaking responsibilities required to communicate your ideas on the job. The following discusses the main characteristics of technical communication:

- **Purpose:** Getting something done within an organization or helping a customer, client, or colleague get something done
- **Your knowledge of topic:** Usually greater than that of the reader



My Tech Comm Lab includes examples of technical communication genres.

- **Audience:** Often several people with differing technical backgrounds
- **Criteria for evaluation:** Clear and simple organization of ideas and supporting detail appropriate to the needs of busy readers
- **Graphic elements:** Frequently used to explain existing conditions and to present alternative courses of action

Contrast these features with those of academic writing, listed earlier. In particular, note the following main differences:

1. Technical communication aims to help people make decisions and perform tasks, whereas academic writing aims only to display your knowledge.
2. Technical communication usually responds to the needs of the workplace, whereas academic writing usually responds to an assignment created by a teacher.
3. Technical communication is created by an informed writer conveying needed information both verbally and visually to an uninformed reader, whereas academic writing is created by a student as the learner for a teacher as the source of knowledge.
4. Technical communication often is read by many readers, whereas academic writing usually aims to satisfy only one person, the teacher.

Finally, technical communication places greater emphasis on techniques of organization and visual cues that help readers find important information as quickly as possible.

Figure 1-1 lists some typical on-the-job writing assignments. Although not exhaustive, the list does include many of the writing projects you will encounter. Figure 1-2 is an

Correspondence: In-House or External

- Memos to your boss and to your subordinates
- Routine letters to customers, vendors, etc.
- “Good news” letters to customers
- “Bad news” letters to customers
- Sales letters to potential customers
- Electronic mail (e-mail) messages to coworkers or customers over a computer network

Short Reports: In-House or External

- Analysis of a problem
- Recommendation
- Equipment evaluation
- Progress report on project or routine periodic report
- Report on the results of laboratory or field work
- Description of the results of a company trip

Long Reports: In-House or External

- Complex problem analysis, recommendation, or equipment evaluation
- Project report on field or laboratory work
- Feasibility study

Other Examples

- Proposal to boss for new product line
- Proposal to boss for change in procedures
- Proposal to customer to sell a product, a service, or an idea
- Proposal to funding agency for support of research project
- Abstract or summary of technical article
- Technical article or presentation
- Operation manual or other manual
- Web site

■ **Figure 1-1** ■
Examples of
technical
communication

■ **Figure 1-2** ■ Short report

MEMORANDUM

DATE: December 6, 2011
TO: Holly Newsome
FROM: Michael Allen
SUBJECT: Printer Recommendation

Introductory Summary

Recently you asked for my evaluation of the Hemphill 5000 printer/fax/scanner/copier currently used in my department. Having analyzed the machine's features, print quality, and cost, I am quite satisfied with its performance.

Features

Among the Hemphill 5000's features, I have found these five to be the most useful:

1. Easy to use control panel
2. Print and copy speed of up to 34 pages per minute for color and black and white
3. Ability to print high-quality documents like brochures & report covers
4. Built-in networking capability
5. Ability to scan documents to or from a USB port

In addition, the Hemphill 5000 offers high-quality copies, color copies and faxes, and it uses high capacity ink cartridges to reduce costs.

Print Quality

The Hemphill 5000 produces excellent prints that rival professional typeset quality. The print resolution is 1200 x 1200 dots per inch, among the highest attainable in printer/fax/scanner/copier combinations. This memo was printed on the 5000, and, as you can see, the quality speaks for itself.

Cost

Considering the features and quality, the 5000 is an excellent network combination printer for workgroups within the firm. At a retail price of \$239, it is also one of the lowest-priced combination printers, yet it comes with a two-year warranty and excellent customer support.

Conclusion

On the basis of my observation, I strongly recommend that our firm continue to use and purchase the Hemphill 5000. Please call me at ext. 204 if you want further information about this excellent machine.



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example of a short technical document. Note that it has the features of technical communication listed previously.

1. It is written to get something done—that is, to evaluate a printer
2. It is sent from someone more knowledgeable about the printer to someone who needs information about it
3. Although the memo is directed to one person, the reader probably will share it with others before making a decision concerning the writer's recommendation

4. It is organized clearly, moving from data to recommendations and including headings
5. It provides limited data to describe the features of the printer

Now that you know the nature and importance of technical communication, the next section examines the ethical context for communication done on the job.

>>> Determining the Purpose of Your Documents

If you have already taken a basic composition course, you will see similarities between rhetorical aims studied in that course and those in technical communication. Writing assignments you have had in school have probably asked you to *inform* your reader about an event or object, to *analyze* a process or idea, or to *argue* the strength or weakness of an interpretation or theory.

Information: When readers pick up a technical communication document, they may want to know how to perform an operation or follow an established procedure. They may want to make an informed decision. Clear, reliable information is the basis of analysis and argument.

Analysis: At first, it may not seem like analysis is an important purpose of workplace writing, but it is essential to problem solving and decision making. You may be asked to analyze options for a supervisor who will make a recommendation to a client or you may be asked to use analysis to make your own recommendation.

Argument: Good argument forms the basis for all technical communication. Some people have the mistaken impression that only recommendation reports and proposals argue their case to the reader, and that all other writing should be objective rather than argumentative. The fact is, every time you commit words to paper, you are arguing your point.

Every piece of your workplace writing should have a specific reason for being. The purpose may be dictated by someone else or selected by you. In either case, it must be firmly understood *before* you start writing. Purpose statements guide every decision you make while you plan, draft, and revise. Your choice of purpose statement will fall somewhere within this continuum:

Neutral, objective statement $\longleftrightarrow$ Persuasive, subjective statement

For example, when reporting to your boss on the feasibility of adding a new wing to your office building, you should be quite objective. You must provide facts that can lead to an informed decision by someone else. If you are an outside contractor proposing to construct such a wing, however, your purpose is more persuasive. You will be trying to convince readers that your firm should receive the construction contract.

When preparing to write, therefore, ask yourself two related questions about your purpose.

>> Question 1: Why Am I Writing This Document?

This question should be answered in just one or two sentences, even in complicated projects. Often the resulting purpose statement can be moved as is to the beginning of your outline and later to the first draft.

>> Question 2: What Response Do I Want from Readers?

The first question about purpose leads inevitably to the second about results. Again, your response should be only one or two sentences long. Although brief, it should pinpoint exactly what you want to happen as a result of your document. Are you just giving data for the file? Will information you provide help others do their jobs? Will your document recommend a major change? Unlike the purpose statement, the results statement may not go directly into your document.

The answers to these two questions about purpose and results should be included on the Planning Form your instructor may ask you to use for assignments. Figure 1-3 on pages 7 and 8 includes a copy of the form along with instructions for using it. The last page of this book contains another copy you can duplicate for use with assignments.

Having established your purpose, you are now ready to consider the next part of the writing process: audience analysis.

>>> Analyzing Your Readers

One cardinal rule governs all on-the-job writing:

Write for your reader, not for yourself.

This rule especially applies to science and technology because many readers may know little about your field. The key to avoiding this problem is to examine the main obstacles readers face and adopt a strategy for overcoming them.

Obstacles for Readers

Readers of all backgrounds often have these four problems when reading any technical document:

1. Constant interruptions
2. Impatience finding information they need
3. A different technical background from the writer
4. Shared decision-making authority with others

If you think about these obstacles every time you write, you will be better able to understand and respond to your readers.

>> Obstacle 1: Readers Are Always Interrupted

As a professional, how often will you have the chance to read a report or other document without interruption? Such times are rare. Your reading time will be interrupted by

PLANNING FORM

Name: _____ Assignment: _____

I. Purpose: Answer each question in one or two sentences.

- A. Why are you writing this document?
- B. What response do you want from readers?

II. Audience

- A. Reader Matrix: Fill in names and positions of people who may read the document

	Decision Makers	Advisers	Receivers
Managers			
Experts			
Operators			
General Readers			

- B. Information on individual readers: Answer these questions about the primary audience for this document. If the primary audience includes more than one reader (or type of reader) and there are significant differences between the readers, answer the questions for each (type of) reader. Attach additional sheets as necessary.

Primary audience:

1. What is this reader's technical or educational background?
2. What main question does this person need answered?
3. What main action do you want this person to take?
4. What features of this person's personality might affect his or her reading?

III. Document

- A. What information do I need to include in the
 1. Abstract?
 2. Body?
 3. Conclusion?
- B. What organizational patterns are appropriate to the subject and purpose?
- C. What style choices will present a professional image for me and the organization I represent?

■ **Figure 1-3** ■ Planning Form for all technical documents

Instructions for Completing the Planning Form

The Planning Form is for your use in preparing assignments in your technical communication course. It focuses only on the planning stage of writing. Complete it before you begin your first draft.

1. Use the Planning Form to help plan your strategy for all writing assignments. Your instructor may or may not require that it be submitted with assignments.
2. Photocopy the form on the back page of this book or write the answers to questions on separate sheets of paper, whatever option your instructor prefers. (Your instructor may ask you to use an electronic version or enlarged, letter-sized copies of the form that are included in the Instructor's Resource Manual.)
3. Answer the two purpose questions in one or two sentences each. Be as specific as possible about the purpose of the documents and the response you want—especially from the decision makers.
4. Note that the reader matrix classifies each reader by two criteria: (a) technical levels (shown on the vertical axis) and (b) relationship to the decision-making process (shown on the horizontal axis). Some of the boxes will be filled with one or more names whereas others may be blank. How you fill out the form depends on the complexity of your audience and, of course, on the directions of your instructor.
5. Note that the "Information on Individual Readers" section can be filled out for one or more readers, depending on what your instructor requires.
6. Answer the document questions in one or two sentences each. Refer to Chapter 1 for information about the ABC format and organizing patterns that can be used in documents.

■ Figure 1-3 ■ Continued

meetings and phone calls, so a report often gets read in several sittings. Aggravating this problem is the fact that readers may have forgotten details of the project.

>> Obstacle 2: Readers Are Impatient

Many readers lose patience with vague or unorganized writing. They think, "What's the point?" or "So what?" as they plod through memos, letters, reports, and proposals. They want to know the significance of the document right away.

>> **Obstacle 3: Readers Lack Your Technical Knowledge**

In college courses, the readers of your writing are professors who usually have knowledge of the subject on which you are writing. In your career, however, you will write to readers who lack the information and background you have. They expect a technically sophisticated response, but in language they can understand. If you write over their heads, you will not accomplish your purpose. Think of yourself as an educator; if readers do not learn from your reports, you have failed in your objective.

>> **Obstacle 4: Most Documents Have More Than One Reader**

Readers usually share decision-making authority with others who may read all or just part of the text. Thus, you must respond to the needs of many individuals—most of them have a hectic schedule, are impatient, and have a technical background different from yours.

Ways to Understand Readers

Obstacles to communication can be frustrating, yet there are techniques for overcoming them. First, you must try to find out exactly what information each reader needs. Think of the problem this way—would you give a speech without learning about the background of your audience? Writing depends just as much, if not more, on such analysis. Follow these four steps to determine your readers' needs:

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See Model Report 11: Position Paper for a model document that clearly addresses its purpose and audience.

>> **Audience Analysis Step 1: Write Down What You Know About Your Reader**

To build a framework for analyzing your audience, you need to write down—not just casually think about—the answers to these questions for each reader:

1. What is this reader's technical or educational background?
2. What main question does this person need the answer to?
3. What main action do you want this person to take?
4. What features of this person's personality might affect his or her reading?

The Planning Form in Figure 1-3 includes these four questions.

>> **Audience Analysis Step 2: Talk With Colleagues Who Have Written to the Same Readers**

Often your best source of information about readers is a colleague where you work. Ask around the office or check company files to discover who else may have written to the same audience. Useful information could be as close as the next office.

>> **Audience Analysis Step 3: Find Out Who Makes Decisions**

Almost every document requires action of some kind. Identify decision makers ahead of time so that you can design the document with them in mind. Know the needs of your *most important* reader.

>> Audience Analysis Step 4: Remember That All Readers Prefer Simplicity

Even if you uncover little specific information about your readers you can always rely on one basic fact: Readers of all technical backgrounds prefer concise and simple writing.

Types of Readers

To complete the audience-analysis stage, this section shows you how to classify readers by two main criteria: knowledge and influence. Specifically, you must answer two questions about every potential reader:

1. How much does this reader already know about the subject?
2. What part will this reader play in making decisions?

Then use the answers to these questions to plan your document. The Planning Form in Figure 1-3 provides a reader matrix by which you can quickly view the technical levels and decision-making roles of all your readers. For complex documents, your audience may include many of the 12 categories shown on the matrix. Also, you may have more than one person in each box—that is, there may be more than one reader with the same background and decision-making role.

Technical Levels

On-the-job writing requires that you translate technical ideas into language that nontechnical people can understand. This task can be very complicated because you often have several readers, each with different levels of knowledge about the topic. If you are to “write for your reader, not for yourself,” you must identify the technical background of each reader. Four categories help you classify each reader’s knowledge of the topic.

>> Reader Group 1: Managers

Many technical professionals aspire to become managers. Once into management, they may be removed from hands-on technical details of their profession. Instead, they manage people, set budgets, and make decisions of all kinds. Thus you should assume that management readers are not familiar with fine technical points, have forgotten details of your project, or both. These managers often need

- Background information
- Definitions of technical terms
- Lists and other format devices that highlight points
- Clear statements about what is supposed to happen next

>> Reader Group 2: Experts

Experts include anyone with a good understanding of your topic. They may be well-educated—as with engineers and scientists—but that is not necessarily the case. Whatever their educational levels, most experts in your audience need

- Thorough explanations of technical details
- Data placed in tables and figures
- References to outside sources used in writing the report
- Clearly labeled appendices for supporting information

>> Reader Group 3: Operators

Because decision makers are often managers or technical experts, these two groups tend to get most of the attention. However, many documents also have readers who are operators. They may be technicians in a field crew, workers on an assembly line, salespeople in a department store, or drivers for a trucking firm—anyone who puts the ideas in your document into practice. These readers expect

- A clear table of contents for locating sections that relate to them
- Easy-to-read listings for procedures or instructions
- Definitions of technical terms
- A clear statement of exactly how the document affects their jobs

>> Reader Group 4: General Readers

General readers often have the least amount of information about your topic or field. For example, a report on the environmental impact of a toxic waste dump might be read by general readers who are homeowners in the surrounding area. Most will have little technical understanding of toxic wastes and associated environmental hazards. These general readers often need

- Definitions of technical terms
- Frequent use of graphics, such as charts and photographs
- A clear distinction between facts and opinions

As with managers, general readers must be assured that (1) all implications of the document have been put down on paper and (2) important information has not been buried in overly technical language.

Decision-Making Levels

Figure 1-3 shows that your readers, whatever their technical level, can also be classified by the degree to which they make decisions based on your document. Use the following three levels to classify your audience during the planning process:

>> First-Level Audience: Decision Makers

The first-level audience, the *decision makers*, must act on the information. If, for example, you are comparing two computer systems for storing records at a hospital, the first-level audience decides which unit to purchase. In other words, decision makers translate information into action. They are usually, but not always, managers within the organization.

>> Second-Level Audience: Advisers

This second group could be called *influencers*. Although they do not make decisions themselves, they read the document and give advice to those who make the decisions. Often, the second-level audience is composed of experts, such as engineers and accountants, who are asked to comment on technical matters. After reading the summary, a decision-making manager may refer the rest of the document to advisers for their comments.

>> Third-Level Audience: Receivers

Some readers do not take part in the decision-making process, but only receive information contained in the document. For example, a report recommending changes in the hiring of fast-food workers may go to the store managers after it has been approved, just so they can put the changes into effect. This third-level audience usually includes readers defined as *operators* in the previous section—that is, those who may be asked to follow guidelines or instructions contained in a report.

Using all this information about technical and decision-making levels, you can analyze each reader's (1) technical background with respect to your document and (2) potential for making decisions after reading what you present. Then you can move on to the research and outline stages of writing.

>>> Planning Your Documents

Given the varied backgrounds and interests of your readers, you must answer one essential question: How can you best organize information to satisfy so many different people?

Figure 1-4 shows you three possible options for organizing information for the technical expertise of a mixed technical audience, but only one is recommended in this book. Writers who choose Option A direct their writing to the most technical people. Writers

■ **Figure 1-4** ■
Options for
organizing
information

Experts	Operators	Managers	General Readers
Option A Organize information for technical readers			
		Option B Organize information for less-technical readers	
Option C Organize information for all readers			

who choose Option B respond to the dilemma of a mixed technical audience by finding the lowest common denominator—that is, they write to the level of the least technical person. Each option satisfies one segment of readers at the expense of the others.

Option C is preferred in technical communication for mixed readers. It encourages you to organize documents so that all readers—both technical and nontechnical—get what they need. The rest of this chapter provides strategies for developing this option. It describes general principles of organization and guidelines for organizing entire documents, individual document sections, and paragraphs.



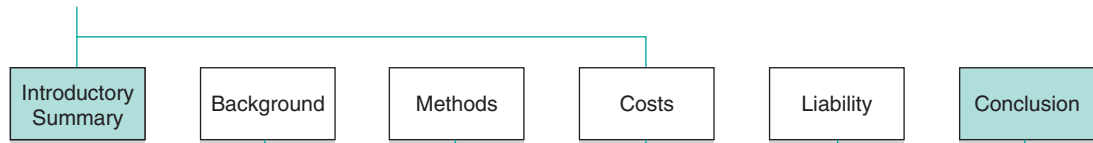
Three Principles of Organization

Good organization starts with careful analysis of your audience. Most readers are busy, and they skip around as they read. Think about how you examine a news organization's Web site or a weekly newsmagazine. You are likely to take a quick look at articles of special interest to you; then you might read them more thoroughly, if there is time. That approach also resembles how your audience treats technical reports and other work-related documents. If important points are buried in long paragraphs or sections, busy readers may miss them. Three principles respond realistically to the needs of your readers:

>> Principle 1: Write Different Parts for Different Readers

The longer the document, the less likely it is that anyone will read it from beginning to end. As shown in Figure 1-5, they use a *speed-read approach* that includes these steps:

- Step 1: Quick scan.** Readers scan easy-to-read sections like executive summaries, introductory summaries, introductions, tables of contents, conclusions, and recommendations. They pay special attention to beginning and end sections, especially in documents longer than a page or two, and to illustrations.
- Step 2: Focused search.** Readers go directly to parts of the document body that give them what they need at the moment. To find information quickly, they search for navigation devices like subheadings, listings, and white space in margins to guide their reading. (See Chapter 3 for a discussion of page design.)
- Step 3: Short follow-ups.** Readers return to the document, when time permits, to read or reread important sections.

STEP 1: Quick Scan**STEP 2: Focused Search****STEP 3: Short Follow-Ups**

Can involve any section, especially the Introductory Summary

■ **Figure 1-5** ■ Sample speed-read approach to short proposal

Your job is to write in a way that responds to this nonlinear and episodic reading process of your audience. Most important, you should direct each section to those in the audience most likely to read that particular section.

>> Principle 2: Emphasize Beginnings and Endings

Suspense fiction relies on the interest and patience of readers to piece together important information. The writer usually drops hints throughout the narrative before finally revealing who did what to whom. Technical communication operates differently. Busy readers expect to find information in predictable locations without having to search for it. Their first-choice locations for important information are as follows:

- The beginning of the entire document
- The beginnings of report sections
- The beginnings of paragraphs

Emphasizing beginnings and endings responds to the reading habits and psychological needs of readers. At the beginning, they want to know where you are heading. They need a simple road map for the rest of the passage. In fact, if you do not provide something important at the beginnings of paragraphs, sections, and documents, readers will start guessing the main point themselves. At the ending, readers expect some sort of wrap-up or transition; your writing should not simply drop off. The following paragraph begins and ends with such information (*italics added*):

Already depleted sea turtle, marine mammal, seabird, and noncommercial fish populations are endangered by incidental capture in fishing gear. Worldwide, about 25 percent of the catch is discarded, either because it is not commercially valuable or because of regulatory requirements that prohibit keeping undersized or nontargeted marine life. Destructive fishing practices, such as bottom trawling and dredging, are damaging vital habitat upon which fish and other living resources depend. *Taken together, overfishing, bycatch, and habitat destruction are changing relationships among species in food webs and altering the function of marine ecosystems.* [Pew Oceans Commission, *America's Living Oceans: Charting a Course for Sea Change* (Arlington, VA: Pew Oceans Commission, 2003), 5, 9.]

The first sentence gives readers an immediate impression of the two topics to be covered in the paragraph. The paragraph body explores details of both topics. Then the last sentence flows smoothly from the paragraph body by reinforcing the main point about over fishing.

>> Principle 3: Repeat Key Points

You have learned that different people focus on different sections of a document. Sometimes no one carefully reads the entire report. For example, managers may have time to read only the summary, whereas technical experts may skip the leadoff sections and go directly to “meaty” technical sections with supporting information. These varied reading patterns require a *redundant* approach to organization—you must repeat important information in different sections for different readers. Your strategic repetition of a major finding, conclusion, or recommendation gives helpful reinforcement to readers always searching for an answer to the “So what?” question as they read. Now we are ready to be more specific about how the three general principles of organization apply to documents, document sections, and paragraphs.

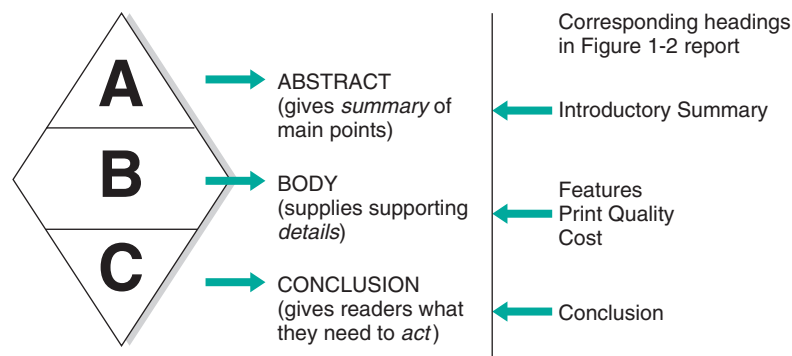
ABC Format for Documents

Technical documents should assume a three-part structure that consists of a beginning, a middle, and an end. This book labels this structure the *ABC Format* (for *Abstract*, *Body*, and *Conclusion*). Visually, think of this pattern as a three-part diamond structure, as shown in Figure 1-6:

- **Abstract:** A brief beginning component is represented by the narrow top of the diamond, which leads into the body.
- **Body:** The longer middle component is represented by the broad, expansive portion of the diamond figure.
- **Conclusion:** A brief ending component is represented by the narrow bottom of the diamond, which leads away from the body.

First and foremost, the ABC format pertains to the organization of entire documents. However, the same beginning-middle-end strategy applies to the smaller units of discourse—document sections and paragraphs.

Figure 1-2 (page 4) includes a memo report that conforms to this structure. The following sections discuss the three ABC components in detail.



■ **Figure 1-6** ■ ABC format for all documents

Document Abstract: The “Big Picture” for Decision Makers

Every document should begin with an overview. As used in this text, *abstract* is defined as follows:



For more examples, see Model Abstract 1: Report and Model Abstract 2: Proposal.

Abstract: brief summary of a document's main points. Although its makeup varies with the type and length of the document, an abstract usually includes (1) a clear purpose statement for the document, (2) the most important points for decision makers, and (3) a list or description of the main sections that follow the abstract. As a capsule version of the entire document, the abstract should answer readers' typical mental questions, such as: "How does this document concern me?" "What's the bottom line?" "So what?"

Abstract information is provided under different headings, depending on the document's length and degree of formality. Some common headings are "Summary," "Executive Summary," "Introductory Summary," "Overview," and "Introduction." The abstract may vary in length from a short paragraph to a page or so. Its purpose, however, is always the same: to provide decision makers with highlights of the document.

For example, assume you are an engineer who has evaluated environmental hazards for the potential purchaser of a shopping mall site. Here is how the summary might read:

As you requested, we have examined the possibility of environmental contamination at the site being considered for the new Klinesburg Mall. Our field exploration revealed two locations with deposits of household trash, which can be easily cleaned up. Another spot has a more serious deposit problem of 10 barrels of industrial waste. However, our inspection of the containers and soil tests revealed no leaks.

Given these limited observations and tests, we conclude that the site poses no major environmental risks and recommend development of the mall. The rest of this report details our field activities, test analyses, conclusions, and recommendations.

You have provided the reader with a purpose for the report, an overview of important information for decision makers, and a reference to the four sections that follow. In doing so, you have answered the following questions, among others, for the readers:

- What are the major risks at the site?
- Are these risks great enough to warrant not buying the land?
- What major sections does the rest of the report contain?

This general abstract, or overview, is mainly for decision makers. Highlights must be brief, yet free of any possible misunderstanding. On some occasions, you may need to state that further clarification is included in the text, even though that point may seem obvious. For example, if your report concerns matters of safety, the overview may not be detailed enough to prevent or eliminate risks. In this case, state this point clearly so that the reader will not misunderstand or exaggerate the purpose of the abstract.

Document Body: Details for All Readers

The longest part of any document is the body. As used in this book, the *body* is defined as follows:

Body: the middle section(s) of the document providing supporting information to readers, especially those with a technical background. Unlike the abstract and conclusion, the body component allows you to write expansively about items, such as (1) the background of the project; (2) field, lab, office, or any other work on which the document is based; and (3) details of any conclusions, recommendations, or proposals that might be highlighted at the beginning or end of the document. The body answers this main reader question: “What support is there for points put forth in the abstract at the beginning of the document?”

Managers may read much of the body, especially if they have a technical background and if the document is short. Yet the more likely readers are technical specialists who (1) verify technical information for the decision makers or (2) use your document to do their jobs. In writing the body, use the following guidelines:

- **Separate fact from opinion.** Never leave the reader confused about where opinions begin and end. Body sections usually move from facts to opinions that are based on facts. To make the distinction clear, preface opinions with phrases, such as “We believe that,” “I feel that,” “It is our opinion that,” and the like. Such wording gives a clear signal to readers that you are presenting judgments, conclusions, and other nonfactual statements. Also, you can reinforce the facts by including data in graphics.
- **Adopt a format that reveals the structure.** Use frequent headings and subheadings to help busy readers locate important information immediately.
- **Use graphics whenever possible.** Use graphics to draw attention to important points. Today more than ever, readers expect visual reinforcement of your text, particularly in more persuasive documents like proposals.

By following these guidelines, which apply to any document, you will make detailed body sections as readable as possible. They keep ideas from becoming buried in text and show readers what to do with the information they find.

Document Conclusion: Wrap-Up Leading to Next Step

Your conclusion deserves special attention, for readers often recall first what they have read last. We define the *conclusion* component as follows:

Conclusion: the final section(s) of the document bringing readers—especially decision makers—back to one or more central points already mentioned in the body. Occasionally, it may include one or more points not previously mentioned. In any case, the conclusion provides closure to the document and often leads to the next step in the writer’s relationship with the reader.

The conclusion component may have any one of several headings, depending on the type and length of the document. Possibilities include “Conclusion,” “Closing,” “Closing

Remarks,” and “Conclusions and Recommendations.” A conclusion component answers the following types of questions:

- What major points have you made?
- What problem have you tried to solve?
- What should the reader do next?
- What will you do next?
- What single idea do you want to leave with the reader?

Whichever alternative you choose, your goal is to return to the main concerns of the most important readers—decision makers. Both the abstract and conclusion, in slightly different ways, should respond to the needs of this primary audience.

>>> Ethics in the Workplace

This section presents a set of ethical guidelines for the workplace, and shows how ethical guidelines, when applied to writing, can have legal applications. Throughout this book are assignments in which your own ethical decisions play an important role.

Ethical Guidelines for Work

As with your personal life, your professional life holds many opportunities for demonstrating your views of what is right or wrong. Most occur daily and without much fanfare, but cumulatively they compose our personal approach to morality. Thus, our belief systems are revealed by the manner in which we respond to this continuous barrage of ethical dilemmas.

Obviously, not everyone in the same organization or profession has the same ethical beliefs, nor should they. After all, each person’s understanding of right and wrong flows from individual experiences, upbringing, religious beliefs, and cultural values. Some *ethical relativists* even argue that ethics only makes sense as a descriptive study of what people do believe, not a prescriptive study of what they should believe. Yet there are some basic guidelines that, in our view, should be part of the decision-making process in every organization. Although they may be displayed in different ways in different cultures, they should transcend national identity, cultural background, and family beliefs. In other words, these guidelines represent what, ideally, should be the *core* values for employees at international companies.

The guidelines in this section are common in many professional codes of ethics. These are general guidelines and provide a good foundation for ethical behavior in the workplace. However, you should also become familiar with the ethical guidelines specific to your employer and professional organizations.

>> Ethics Guideline 1: Be Honest

First, you should relate information accurately and on time—to your colleagues, to customers, and to outside parties, such as government regulators. This guideline also means

you should not mislead listeners or readers by leaving out important information that relates to a situation, product, or service, including information about any conflicts of interest. You should interpret data carefully and present estimates as accurately as possible. In other words, give those with whom you communicate the same information that you would want presented to you.

>> **Ethics Guideline 2: Do No Harm**

Technical communicators often work in fields that affect public health and safety. You should avoid practices, inaccuracies, or mistakes that can harm people or property.

You should also support a positive and constructive work atmosphere. One way to achieve such a working environment is to avoid words or actions calculated to harm others. For example, avoid negative, rumor-laden conversations that hurt feelings, spread unsupported information, or waste time.

>> **Ethics Guideline 3: Be Fair**

You should treat those around you fairly, regardless of differences in race, religion, disability, age, or gender. You should also be aware of, and respect, differences in culture. This is especially important as business becomes more global.

>> **Ethics Guideline 4: Honor Intellectual Property Rights**

Of course you should follow copyright and patent laws, but you should also respect the work that others have put into developing and presenting their ideas. Credit others for ideas, text, or images that you have used.

When collaborating with others, show appreciation for their contribution, and welcome their input. Offer and accept feedback that will make the final product stronger.

>> **Ethics Guideline 5: Respect Confidentiality**

Remember that you are acting on the part of both your employer and your clients. Disclose sensitive information only with permission, and obtain written releases before you share materials. This is especially important for contract and freelance workers, who must have a portfolio of accomplishments to share with prospective clients. If you share confidential information with a prospective client, you show that you can not be trusted with sensitive material.

>> **Ethics Guideline 6: Be Professional**

When you are working, you represent your profession. Not only does this mean that you should act in an honorable manner, but also that you should meet deadlines with quality work. One way to do this is to keep current on developments in your field. Join a professional organization like the Association of Computing Machinery's Special Interest Group on the Design of Communication (ACM SIGDOC), read journal and magazine articles in your field, and participate in continuing education activities.

In your career, you should develop and apply your own code of ethics, making sure that it follows these guidelines and the guidelines of your employer and your professional organization.

Ethics and Legal Issues in Writing

Writing—whether on paper, audiotape, videotape, or computer screen—presents a special ethical challenge for demonstrating your personal code of ethics. Some countries, such as the United States, have a fairly well-developed legal context for writing, which means you must pay great attention to detail as you apply ethical principles to the writing process. This section highlights some common guidelines.

■ Acknowledge Sources for Information Other Than Common

Knowledge You are obligated to provide sources for any information other than common knowledge. *Common knowledge* is usually considered to be factual and non-judgmental information that could be found in general sources about a subject. The sources for any other types of information beyond this definition must be cited in your document.

■ Seek Written Permission Before Borrowing Extensive Text

Generally, it is best to seek written permission for borrowing more than a few hundred words from a source, especially if the purpose of your document is profit. This so-called “fair use” is, unfortunately, not clearly defined and subject to varying interpretations. It is best to (1) consult a reference librarian or other expert for an up-to-date interpretation of the application of fair use to your situation and (2) err on the side of conservatism by asking permission to use information, if you have any doubt. This probably has not been an issue in papers you have written for school, because they were for educational use and were not going to be published. However, this issue should be addressed in any writing you do outside of school.

■ Seek Written Permission Before Borrowing Graphics

Again, you probably have not been concerned about this issue in projects you have created in school, but you must seek permission for any graphics you borrow for projects created outside of school. This guideline applies to any nontextual element, whether it is borrowed directly from the original or adapted by you from the source. Even if the graphic is not copyrighted, such as in an annual report from a city or county, you should seek permission for its use.

■ Seek Legal Advice When You Cannot Resolve Complex Questions

Some questions, such as the use of trademarks and copyright, fall far outside the expertise of most of us. In such cases it is best to consult an attorney who specializes in such law. Remember that the phrase *Ignorance is bliss* has led many a writer into problems that could have been prevented by seeking advice when it was relatively cheap—at the beginning. Concerning U.S. copyrights in particular, you might first want to consult free information provided by the U.S. Copyright Office at its Web site (www.copyright.gov).



In the final analysis, acting ethically on the job means thinking constantly about the way in which people are influenced by what you do, say, and write. Also, remember that what you write could have a very long shelf life, perhaps to be used later as a reference for legal proceedings. Always write as if your professional reputation could depend on it, because it just might.

>>> Chapter Summary

Technical communication refers to the many kinds of writing and speaking you will do in your career. In contrast to most academic writing, technical communication aims to get something done (not just to demonstrate knowledge), relays information from someone (you) more knowledgeable about the topic to someone (the reader) less knowledgeable about it, is read by people from mixed technical and decision-making levels, presents ideas clearly and simply, and often uses data and graphics to provide support.

Good technical communication calls on special skills, especially in organization. Writers should follow three guidelines for organizing information: (1) Write different parts of the document for different readers, (2) place important information at the beginnings and endings, and (3) repeat key points throughout the document.

This chapter recommends the ABC format for organizing technical documents. This format includes an *Abstract* (summary), a *Body* (supporting details), and a *Conclusion* (wrap-up and transition to next step). The abstract section is particularly important because most readers give special attention to the start of a document.

Your technical communication always exists within an ethical context. Use the general ethics guidelines presented in this chapter, but also become familiar with the ethical guidelines of your field of professional practice.

Chapter | 3 | Visual Design



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As with the organizing principles discussed in Chapter 1, good visual design can help your readers find the information that they need. This chapter covers the visual elements of technical documents, another basic building block in technical communication. An operating definition is as follows:

Visual design: A term that refers to creating clear, readable, and visually interesting documents

through judicious use of page design, including white space, headings, lists, font choice, and of graphic design elements.

This chapter presents guidelines and examples for visual design, as well as commentary on the use of computers in the design process.

>>> Elements of Page Design

In the workplace, readers are busy, and few take the time to read a document from cover to cover. Some documents, such as manuals, are used as reference works, only consulted as a last resort. Your challenge is to make your documents inviting by making pages interesting to the eye. A document may offer your reader great opportunities but never get read. Why? Because it does not look inviting.

Good organization, as pointed out in Chapter 1, can fight readers' indifference by giving information when and where they want it. However, to get and keep readers interested, you must use effective page design—on each page of your document. Each page needs the right combination of visual elements to match the needs of your readers and the purpose of the document.

Good page design can work with good organization to help readers find information that they need in a document. Readers can recognize important information by its location on a page, by use of contrast, or by repetition of identically formatted elements, such as guidelines or warning boxes.

Many firms know the benefits of visual imaging so they develop company style sheets for frequently used documents, including letters, memos, various types of reports, and proposals. Once developed, these style sheets are assembled into a style manual and distributed for general use. To make universal use easier, they are often loaded as templates or styles into the company's text editing software. If the company you work for does not use a style manual, use the elements of page design shown in this chapter to develop your own style sheets. More information about templates and style sheets appears on pages 66 to 69 at the end of this chapter.

White Space

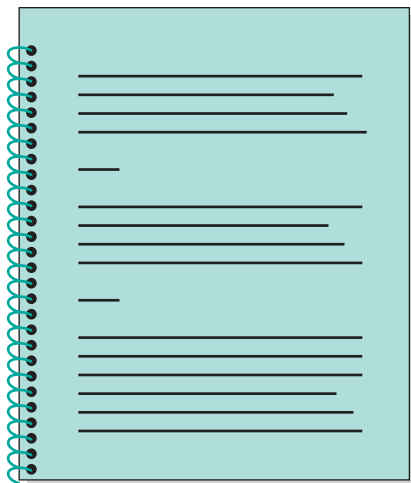
The term *white space* simply means the open places on the page with no text or graphics—literally, the white space (assuming you are using white paper). Experts have learned that readers are attracted to text because of the white space that surrounds it, as with a newspaper advertisement that includes a few lines of copy in the middle of a white page. Readers connect white space with important information.

In technical communication you should use white space in a way that (1) attracts attention, (2) guides the eye to important information on the page, (3) relieves the boredom of reading text, and (4) helps readers organize information. Here are some opportunities for using white space effectively:

1. **Margins:** Most readers appreciate generous use of white space around the edges of text. Marginal space tends to frame your document, so the text does not appear to push the boundaries of the page. Good practice is to use 1- to 1 1/2-inch margins, with more space on the bottom margin. When the document is bound, the margin on the edge that is bound should be larger than the outside margin to account for the space taken by the binding (Figure 3-1.)

2. **Hanging Indents:** Some writers place headers and subheads at the left margin and indent the text block an additional inch or so, as shown in Figure 3-2. Headers and subheads force the readers' eyes and attention to the text block. Another common use of hanging indents is bulleted and numbered lists.

3. **Line Spacing:** When choosing single, double, or 1 1/2 line spacing, consider the document's degree of formality. Letters, memos, short reports, and other documents read in one sitting are usually single-spaced. Longer documents, especially if they are formal, are usually 1 1/2-space or double-spaced, sometimes with extra spacing between paragraphs. Manuscripts or documents that will be typeset professionally are usually double-spaced.

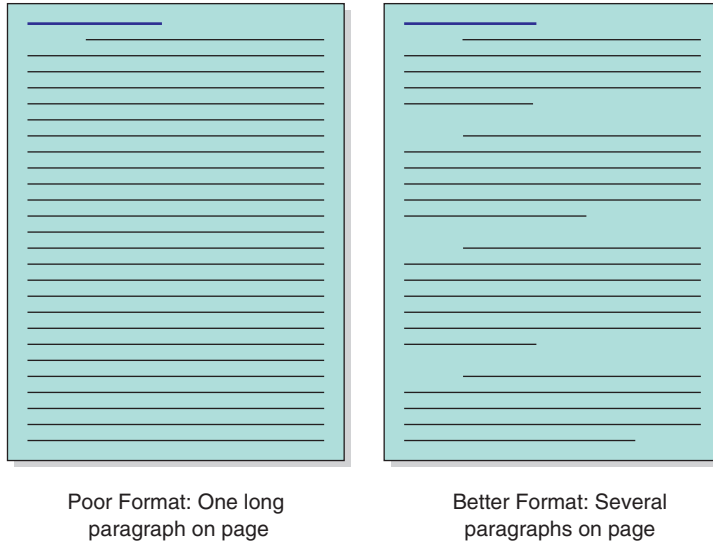


■ **Figure 3-1** ■ Use of white space: margins



■ **Figure 3-2** ■ Use of white space: hanging indents

■ **Figure 3-3** ■ Use of white space: paragraphs



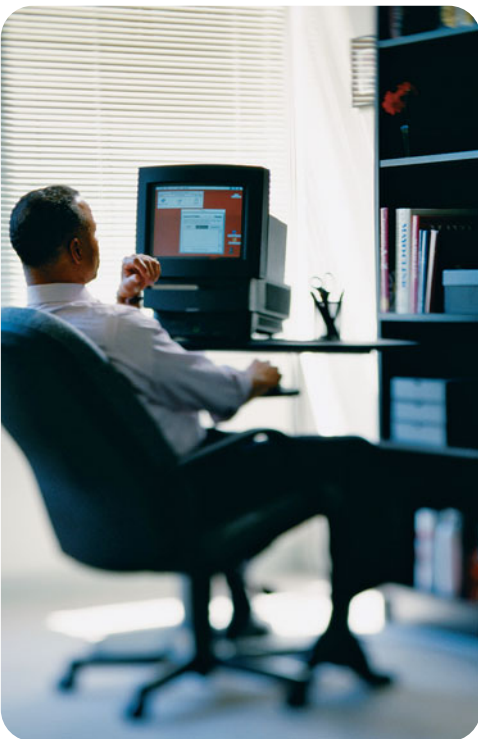
4. Paragraph Length: New paragraphs give readers a chance to regroup as one topic ends and another begins. These shifts also have a visual impact. The amount of white space produced by paragraph lengths can shape reader expectations. For example, two long paragraphs suggest a heavier reading burden than do three or four paragraphs of differing lengths. Thus, it is helpful to break complex information into shorter paragraphs. Many readers skip long paragraphs, so vary paragraph lengths and avoid putting more than 10 lines in any one paragraph (Figure 3-3).

5. Heading Space and Ruling: White space helps the reader connect related information immediately. Always have slightly more space above a heading than below it. That extra space visually connects the heading with the material into which it leads. In a double-spaced document, for example, you would add a third line of space between the heading and the text that came before it. In addition, some writers add a horizontal line across the page above headings, to emphasize the visual break.

In summary, well-used white space can add to the persuasive power of your text. As with any design element, however, it can be overused and abused. Make sure there is a reason for every decision you make with regard to white space on your pages.

Lists

Technical communication benefits from the use of lists. Readers welcome your efforts to cluster items into lists for easy reading. In fact, almost any group of three or more related points can be made



into a bulleted or numbered listing. Following are some points to consider as you apply this important feature of page design:

1. **Typical uses:** Lists emphasize important points and provide a welcome change in format. Because they attract more attention than text surrounding them, they are usually reserved for these uses:

Examples

Reasons for a decision

Conclusions

Recommendations

Steps in a process

Cautions or warnings about a product

Limitations or restrictions on conclusions

2. **Number of items:** The best lists subscribe to the often-quoted rule that people retain no more than five to nine items in their short-term memory. A listing of more than nine items may confuse rather than clarify an issue. Consider placing 10 or more items in two or three groupings, or grouped lists, as you would in an outline. This format gives the reader a way to grasp information being presented.

3. **Use of bullets and numbers:** The most common visual clues for listings are numbers and *bullets* (enlarged dots or squares like those used in the following listing). Following are a few pointers for choosing one or the other:

- **Bullets.** Best in lists of five or fewer items, unless there is a special reason for using numbers.
- **Numbers.** Best in lists of more than five items or when needed to indicate an ordering of steps, procedures, or ranked alternatives. Remember that your readers sometimes infer sequence or ranking in a numbered list.

4. **Format on page:** Every listing should be easy to read and pleasing to the eye. The following specific guidelines cover practices preferred by most readers:

- **Indent the listing.** Although there is no standard list format, readers prefer lists that are indented farther than the standard left margin. Five spaces are adequate.
- **Hang your numbers and bullets.** Visual appeal is enhanced by placing numbers or bullets to the left of the margin used for the list, as done with the items here.
- **Use line spaces for easier reading.** When one or more listed items contain over a line of text, an extra line space between listed items can enhance readability.
- **Keep items as short as possible.** Depending on purpose and substance, lists can consist of words, phrases, or sentences—such as the list you are reading.

Whichever format you choose, pare down the wording as much as possible to retain the impact of the list format.

5. **Parallelism and lead-ins:** Make the list easy to read by keeping all points grammatically parallel and by including a smooth transition from the lead-in to the listing itself. (The term *lead-in* refers to the sentence or fragment preceding the listing.) *Parallel* means that each point in the list is in the same grammatical form, whether a complete sentence, verb phrase, or noun phrase. If you change form in the midst of a listing, you take the chance of upsetting the flow of information.

Example: To complete this project, we plan to do the following:

- Survey the site
- Take samples from the three boring locations
- Test selected samples in our lab
- Report the results of the study

The listed items are in verb form (note the introductory words *survey*, *take*, *test*, and *report*).

6. **Punctuation and capitalization:** Although there are acceptable variations on the punctuation of lists, preferred usage includes a colon before a listing, no punctuation after any of the items, and capitalization of the first letter of the first word of each item. Refer to the alphabetized Handbook under “Punctuation: Lists” for alternative ways to punctuate lists.

In-Text Emphasis

Sometimes you want to emphasize an important word or phrase within a sentence. Computers give you these options: underlining, boldface, italics, and caps. The least effective are FULL CAPS and underlining. Both are difficult to read within a paragraph and distracting to the eye. The most effective highlighting techniques are *italics* and **boldface**; they add emphasis without distracting the reader.

Whatever typographical techniques you select, use them sparingly. They can create a busy page that leaves the reader confused about what to read. Excessive in-text emphasis also detracts from the impact of headings and subheadings, which should be receiving significant attention.



The Model Sales Proposal shows how elements of page design can call attention to important information.

>>> Elements for Navigation

As noted in Chapter 1, your audience will be busy and rarely read a longer report or document from the first page to the last. One way that you can help readers can locate the information they need is by using visual design to create navigational tools in your documents. You may be used to thinking of navigational tools in electronic texts such as Web pages or even PDFs, but print documents also use navigational devices such as tables of contents, indexes, headings, running headers and footers, and even color coding.

Headings

Headings are brief labels used to introduce each new section or subsection of text. They serve as (1) a signpost for the reader who wants to know the content, (2) a grabber to entice readers to read documents, and (3) a visual oasis of white space where the reader gets relief from text. Following are some general guidelines:

1. **Use your outline to create headings and subheadings.** A well-organized outline lists major and minor topics. With little or no change in wording, topics can be converted to headings and subheadings within the document.

2. **Use substantive wording.** Headings give readers an overview of the content that follows. They can determine whether readers read or skip the text. Strive to use concrete rather than abstract nouns, even if the heading must be a bit longer.

Original: “Background”

Revised: “How the Simmons Road Project Got Started” or “Background on Simmons Road Project”

Original: “Costs”

Revised: “Production Costs of the FastCopy 800” or “Producing the FastCopy 800: How Much?”

3. **Maintain parallel form in wording.** Headings of equal value and degree should have parallel grammatical form:

A. *Headings That Lack Parallel Form*

Scope of Services

How Will Fieldwork Be Scheduled?

Establish Contract Conditions

B. *Revised Headings with Parallel Form*

Scope of Services

Schedule for Fieldwork

Conditions of Contract

4. **Establish clear hierarchy in headings.** Whatever typographical techniques you choose, your readers must be able to distinguish one heading level from another.

■ **Consider using decimal headings for long documents.** Decimal headings include a hierarchy of numbers for every heading and subheading listed in the table of contents. Many an argument has been waged over their use. People who like them say that they help readers find their way through documents and refer to subsections in later discussions. People who dislike them say that they are cumbersome and give the appearance of bureaucratic writing.

Unless decimal headings are expected by your reader, use them only with formal documents that are fairly long. Following is the normal progression of numbering in decimal headings for a three-level document:

```
1.0 xxxxxxxxxxxxxxxx
    1.1 xxxxxxxxxxxxxx
        1.1.1 xxxxxxxxxxxxxx
        1.1.2 xxxxxxxxxxxxxx
    1.2 xxxxxxxxxxxxxx
        1.2.1 xxxxxxxxxxxxxx
        1.2.2 xxxxxxxxxxxxxx
2.0 xxxxxxxxxxxxxxxx
    2.1 xxxxxxxxxxxxxx
        2.1.1 xxxxxxxxxxxxxx
        2.1.2 xxxxxxxxxxxxxx
    2.2 xxxxxxxxxxxxxx
3.0 xxxxxxxxxxxxxxxx
```

Headers and Footers

As noted earlier, running headers and footers are important navigation devices in documents, and most word processing programs make the creation of headers and footers easy. In addition to being able to insert automatic page numbering, you can insert other information, such as short titles, your name, or your organization's name, on each page. You can decide where to position that information and you can hide it on selected pages. Some organizations put information, such as the computer filename or project identification number in document footers.

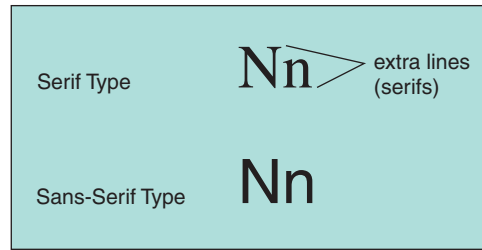
Color

Use of color, like fonts, focuses your reader's attention on important details. When used indiscriminately—inserted into a document just to show that color can be used—it is distracting.

Limit your use of color in routine documents for two reasons: (1) When you use professional printers, printing in color can be very expensive and (2) when you are using desktop printers, printing color documents can be very slow.

>>> Fonts

Your choice of size and type of font can affect how easy it is for your readers to find information, as well as how they interpret your tone and professionalism.



■ **Figure 3-4** ■ Font types

Font Types

Your choice of fonts may be either prescribed by your employer or determined by you on the basis of (1) the purpose of the document, (2) the image you want to convey, and (3) your knowledge of the audience.

Font types are classified into two main groups:

- **Serif fonts:** Characters have “tails” at the ends of the letterlines.
- **Sans-serif fonts:** Characters do not have tails (Figure 3-4).

If you are able to choose your font, the obvious advice is to use the one that you know is preferred by your readers. If you have no reader-specific guidelines, following are four general rules:

1. **Use serif fonts for regular text in your documents.** The tails on letters make letters and entire words more visually interesting to the reader’s eye, and they reduce eye fatigue.
2. **Consider using another typeface—sans serif—for headings.** Headings benefit from a clean look that emphasizes the white space around letters. Sans-serif type helps attract attention to these elements of organization within your text.
3. **Avoid too many font variations in the same document.** Your rule of thumb might be to use no more than two fonts per document: one for text and another for headings and subheadings.
4. **Avoid unusual or novelty fonts.** To avoid problems in documents that are transmitted electronically, use fonts that are available on most computers. You should also avoid novelty fonts like Comic Sans or Broadway; these do not communicate a professional image.

Type Size

Most technical writing is printed in 10- or 12-point type. When you are choosing type size, however, be aware that the actual size of letters varies among font types. Some 12-point type appears larger than other 12-point type. Differences stem from the fact that your selection of a font affects (1) the thickness of the letters, (2) the size of lowercase letters, and (3) the length and style of the parts of letters that extend above and below the line. Before selecting your type size, run samples on your printer so that you are certain of how your copy will appear in final form.

With basics of page design foundation, you can decide to include graphics to clarify your document. The next sections provide general and specific guidelines for various graphics.

>>> General Guidelines for Graphics



Model Report 12:
Final Report uses
graphics effectively.

A few basic guidelines apply to all graphics. Keep the following fundamentals in mind as you move from one type of illustration to another.

>> Graphics Guideline 1: Determine the Purpose of the Graphic

Graphics, like text, should only be used if they serve a purpose. Ask yourself the following questions:

- What kind of visual information does your audience need to better understand the object, process, problem, or solution?
- What type of graphic can be used to present the data in the most interesting and informative way?
- Are any special symbols, colors, or font styles needed to reinforce the data?

>> Graphics Guideline 2: Evaluate the Accuracy and Validity of the Data

Unless the information you plan to include in your document is accurate, you run the risk of presenting information that could damage your credibility as well as the credibility of the document.

- Check the accuracy of information
- Make sure the source is reputable
- Ensure that data are not distorted by flawed scales or images

>> Graphics Guideline 3: Refer to All Graphics in the Text

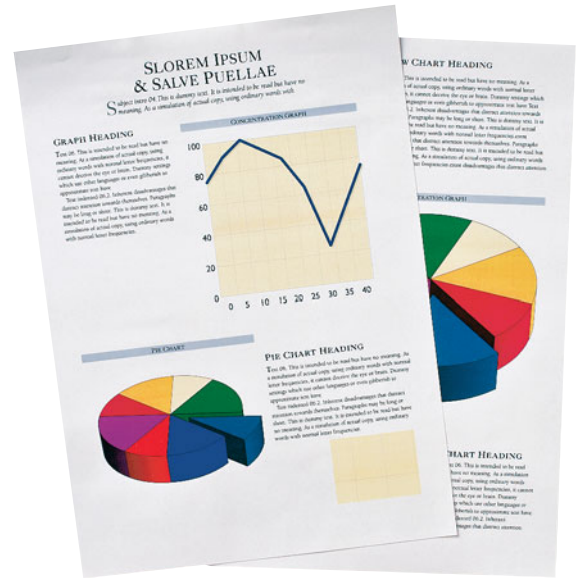
With a few exceptions—such as cover illustrations used to grab attention—graphics should be accompanied by clear references within your text.

- Include the graphic number in Arabic, not Roman, when you are using more than one graphic
- Include the title, and sometimes the page number, if either is needed for clarity or emphasis
- Incorporate the reference smoothly into text wording
- Highlight significant information being communicated by the graphic

Following are two ways to phrase and position a graphics reference. In Example 1, there is the additional emphasis of the graphic's title, whereas in Example 2, the title is left out. Also, note that you can draw more attention to the graphic by placing the reference at the start of the sentence in a separate clause, or you can relegate the

reference to a parenthetical expression at the end or middle of the passage. Choose the option that best suits your purposes.

- **Example 1:** In the past five years, 56 businesses in the county have started in-house recycling programs. The result has been a dramatic shift in the amount of property the county has bought for new waste sites, as shown in Figure 5.
- **Example 2:** As shown in Figure 5, the county has purchased much less land for landfills during the past five years. This dramatic reduction results from the fact that 56 businesses have started in-house recycling programs.



>> Graphics Guideline 4: Think About Where to Put Graphics

In most cases, locate a graphic close to the text in which it is mentioned. This immediate reinforcement of text by an illustration gives graphics their greatest strength. Variations of this option, as well as several other possibilities, include the following:

- **Same page as text reference:** A simple visual, such as an informal table, should go on the same page as the text reference if you think it too small for a separate page.
- **Page opposite text reference:** A complex graphic, such as a long table, that accompanies a specific page of text can go on the page opposite the text—that is, on the opposite page of a two-page spread. Usually this option is exercised *only* in documents that are printed on both sides of the paper throughout.
- **Page following first text reference:** Most text graphics appear on the page after the first reference. If the graphic is referred to throughout the text, it can be repeated at later points. (*Note:* Readers prefer to have graphics positioned exactly where they need them, rather than their having to refer to another part of the document.)
- **Attachments or appendices:** Graphics can go at the end of the document in two cases: first, if the text contains so many references to the graphic that placement in a central location, such as an appendix, would make it more accessible; and second, if the graphic contains less important supporting material that would only interrupt the text.

>> Graphics Guideline 5: Position Graphics Vertically When Possible

Readers prefer graphics they can view without having to turn the document sideways. However, if the table or figure cannot fit vertically on a standard $8\frac{1}{2} \times 11$ -inch page, either use a foldout or place the graphic horizontally on the page. In the latter case, position the illustration so that the top is on the left margin. (In other words, the page must be turned clockwise to be viewed.)

>> Graphics Guideline 6: Avoid Clutter

Let simplicity be your guide. Readers go to graphics for relief from or reinforcement of the text. They do not want to be bombarded by visual clutter. Omit information that is not relevant to your purpose while still making the illustration clear and self-contained. Also, use enough white space so that the readers' eyes are drawn to the graphic. The final section of this chapter discusses graphics clutter in more detail.

>> Graphics Guideline 7: Provide Titles, Notes, Keys, and Source Data

Graphics should be as self-contained and self-explanatory as possible. Moreover, they must include any borrowed information. Follow these basic rules for format and acknowledgment of sources:

- **Title:** Follow the graphic number with a short, precise title—either on the line beneath the number or on the same line after a colon (e.g., “Figure 3: Salary Scales”).
- **Tables:** The number and title go at the top. (As noted in Table Guideline 1 on page 55, one exception is informal tables, which have no table number or title.)
- **Figures:** The number and title usually go below the illustration. Center titles or place them flush with the left margin.
- **Notes for explanation:** When introductory information for the graphic is needed, place a note directly underneath the title or at the bottom of the graphic.
- **Keys or legends for simplicity:** If a graphic needs many labels, consider using a legend or key, which lists the labels and corresponding symbols on the graphic. For example, a pie chart might have the letters *A*, *B*, *C*, *D*, and *E* printed on the pie pieces and a legend at the top, bottom, or side of the figure listing what the letters represent.
- **Source information at the bottom:** You have an ethical, and sometimes legal, obligation to cite the person, organization, or publication from which you borrowed information for the figure. Either (1) precede the description with the word *Source* and a colon or (2) if you borrowed just part of a graphic, introduce the citation with *Adapted from*.

As well as citing the source, it is sometimes necessary to request permission to use copyrighted or proprietary information, depending on how you use it and how much you are using. (A prominent exception is most information provided by the federal government; most government publications are not copyrighted.) Consult a reference librarian for details about seeking permission.

>>> Specific Guidelines for Six Graphics

Illustrations come in many forms. Almost any nontextual part of your document can be placed under the umbrella term *graphic*. Among the many types, the following are often used in technical communication: (1) tables, (2) pie charts, (3) bar charts, (4) line charts, (5) flowcharts, and (6) technical drawings. This section of the chapter highlights their different purposes and gives guidelines for using each type.

Tables

Tables present readers with raw data, usually in the form of numbers but sometimes in the form of words. Tables are classified as either *informal* or *formal*:

- **Informal tables:** Limited data arranged in the form of either rows or columns
- **Formal tables:** Data arranged in a grid, always with both horizontal rows and vertical columns

The following five guidelines help you design and position tables within the text of your documents:

>> Table Guideline 1: Use Informal Tables as Extensions of Text

Informal tables are usually merged with the text on a page, rather than isolated on a separate page or attachment. As Figure 3-5 shows, an informal table usually has (1) no table number or title and (2) few if any headings for rows or columns. Also, it is not included in the list of illustrations in a formal document.

FTC staff then posted sets of three of these newly-created email addresses – consisting of an Unfiltered Address, an address at Filtered ISP 1, and an address at Filtered ISP 2 – on 50 Internet locations. The 50 Internet locations included websites controlled by the FTC⁵ and several popular message boards, blogs, chat rooms, and USENET groups which had high hit/visit rates, according to ranking websites such as www.message-boards.com and Google popularity searches.⁶ All of the 150 addresses were posted during a three day period in July 2005.

Graphic 1

Locations On Which E-mail Addresses
Were Posted

Type	Number
FTC Web site Pages	12
Message Boards	12
Blogs	12
Chat Rooms	12
USENET Groups	2

■ **Figure 3-5** ■
Informal table in a report

Source: Federal Trade Commission, *Email Address Harvesting and the Effectiveness of Anti-Spam Filters: A Report by the Federal Trade Commission's Division of Marketing Practices* (July 2010): 2.

>> Table Guideline 2: Use Formal Tables for Complex Data Separated from Text

Formal tables may appear on the page of text that includes the table reference, on the page following the first text reference, or in an attachment or appendix. In every case, you should do the following:

1. Extract important data from the table and highlight them in the text
2. Make every formal table as clear and visually appealing as possible by doing the following:
 - Use color to designate positive or negative totals, increases, or decreases or very important points.
 - Use gray screens (no denser than 10%–25%) to subordinate less-important data that appear on the table.

>> Table Guideline 3: Use Plenty of White Space

Used around and within tables, white space guides the eye through a table much better than black lines.

>> Table Guideline 4: Follow Usual Conventions for Dividing and Explaining Data

Figure 3-6 shows a typical formal table. It satisfies the overriding goal of being clear and self-contained. To achieve that objective in your tables, use the following guidelines:

1. **Titles and numberings:** Give a title to each formal table, and place the title and number above the table. Number each table if the document contains two or more tables.
2. **Headings:** Create short, clear headings for all columns and rows.
3. **Abbreviations:** Include in the headings any necessary abbreviations or symbols, such as *lb* or *%*. Spell out abbreviations and define terms in a key or footnote if the reader may need such assistance.

■ **Figure 3-6** ■
Example of a formal table

Investment Type	Book Value	Market Value	% of Total Market Value
Temporary Securities	\$ 434,084	434,084	5.9%
Bonds	3,679,081	3,842,056	52.4
Common Stocks	2,508,146	3,039,350	41.4
Mortgages	18,063	18,063	0.3
Real Estate	1,939	1,939	nil
Totals	\$6,641,313	\$7,335,492	100.0%

Note: This table contrasts the book value versus the market value of the Employee Retirement Fund, as of December 31, 2010.

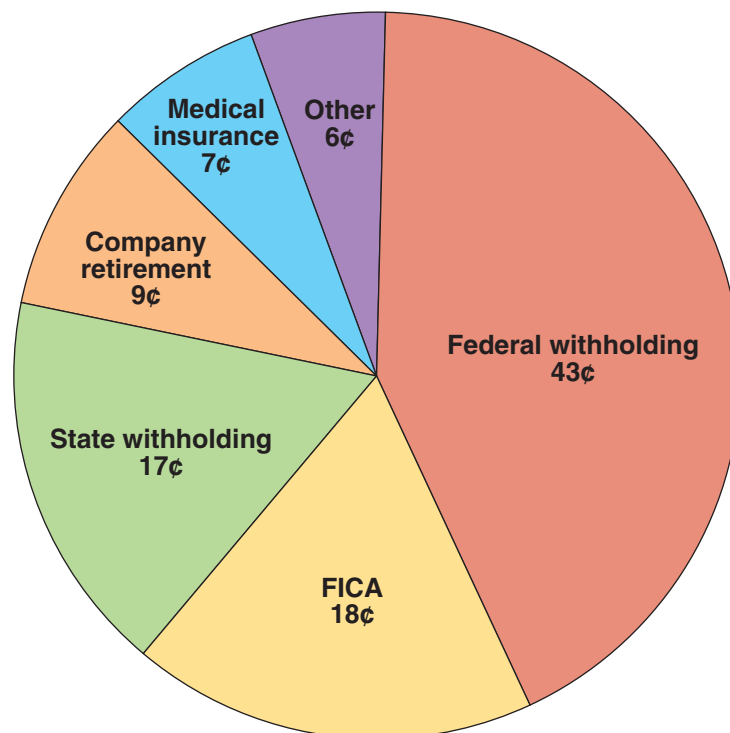
4. **Numbers:** For ease of reading, round off numbers when possible. Also, align multi-digit numbers on the right edge, or at the decimal when shown.
5. **Notes:** Place any necessary explanatory headnotes either between the title and the table (if the notes are short) or at the bottom of the table.
6. **Footnotes:** Place any necessary footnotes below the table.
7. **Sources:** Place any necessary source references beneath the footnotes.
8. **Caps:** Use uppercase and lowercase letters rather than all caps.

Pie Charts

Familiar to most readers, *pie charts* show approximate relationships between the parts and the whole. Their simple circles with clear labels can provide comforting simplicity within even the most complicated report. Yet, the simple form keeps them from being useful when you must reveal detailed information or changes over time. Following are specific guidelines for constructing pie charts:

>> Pie Chart Guideline 1: Use Pie Charts Especially for Percentages and Money

Pie charts catch the readers' eyes best when they represent items divisible by 100—as with percentages and dollars. As Figure 3-7 shows, using pie charts for money breakdowns is made even more appropriate by the coinlike shape of the chart. *In every case, make sure your percentages or cents add up to 100.*



■ **Figure 3-7** ■ Pie chart showing money breakdown for average deductions from employees' paychecks

>> Pie Chart Guideline 2: Use No More Than Six or Seven Divisions

To make pie charts work well, limit the number of pie pieces to no more than six or seven. This approach lets the reader grasp major relationships without having to wade through the clutter of tiny divisions that are difficult to label and read.

>> Pie Chart Guideline 3: Move Clockwise from 12:00, from Largest to Smallest Wedge

Readers prefer pie charts oriented like a clock, with the first wedge starting at 12:00. Move from the largest to the smallest wedge to provide a convenient organizing principle, as in Figure 3-7.

>> Pie Chart Guideline 4: Draw and Label Carefully

The most common pie chart errors are (1) wedge sizes that do not correspond correctly to percentages or money amounts and (2) pie sizes that are too small to accommodate the information placed in them.

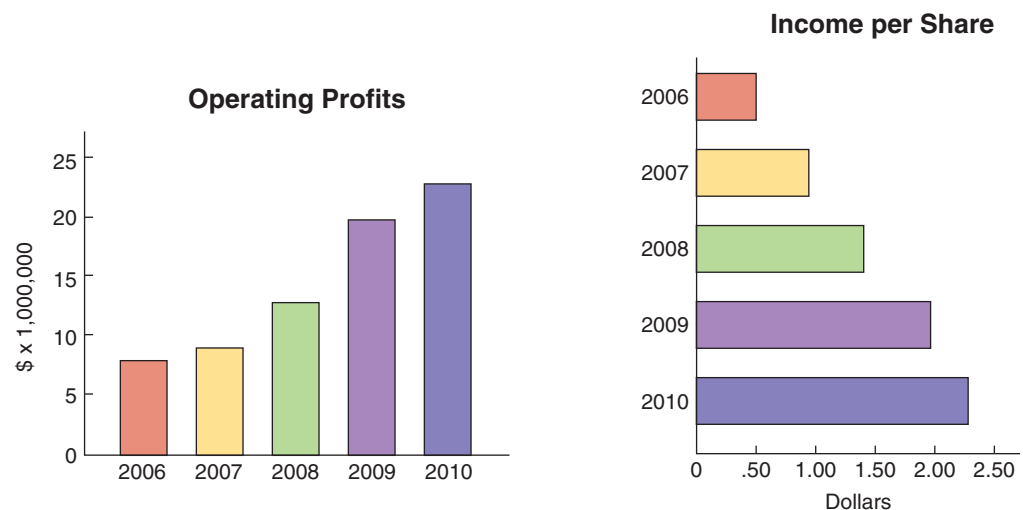
Bar Charts

Like pie charts, bar charts are easily recognized, because they are seen everyday in newspapers and magazines. Unlike pie charts, however, bar charts can accommodate a good deal of technical detail. Comparisons are provided by means of two or more bars running horizontally or vertically on the page. Use the following five guidelines to create effective bar charts:

>> Bar Chart Guideline 1: Use a Limited Number of Bars

Although bar charts can show more information than pie charts, both illustrations have their limits. Bar charts begin to break down when there are so many bars that information is not easily grasped. The maximum bar number can vary according to chart size, of course. Figure 3-8 shows two multi-bar charts. The impact of the charts is enhanced by the limited number of bars.

■ **Figure 3-8** ■ Bar charts



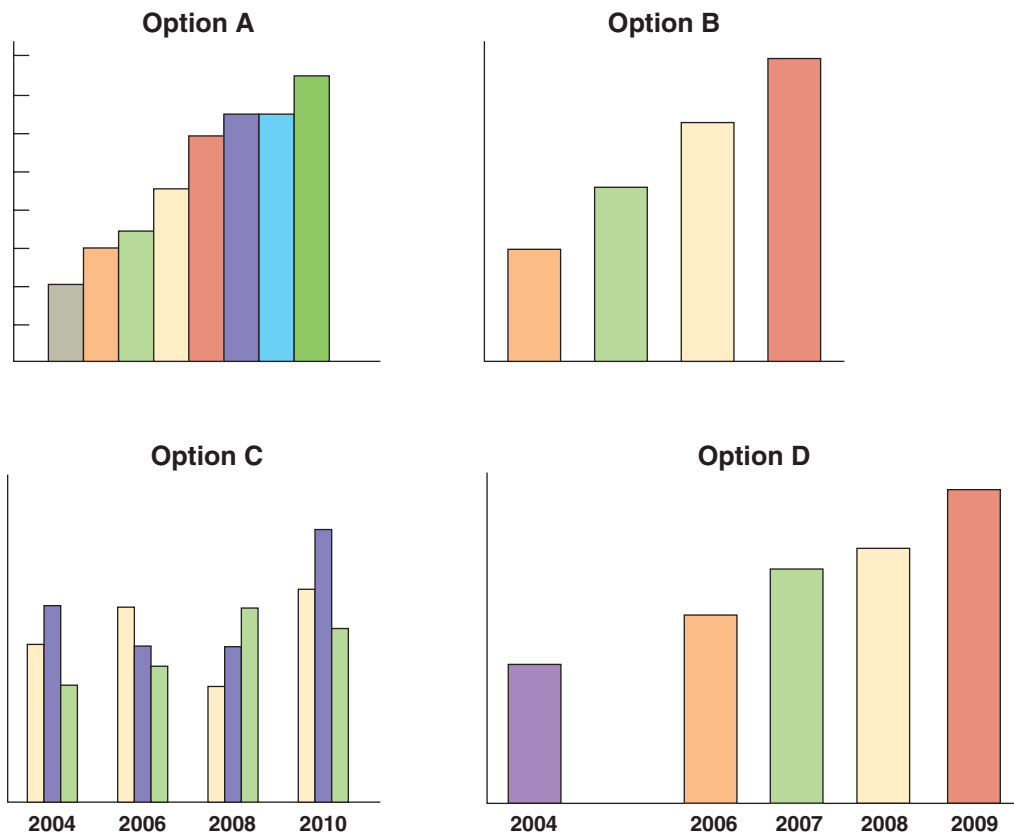
>> Bar Chart Guideline 2: Show Comparisons Clearly

Bar lengths should be varied enough to show comparisons quickly and clearly. Avoid using bars that are too close in length, because then readers must study the chart before understanding it. Such a chart lacks immediate visual impact.

>> Bar Chart Guideline 3: Keep Bar Widths Equal and Adjust Space between Bars Carefully

Although bar length varies, bar width must remain constant. As for distance between the bars, following are four options (along with examples in Figure 3-9):

- **Option A: Use no space** when there are close comparisons or many bars, so that differences are easier to grasp.
- **Option B: Use equal space, but less than bar width** when bar height differences are great enough to be seen in spite of the distance between bars.
- **Option C: Group related bars** to emphasize related data.
- **Option D: Use variable space** when gaps between some bars are needed to reflect gaps in the data.



■ **Figure 3-9** ■ Bar chart variations

>> Bar Chart Guideline 4: Carefully Arrange the Order of Bars

The arrangement of bars is what reveals meaning to readers. Following are two common approaches:

- **Sequential:** Used when the progress of the bars shows a trend
- **Ascending or descending order:** Used when you want to make a point by the rising or falling of the bars

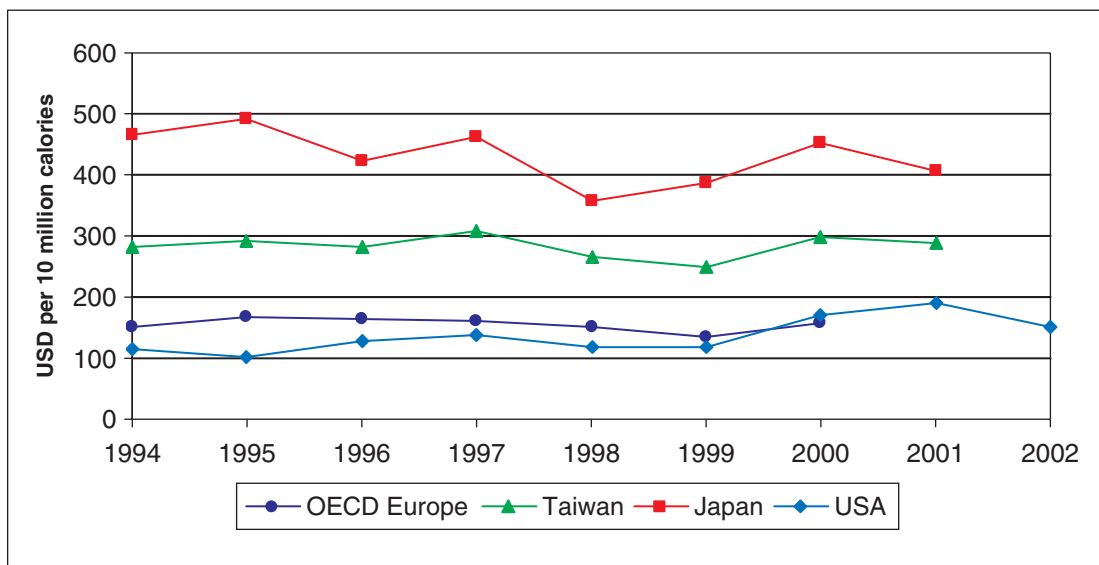
Line Charts

Line charts are a common graphic and work by using vertical and horizontal axes to reflect quantities of two different variables. The vertical (or *y*) axis usually plots the dependent variable; the horizontal (or *x*) axis usually plots the independent variable. (The dependent variable is affected by changes in the independent variable.) Lines then connect points that have been plotted on the chart.

>> Line Chart Guideline 1: Use Line Charts for Trends

Readers are affected by the direction and angle of the chart's line(s), so take advantage of this persuasive potential. In Figure 3-10, for example, the writer wants to show changes in gas prices over time. Including a line chart in the study gives immediate emphasis to the recent downward trend in book purchases.

Figure A9-3: Industrial Sector Gas Prices in the United States, OECD Europe, Japan, and Taiwan, 1994-2002, in 2003 Dollars



■ **Figure 3-10** ■ Line chart using multiple lines

Source: *The Economic Future of Nuclear Power: A Study Conducted at The University of Chicago* (August 2004): A9-A13.

>> Line Chart Guideline 2: Locate Line Charts with Care

Given their strong impact, line charts can be especially useful as attention-grabbers. Consider placing them (1) on cover pages (to engage reader interest in the document), (2) at the beginning of sections that describe trends, and (3) in conclusions (to reinforce a major point of your document).

>> Line Chart Guideline 3: Strive for Accuracy and Clarity

Be sure that the line or lines on the graph truly reflect the data from which you have drawn. Also, select a scale that does not mislead readers with visual gimmicks. Following are some specific suggestions to keep line charts accurate and clear:

- Start all scales from zero to eliminate the possible confusion of breaks in amounts.
- Select a vertical-to-horizontal ratio for axis lengths that is pleasing to the eye (three vertical to four horizontal is common).
- Make chart lines as thick as, or thicker than, the axis lines.

>> Line Chart Guideline 4: Do Not Place Numbers on the Chart Itself

Line charts derive their main effect from the simplicity of lines that show trends. Avoid cluttering the chart with a lot of numbers that only detract from the visual impact.

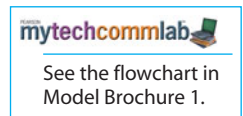
>> Line Chart Guideline 5: Use Multiple Lines With Care

Like bar charts, line charts can show multiple trends. Simply add another line or two. To help readers quickly distinguish between lines, assign a differently shaped data point to each line. If you place too many lines on one chart, however, you run the risk of confusing the reader with too much data. Use no more than four or five lines on a single chart.

Flowcharts

Flowcharts tell a story about a process, usually by stringing together a series of boxes and other shapes that represent separate activities (Figure 3-11).

Some flowcharts use standardized symbols to represent steps in the decision-making process (Figure 3-12). Although these symbols were originally used for programming, they are now used to represent a wide range of processes. Because they have a reputation for being hard to read, you must take extra care in designing flowcharts.



>> Flowchart Guideline 1: Present Only Overviews

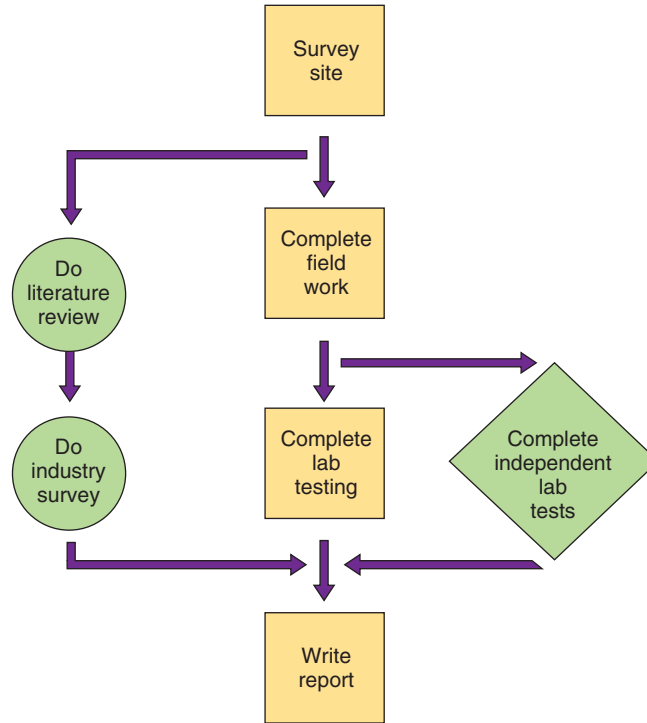
Flowcharts should give only a capsule version of the process, not all details. Reserve your list of particulars for the text or the appendices, where readers expect it.

>> Flowchart Guideline 2: Limit the Number of Shapes

Flowcharts rely on rectangles and other shapes to tell a story. Different shapes represent different types of activities. Some flowcharts use icons and images to present information. This variety helps in describing a complex process, but it can also produce

■ **Figure 3-11** ■

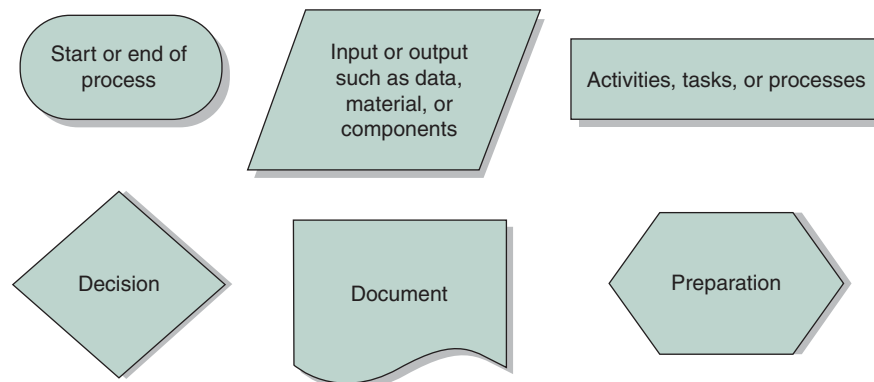
Flowchart for basic site survey project



confusion. For the sake of clarity and simplicity, limit the number of different shapes in your flowcharts.

>> **Flowchart Guideline 3: Provide a Legend When Necessary**

Simple flowcharts often need no legend. The few shapes on the chart may already be labeled by their specific steps. When charts get more complex, however, include a legend that identifies the meaning of each shape used.

■ **Figure 3-12** ■ Selection of standard flowchart symbols

>> Flowchart Guideline 4: Run the Sequence from Top to Bottom or from Left to Right

Long flowcharts may cover the page with several columns or rows; however, they should always show some degree of uniformity by assuming either a basically vertical or horizontal direction.

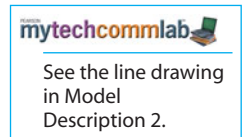
>> Flowchart Guideline 5: Label All Shapes Clearly

Besides a legend that defines meanings of different shapes, the chart usually includes a label for each individual shape or step. Follow one of these approaches:

- Place the label inside the shape.
- Place the label immediately outside the shape.
- Put a number in each shape and place a legend for all numbers in another location (preferably on the same page).

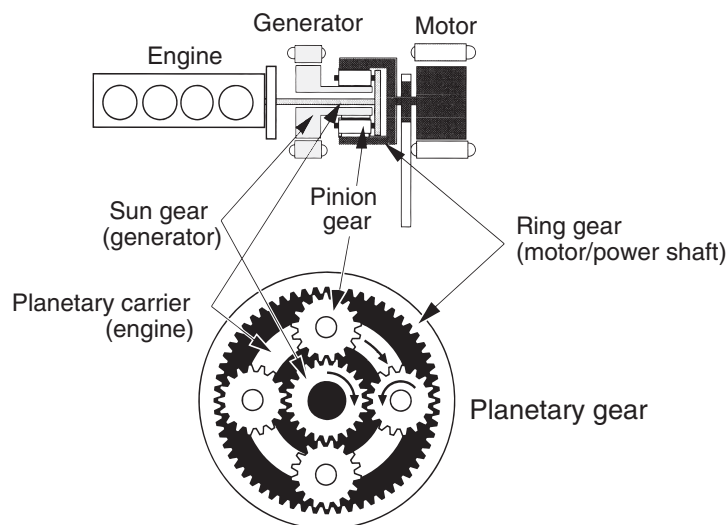
Technical Drawings

Technical drawings are important tools that can accompany documents, such as instructions, reports, sales orders, proposals, brochures, and posters. They are preferred over photographs when specific views are more important than photographic detail.



>> Drawing Guideline 1: Choose the Right Amount of Detail

Keep drawings as simple as possible. Use only the level of detail that serves the purpose of your document and satisfies your readers' needs. For example, Figure 3-13 uses an exploded view to show a gear and its placement in a hybrid engine.



■ **Figure 3-13** ■
Technical drawing
(exploded view)

Source: Toyota Motor Corporation., Toyota Hybrid System (May 2003): 10.

>> Drawing Guideline 2: Label Parts Well

Place labels on every part you want your reader to see. (Conversely, you can also choose not to label those parts that are irrelevant to your purpose.)

When you label parts, use a typeface large enough for easy reading. Also, arrange labels so that they are easy to locate and do not detract from the importance of the drawing itself.

>> Drawing Guideline 3: Choose the Most Appropriate View

Drawings offer you a number of options for perspective or view:

- **Exterior view** shows surface features with either a two- or three-dimensional appearance.
- **Cross-section view** shows a “slice” of the object so that interiors can be viewed.
- **Cutaway view** is similar to a cross section view, but only part of the exterior is removed to show the inner workings of the object.
- **Exploded view** shows relationship of parts to each other by “exploding” the mechanism.

mytechcomm lab

See the cutaway drawing in Model Description 3: Flat Plate Solar Collector.

>> Drawing Guideline 4: Use Legends When There Are Many Parts

In complex drawings, avoid cluttering the illustration with many labels. Instead number parts and place labels with corresponding numbers in a list, or legend, at one side of the illustration.

>>> Misuse of Graphics

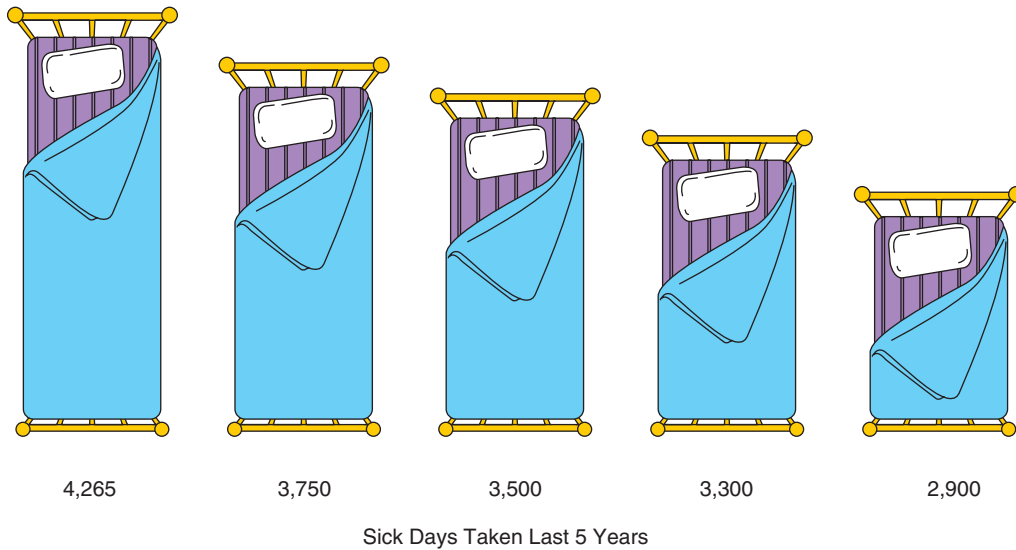
Technology has revolutionized the world of graphics by placing sophisticated tools in the hands of many writers. This largely positive event has a dark side in that many graphics distort data and misinform the reader. This final chapter section shows what can happen to graphics when sound design principles are not applied.

Description of the Problem

Through clutter or distortion, graphics can oversimplify data, be confusing, or be misleading. To avoid misleading or confusing graphics, it is important to proofread and edit them carefully. If you are using charts or tables, it may be useful to ask someone else to look at them carefully to see if they interpret the graphics in the way that you intended. One of the most serious problems with graphics is that charts are often disproportional to the actual differences in the data represented. The next subsection shows some specific ways that this error has worked its way into contemporary graphics.

Examples of Distorted Graphics

This section gets at the problem of misrepresentation by showing several examples and describing the errors involved. None of the examples commits major errors, yet each one fails to represent the data accurately.



■ **Figure 3-14** ■
Faulty comparisons
on modified bar
chart

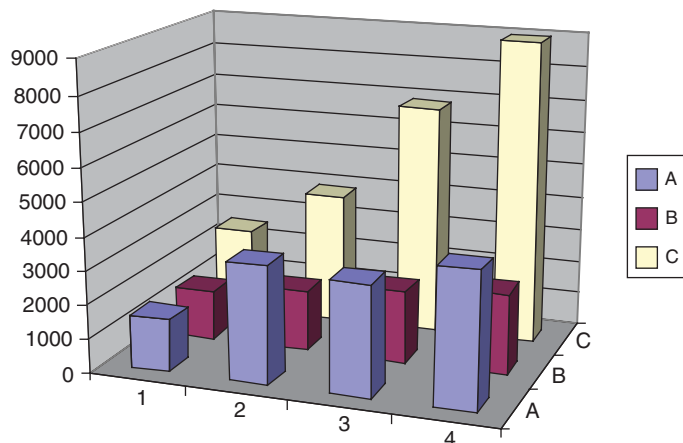
>> Example 1: Confusing Bar Charts

Figure 3-14 accompanied a report about sick days taken in an organization. The problem here is that the chart's decoration—the beds—inhibits rather than promotes clear communication. Although the writer intends to use bed symbolism in lieu of precise bars, the height of the beds does not correspond to the actual decrease in sick days. A revised graph appropriate for a report should include a traditional bar chart without the beds.

Another problem with bar charts is that they become busy or confusing, especially when the options for creating charts in spreadsheet programs are not used carefully. Figure 3-15 uses several stacked bars, which make it hard to compare the data clearly. It also uses too many colors, as well as colors that are similar, and that would not reproduce well on a black-and-white copier.

>> Example 2: Chartjunk That Confuses the Reader

Figure 3-16 concludes a report from a county government to its citizens. Whereas the dollar backdrop is meant to reinforce the topic—that is, the use to which tax funds are



■ **Figure 3-15** ■
Confusing bar
chart



■ **Figure 3-16** ■ Chartjunk that confuses the reader

put—in fact, it impedes communication. Readers cannot quickly see comparisons. Instead, they must read the entire list below the illustration, mentally rearranging the items into some order.

At the very least, the expenditures should have been placed in sequence, from least to greatest percentage or vice versa. Even with this order, however, one could argue that the dollar bill is a piece of what Edward Tufte, an expert in the visual display of data, calls *chartjunk*.

>> Example 3: Confusing Pie Charts

The pie chart in Figure 3-17 (1) omits percentages that should be attached to each of the budgetary expenditures; (2) fails to move in a largest-to-smallest clockwise sequence; (3) includes too many divisions, many of which are about the same size and thus difficult to distinguish; and (4) introduces a third dimension that adds no value to the graphic.

Pie charts should be perfect circles, should have percentages on the circle, and should move in large-to-small sequence from the 12:00 position.

Chapter | 4 | Letters, Memos, and Electronic Communication



>>>Chapter Outline

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- Guidelines for E-mail 86

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Learning Portfolio 90

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You may write more letters, memos, and e-mails in your career than any other type of document. Generally termed “correspondence,” letters, memos, and e-mails are short documents written to accomplish a limited purpose. Letters are directed outside your organization, memos are directed within your organization, and e-mail can be directed to either an external or an internal audience.

Your ability to write good memos, letters, and e-mails depends on a clear sense of purpose, under-

standing of reader needs, and close attention to formats. This chapter prepares you for this challenge by presenting sections that cover general rules that apply to all workplace correspondence and specific formats for positive correspondence, negative correspondence, and neutral correspondence. Also included is advice specific to letter, memoranda, and e-mail formats. Job letters and resumes are discussed in a separate chapter on the job search (Chapter 10).

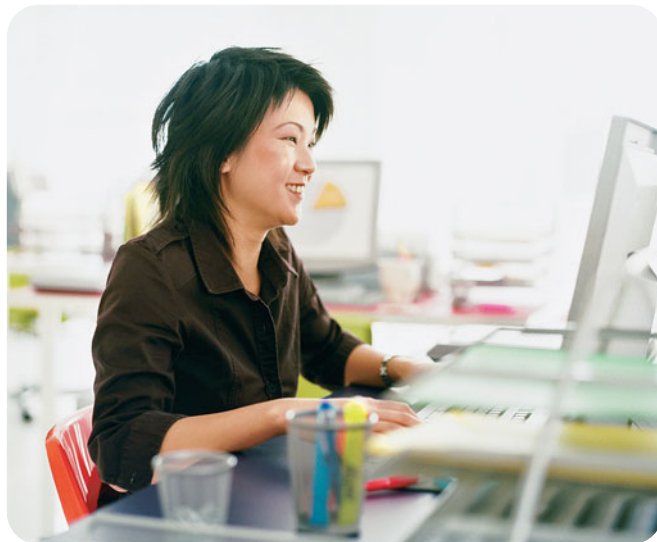
>>> General Guidelines for Correspondence

Letters convey your message to readers outside your organization, just as memos are an effective way to get things done within your own organization, and e-mail is a way to communicate quickly with readers inside and outside of your organization. By applying the guidelines in this chapter, you can master the craft of writing effective correspondence. Refer to the Figure 4-2 on page 83, Figure 4-3 on page 84, and Figure 4-4 on page 87 in this chapter for examples that demonstrate the guidelines that follow.

>> Correspondence Guideline 1: **Know Your Purpose**

Before beginning your draft, write down your purpose in one clear sentence. This *purpose sentence* often becomes one of the first sentences in the document. Following are some samples:

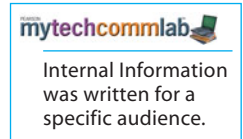
- **Letter purpose sentence:** “As you requested yesterday, I’m sending samples of the new candy brands you are considering placing your office vending machines.”
- **Memo purpose sentence:** “This memo explains the organization’s new policy for selecting rental cars on business trips.”
- **E-mail purpose sentence:** “I have attached the most recent draft of the proposal for the PI Corp. pipeline project.”



Some purpose statements are implied; others are stated. An implied purpose statement occurs in the first paragraph of Figure 4-2. That paragraph indicates that the letter is a response to an earlier letter that requested a change in the construction schedule. Figure 4-3 shows a more obvious purpose statement in the third sentence. The third sentence in Figure 4-4 tells the reader exactly what will follow.

>> Correspondence Guideline 2: Know Your Readers

Who are you trying to inform or influence? Pay particular attention when correspondence will be read by more than one person. If these readers are from different technical levels or different administrative levels within an organization, the challenge increases. A complex audience compels you to either (1) reduce the level of technicality to that which can be understood by all readers or (2) write different parts of the document for different readers.



>> Correspondence Guideline 3: Follow Correct Format

Most organizations adopt letter and memo formats that must be used uniformly by all employees. Later in this chapter, we offer specific guidelines for formatting letters, memos, and e-mail, but the following are some general formatting guidelines:

- **Multiple-page headings:** Each page after the first page of a letter or memo often has a heading that includes the name of the person or company receiving the letter or memo, the date, and the page number. Some organizations may prefer an abbreviated form such as “Jones to Bingham, 2,” without the date.

- **Subject line:** Memos and e-mails always include subject lines, but they are sometimes used in letters as well. Give the subject line special attention, because it telegraphs meaning to the audience immediately. In fact, readers use it to decide when, or if, they will read the complete correspondence. Be brief, but also engage interest. For example, the subject line of the Figure 4-2 memo could have been “Changes.” Yet, that brevity would have sacrificed reader interest. The actual subject line, “Copy Center Changes,” conveys more information and shows readers that the contents of the memo will make their lives easier.

- **Enclosure notation:** If attachments or enclosures accompany a letter or memo, type the singular or plural form of “Enclosure” or “Attachment” one or two lines beneath the reference initials. Some writers also list the item itself (e.g., Enclosure: Code of Ethics). If an e-mail includes an attachment, the filename and file type (e.g. PDF) should be included in the first paragraph.

- **Copy notation:** If the correspondence has been sent to anyone other than the recipient, type “Copy” or “Copies” one or two lines beneath the enclosure notation, followed by the name(s) of the person or persons receiving copies (e.g., Copy: Preston Hinkley). E-mail inserts this information automatically. If you are sending a copy but do not want the original letter or memo to include a reference to that copy, write “bc” (for *blind copy*) and the person’s name only on the copy—not on the original (e.g., bc: Mark Garibaldi). (Note: Send blind copies only when you are certain it is appropriate and ethical to do so.)

>> Correspondence Guideline 4: Follow the ABC Format for All Correspondence

Correspondence subscribes to the same three-part ABC (Abstract/Body/Conclusion) format used throughout this book. According to the ABC format, your correspondence is composed of these three main sections:

- **Abstract:** The abstract introduces the purpose and usually gives a summary of main points to follow. It includes one or two short paragraphs.

- **Body:** The body contains supporting details and thus makes up the largest part of a letter or memo. You can help your readers by using such techniques as:

Deductive patterns for paragraphs: In this general-to-specific plan, your first sentence should state the point that helps the reader understand the rest of the paragraph. This pattern avoids burying important points in the middle or end of the paragraph, where they might be missed. Fast readers tend to focus on paragraph beginnings and expect to find crucial information there. Note how most paragraphs in Figure 4-3 follow this format.

Personal names: If they know you, readers like to see their names in the body of the letter or memo, or in the salutation of an e-mail. Your effort here shows concern for the reader's perspective, gives the correspondence a personal touch, and helps strengthen your personal relationship with the reader. (See the last paragraph in Figure 4-2.) Of course, the same technique can sometimes backfire, because it is an obvious ploy to create an artificially personal relationship.

Lists that break up the text: Listed points are a good strategy for highlighting details. Readers are especially attracted to groupings of three items, which create a certain rhythm, attract attention, and encourage recall. Use bullets, numbers, dashes, or other typographic techniques to signal the listed items. For example, the bulleted list in Figure 4-3 draws attention to three important points about that the writer wants to emphasize. Because some e-mail systems cannot read special characters like bullets, use asterisks or dashes for lists in e-mail.

Strongest points first or last: If your correspondence presents support or makes an argument, include the most important points at the beginning or at the end—not in the middle. For example, the first paragraph in Figure 4-3 announces the changes in the Copy Center, and the last paragraph tells readers when the changes will take place.

Headings to divide information: One-page letters and memos, and even e-mail, sometimes benefit from the emphasis achieved by headings. The three headings in Figure 4-3 quickly steer the reader to main parts of the document.

- **Conclusion:** Readers remember first what they read last. The final paragraph of your correspondence should leave the reader with an important piece of information—for example, (1) a summary of the main idea or (2) a clear statement of what will happen next. The Figure 4-1 letter promises a successful outcome, whereas the Figure 4-4 e-mail ends with a reminder that the sender has submitted a proposal to the recipient.

>> Correspondence Guideline 5: Use the 3Cs Strategy for Persuasive Messages

The ABC format provides a way to organize all letters and memos. Another pattern of organization for you to use is the *3Cs strategy*—especially when your correspondence has a persuasive objective. This strategy has three main goals:

- **Capture** the reader's interest with a good opener, which tells the reader what the letter, memo, or e-mail can do for him or her.

- **Convince** the reader with supporting points, all of which confirm the opening point that this document will make life easier.
- **Contact** solidifies your relationship with the reader with an offer to follow up on the correspondence.

Note, how each of the samples in this chapter uses the 3Cs strategy.

>> Correspondence Guideline 6: Stress the “You” Attitude

As noted earlier, using the reader’s name in the body helps convey interest. However, your efforts to see things from the reader’s perspective must go deeper than a name reference. For example, you should perform the following tasks:

- Anticipate questions your reader might raise and then answer these questions. You can even follow an actual question (“And how will our new testing lab help your firm?”) with an answer (“Now TestCorp’s labs can process samples in 24 hours.”).
- Replace the pronouns *I*, *me*, and *we* with *you* and *your*. Of course, you must use first-person pronouns at certain points in a letter, but many pronouns should be second person. The technique is quite simple. You can change almost any sentence from writer-focused prose (“We feel that this new service will . . .”) to reader-focused prose (“You’ll find that this new service will . . .”).

Figure 4-2 shows this *you* attitude by emphasizing how the sender’s organization can help the recipient with the scheduling change. Figure 4-4 shows it by responding directly to the sender’s concerns.

>> Correspondence Guideline 7: Use Attachments for Details

Keep text brief by placing details in attachments, which readers can examine later, rather than bogging down the middle of the letter or memo.

>> Correspondence Guideline 8: Be Diplomatic

Without a tactful tone, all your planning and drafting will be wasted. Choose words that persuade and cajole, not demand. Be especially careful of memos written to subordinates. If you sound too authoritarian, your message may be ignored—even if it is clear that what you are suggesting will help the readers. Generally speaking, negative (or “bad news”) letters often use the passive voice, whereas positive (or “good news”) letters often use the active voice.

For example, the letter in Figure 4-2 would fail in its purpose if it flatly refused to make changes and offered no alternatives.

>> Correspondence Guideline 9: Respond Quickly

A letter, memo, or e-mail that comes too late fails in its purpose, no matter how well written. Mail letters within 48 hours of your contact with, or request from, the reader. Send memos in plenty of time for your reader to make the appropriate adjustments in schedule, behavior, and so forth. Usually respond to e-mails the same day you receive them.



Correspondence can be broadly categorized as positive, negative, and neutral. For examples of correspondence that meets specific purposes, see the Model letters, memos, and e-mail.



Claim Response is an effective example of a positive letter.

Positive Correspondence

Everyone likes to give good news; fortunately, you will often be in the position of providing it when you write. Following are some sample situations:

- Replying to a question about products or services
- Recommending a colleague for a promotion or job
- Responding favorably to a complaint or an adjustment

The trick is to recognize the good-news potential of many situations. This section gives you an all-purpose format for positive correspondence.

ABC Format: Positive Correspondence

- **ABSTRACT:** Bridge between this correspondence and last communication with person
 - Clear statement of good news you have to report
- **BODY:** Supporting data for main point mentioned in abstract
 - Clarification of any questions reader may have
 - Qualification, if any, of the good news
- **CONCLUSION:** Statement of eagerness to continue relationship, complete project, etc.
 - Clear statement, if appropriate, of what step should come next

ABC Format for Positive Correspondence

All positive correspondence follows one overriding rule. You must always:

State good news immediately!

Any delay gives readers the chance to wonder whether the news will be good or bad, thus causing momentary confusion. On the left is a complete outline for positive letters that corresponds to the ABC format.



Complaint is a carefully written negative letter.

Negative Correspondence

It would be nice if all your correspondence could be as positive as that just described. Unfortunately, the real world does not work like that. You will have many opportunities to display both tact and clarity in relating negative information. Following are a few cases:

ABC Format: Negative Correspondence

- **ABSTRACT:** Bridge between your correspondence and previous communication
 - General statement of purpose or appreciation—in an effort to find common bond or area of agreement
- **BODY:** Strong emphasis on what can be done, when possible
 - Buffered yet clear statement of what cannot be done, with clear statement of reasons for negative news
 - Facts that support your views
- **CONCLUSION:** Closing remarks that express interest in continued association
 - Statement, if appropriate, of what will happen next

- Explaining delays in projects or delivery of services
- Registering complaints about products or services
- Giving bad news about employment or performance

This section gives you a format to follow in writing sensitive correspondence with negative information.

ABC Format for Negative Correspondence

One main rule applies to all negative correspondence:

Buffer the bad news, but still be clear.

Despite the bad news, you want to keep the reader's goodwill. Spend time at the beginning building your relationship with the reader by introducing less controversial

information—before you zero in on the main message. On the right is an overall pattern to apply in all negative correspondence.

Neutral Correspondence

Some correspondence expresses neither positive nor negative news. It is simply the routine correspondence written every day to keep businesses and other organizations operating. Some situations follow:

- Requesting information about a product or service
- Inviting the reader to an event
- Sending solicited or unsolicited items through the mail

Use the following outline in writing your neutral letters.

ABC Format for Neutral Correspondence

Because the reader usually has no personal stake in the news, neutral correspondence requires less emphasis on tone and tact than other types, yet it still requires careful planning. In particular, always abide by this main rule:

Be absolutely clear about your inquiry or response.

Neutral correspondence operates a bit like positive correspondence. You must make your point early, without giving the reader time to wonder about your message. Neutral correspondence varies greatly in specific organization patterns. The *umbrella plan* suggested here emphasizes the main criterion of clarity.

>>> Letters

Letters are to your clients and vendors what memos are to your colleagues. They relay information quickly and keep business flowing. Here is a working definition:

Letter: A document that conveys information to a member of one organization from someone outside that same organization. Letters usually cover one major point and fit on one page. This chapter classifies letters into these three groups, according to type of message: (1) positive, (2) negative, and (3) neutral.

mytechcommlab

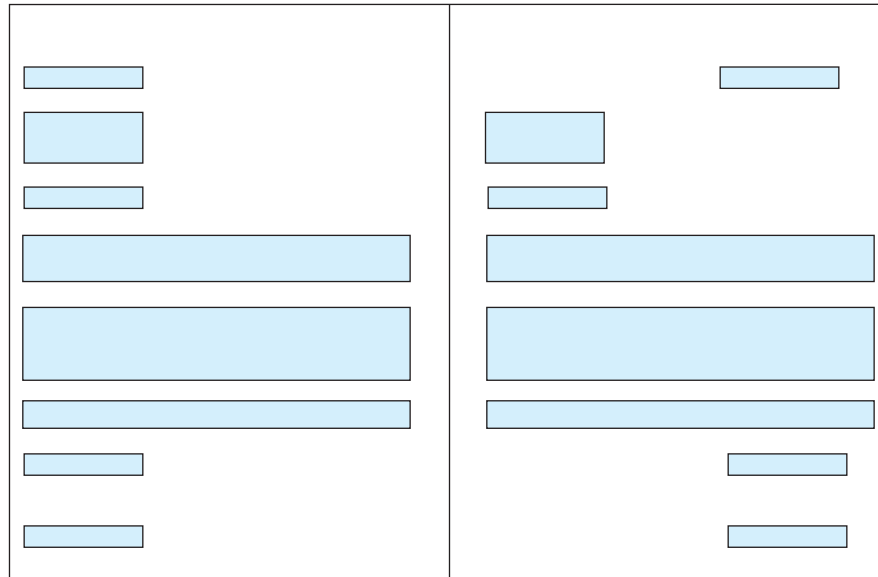
Memo 10: Web site Request is a good example of a neutral memo.

ABC Format: Neutral Correspondence

- **ABSTRACT:** Bridge or transition between correspondence and previous communication, if any
 - Precise purpose of correspondence (e.g., request, invitation, response to invitation)
- **BODY:** Details that support the purpose statement (e.g., a description of item(s) requested, the requirements related to the invitation, a description of item(s) being sent)
- **CONCLUSION:** Statement of appreciation
 - Description of actions that should occur next



■ **Figure 4-1** ■ Block and modified block style for letters



There are two main letter formats—block and modified block. Figure 4-1 shows the basic page design of each; Figure 4-2 is in block style. As noted, you usually follow the preferred format of your own organization. Whatever format you choose, most or all business letters include the following sections: address of the sender (usually in letterhead), date, address of the receiver, greeting, body, signature, and reference to any attachments and/or those individuals being copied.

Addresses on envelopes and in letters should use the format recommended by the United States Postal Service. Addresses should include no more than four lines, and should not include punctuation such as commas or periods.

>>> Memoranda

Even though e-mail has become common in the workplace, *memoranda* (the plural of *memorandum*, also called *memos*) are still important. Even if you work in an organization that uses e-mail extensively, you will still compose print messages that convey your point with brevity, clarity, and tact. Here is a working definition:

Memorandum: A document written from a member of an organization to one or more members of the same organization. Abbreviated *memo*, it usually covers just one main point and no more than a few. Readers prefer one-page memos.

With minor variations, all memos look much the same. The obligatory “Date/To/From/Subject” information hangs at the top left margin, in whatever order your organization requires. Figure 4-3 shows one basic format. These four lines allow you to dispense with lengthy introductory passages seen in more formal documents. Note that the sender signs his or her initials after or above the typed name in the “From” line.

12 Post Street
Houston Texas 77000
713.555.1381

July 23, 2011

The Reverend Mr John C Davidson
Maxwell Street Church
Canyon Valley
Texas 79195

Dear Reverend Davidson:

Thanks for your letter asking to reschedule the foundation project at your church from mid-August to late August, because of the regional conference. I am sure you are proud that Maxwell was chosen as the conference site.

One reason for our original schedule, as you may recall, was to save the travel costs for a project crew going back and forth between Houston and Canyon Valley. Because **Delvientos Construction** has several other jobs in the area, we had planned not to charge you for travel.

We can reschedule the project, as you request, to a more convenient date in late August, but the change will increase project costs from \$1,500 to \$1,800 to cover travel. At this point, we just do not have any other projects scheduled in your area in late August that would help defray the additional expenses. Given our low profit margin on such jobs, that additional \$300 would make the difference between our firm making or losing money on the foundation repair at your church.

I'll call you next week, Reverend Davidson, to select a new date that would be most suitable. **Delvientos Construction** welcomes its association with the Maxwell Street Church and looks forward to a successful project in late August.

Sincerely,

Nancy Slade

Project Manager

NS/mh

File #34678

Implied purpose response to previous letter.

Opens on positive note.

Reminds him about original agreement—in tactful manner

Phrases negative message as *positively* as possible, giving rationale for necessary change.

Closes with promise of future contact.

Makes it clear what will happen next. Ends on positive note.

Writer's initials/typists initials.

■ **Figure 4-2** ■ Negative letter in block style

Memorandum

Date: August 1, 2011

To: All Employees

From: Gini Preston
Chair
Copy Services Committee

Subject: Copy Center Changes

Gives brief purpose statement and overview of contents.

With the purchase of two new copiers and a mechanical folder, the Copy Center is able to expand its services. At the same time, we have had to reduce the paper stock that we keep on hand because of space limitations. This memo highlights the services and products now available at the Copy Center.

Color copies

With our new equipment, color copies do not require additional time to process. However, because color copies are expensive, please limit your use of them. If you have a document that includes both color and black-and-white pages, submit them as separate jobs so that the color copier is used only for color copies.

Special stock

The Copy Center now stocks only two colors of paper in addition to white paper: blue and goldenrod. Cover stock is available only in white and blue. We continue to stock transparencies. Although we are no longer stocking other kinds of paper, we are still able to meet requests for most special stock:

Emphasizes need for special handling of requests for special paper.

- **Stocks available with 24-hour notice:** We can purchase 11 x 17 inch paper, cover stock and regular stock in a variety of colors, and specialized paper such as certificates and NCR (carbonless copy) paper. Departments will be charged for all special stock.
- **Coated stock:** Our copiers do not produce quality copies on coated stock (paper or cover stock with a slick coating, like magazine paper). We will continue to outsource jobs that use coated stock to KDH Printing. Please allow at least one week for jobs that use coated stock.

Bindery services

With our new equipment, collating and stapling of large jobs no longer require additional time. The following bindery services are also available in house, but may require additional time:

Makes it clear when changes will take place.

- Perfect and spiral binding
- Folding
- Cutting and hole punching. (The paper cutter and paper drill can be used on up to 500 sheets at a time.)

The new equipment will be available August 15. Your efforts to make the most efficient use of Copy Center resources help improve the quality of your documents and the productivity of the company.

Invites contact.

Feel free to call me at ext. 567 if you have any questions.

■ Figure 4-3 ■ Memorandum: changes in services

Abide by this one main rule in every memo-writing situation:

Be clear, brief, and tactful.

Because many activities are competing for their time, readers expect information to be related as quickly and clearly as possible; however, you must be sure not to sacrifice tact and sensitivity as you strive to achieve conciseness.

>>> E-Mail

Electronic communication (*e-mail*) has become the preferred means of communication for many people in their professional lives.

E-mail is an appropriate reflection of the speed at which we conduct business today. Following are some of the obvious advantages that using e-mail provides:

- It gets to the intended receiver quickly.
- Its arrival can be confirmed easily.
- Your reader can reply to your message quickly.
- It's cheap to use—once you have invested in the hardware and software.
- It permits cheap transmission of multiple copies and attachments. Any medium so widely used deserves special attention in a chapter on correspondence. Here is a working definition:

E-mail: A document written often in an informal style either to members of one's own organization or to an external audience. E-mail messages often cover one main point. Characterized by the speed with which it is written and delivered, an e-mail can include more formal attachments to be read and possibly printed by the audience.



Adding to the ease of transmission is the fact that e-mail allows you to create mailing lists. One address label can be an umbrella for multiple recipients, saving you much time.

Of course, the flip side of this ease of use: E-mail is *not* private. Every time you send an e-mail, remember that it may be archived or forwarded, and may end up being read by “the world.” Either by mistake or design, many supposedly private e-mails often are received by unintended readers.

Appropriate Use and Style for E-mail

Simply understanding that e-mail *should* have a format puts you ahead of many writers, who consider e-mail a license to ramble on without structure. Yes, e-mail is casual and quick, but that does not make it formless. Remember, the configurations of some computers make reading a screen harder on the eyes than reading print memos.



So give each e-mail a structure that makes it simple for your reader to find important information. Computers and e-mail systems handle formatting of texts and special characters differently, so you should format your e-mail so that it can be read on any computer. Use your system's default font, and avoid highlighting, color, bold, italics, and underlining. E-mails are generally short, no longer than what can be seen on a computer screen all at once, and paragraphs should be short. Some e-mail systems cannot translate tabs, so use lines of white space between paragraphs. Figure 4-4 shows how one writer clearly separates topics within an e-mail.

Guidelines for E-mail

When writing e-mail, you should try to strike a balance between speed of delivery on the one hand and quality of the communication to your reader on the other hand. In fact, the overriding rule for e-mail is as follows:

Do not send it too quickly!

By taking an extra minute to check the style and tone of your message, you have the best chance of sending an e-mail that will be well received.

>> E-mail Guideline 1: Use Style Appropriate to the Reader and Subject

E-mail sent early in a relationship with a client or other professional contact should be somewhat formal. It should be written more like a letter, with a salutation, closing, and complete sentences. E-mail written once a professional relationship has been established can use a more casual style. It can resemble conversation with the recipient on the phone. Sentence fragments and slang are acceptable, as long as they contribute to your objectives and are in good taste. Most important, avoid displaying a negative or angry tone. Do not push the *Send* button unless an e-mail produces a constructive exchange.

>> E-mail Guideline 2: Be Sure Your Message Indicates the Context to Which It Applies

Tell your readers what the subject is and what prompts you to write your message. If you are replying to a message, be sure to include the previous message or summarize the message to which you are replying. Most e-mail software packages include a copy of the message to which you are replying. However, you should make sure that you include only the messages that provide the context for your reader. Long strings of forwarded e-mail make it difficult to find the necessary information.

X-Sender: mckinley@mail2.AlvaCon.com

Date: Tue, Nov 11, 2008 09:25:30 -0800

To: pcarmich@advantage.com

From: Mike McKinley <mckinley@mail2.AlvaCon.com>

Subject: Our Recent Visit

Mime-Version: 1.0

Dear Paul,

I enjoyed meeting you also and visiting with your staff. I particularly enjoyed meeting Harold Black, for he will be very valuable in developing the plans for the possible water purification plant.

My responses to your concerns are below.

YOU WROTE:

>If Advantage, Inc., does decide to build the water purification plant, we would be very interested in having Delvientos Construction's Mary Stevens as the project manager.

REPLY:

That certainly will be a possibility; Mary is one of our best managers.

YOU WROTE:

>After you left, I called the city administration here in Murrayville. Delvientos Construction does not need a business license for your work here, but, of course, you will need the necessary construction permits.

REPLY:

Thanks for taking care of this matter—I had not thought of that. We will supply the details to you for applying for the construction permits if you accept our proposal.

Positive opening, with reference to specific events.

Clearly indicates purpose: to respond to an earlier e-mail.

Clear indication of original e-mail text and of reply.

Indicates next action.

Closes with references to pending proposal.

■ **Figure 4-4** ■ An e-mail message that separates different topics for reply

>> **E-mail Guideline 3: Choose the Most Appropriate Method for Replying to a Message**

Short e-mail messages may only require that you include a brief response at the beginning or end of the e-mail to which you are responding. For complex, multi-topic messages, however, you may wish to split your reply by commenting on each point individually (Figure 4-4).

>> **E-mail Guideline 4: Format Your Message Carefully**

Because e-mail messages frequently replace more formal print-based documents, they should be organized and formatted so that the readers can locate the information you want to communicate easily.

- Use headings to identify important chunks of information.
- Use lists to display a series of information.
- Use sufficient white space to separate important chunks of information.
- Use separators to divide one piece of information from another.

>> **E-mail Guideline 5: Chunk Information for Easy Scanning**

Break the information into coherent chunks dealing with one specific topic, including all the details that a reader needs to get all of the essential information. Depending on the nature of the information, include specific topic, time, date, location, and necessary prerequisites and details.

>> **E-mail Guideline 6: When Writing to Groups, Give Readers a Method to Abstain from Receiving Future Notices**

E-mail can easily become invasive and troublesome for recipients. You will gain favor—or at least not lose favor—if you are considerate and allow recipients to decide what e-mail they wish to receive.

>> **E-mail Guideline 7: When Writing to Groups, Suppress the E-mail Addresses of Recipients—Unless the Group Has Agreed to Let Addresses Be Known**

It is inappropriate to reveal the e-mail addresses of group members to other group members. Use the “BC” line to suppress group members’ addresses.

>> **E-mail Guideline 8: When Composing an Important Message, Consider Composing It in Your Word Processor**

Important e-mail messages should be not only clear in format, but also correct in mechanics. Because e-mail software may not have a spelling checker, compose important messages in your word processor and use your spelling checker to check accuracy. Then either cut and paste it into an e-mail message or attach it as a file.

E-mail communication is often considered less formal and, therefore, less demanding in its format and structure than print-based messages, such as memoranda and letters. However, because e-mail messages have become so pervasive a means of communication, you should consider constructing them as carefully as you would a memorandum or a letter.

Another reason to exercise great care is that e-mail, like conventional documents, can be used in legal proceedings and other formal contexts.

>>> Memoranda Versus E-mail

Although e-mail has become the most common form of internal correspondence in the workplace, there are times when a memo is a better option. Send a memo instead of an e-mail in the following situations:

- The document is longer than can be viewed easily on a computer screen.
- The document must include symbols, special characters, or other formatting that may not be available through all computer systems.
- The document includes graphics.
- The document must be posted in print form.
- The document contains sensitive information, including information about clients, projects, or personnel.

>>> Chapter Summary

Correspondence keeps the machinery of business, industry, and government moving. Letters usually are sent to readers outside your organization, whereas memos are sent to readers inside. E-mail can be sent to internal or external audiences. In all types of correspondence, abide by these rules:

1. Know your purpose.
2. Know your readers.
3. Follow correct format.
4. Follow the ABC format for all letters and memos.
5. Use the 3Cs strategy for persuasive messages.
6. Stress the *you* attitude.
7. Use attachments for details.
8. Be diplomatic.
9. Edit carefully.

Besides following these basics, you must follow specific strategies for positive, negative, and neutral correspondence. In correspondence with a positive message, the good news always goes first. In correspondence with a negative message, work on maintaining goodwill by placing a buffer before the bad news. Neutral correspondence, such as requests for information, should be absolutely clear in its message. Letters develop and maintain professional relationships with those outside of your organization. Memos should strive for brevity, clarity, and tact. Your relationship with both superiors and subordinates can depend in part on how well you write memoranda. E-mail messages give you the additional flexibility of adopting an informal and more conversational style. Use e-mail when speed and informality are desired.

Chapter | 6 | Process Explanations and Instructions



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Instructions and process explanations share an important common bond. Both must accurately describe a series of steps leading toward a specific result. Yet they differ in purpose, audi-

ence, and format. This chapter explores these similarities and differences and gives specific guidelines for developing both types.

>>> Process Explanations versus Instructions

Use a process explanation like the one in Figure 6–1 to help readers *understand* what has been, is being, or will be done. Instructions like the ones in Figure 6–2 show readers how to perform the *process* themselves.

Process explanations are appropriate when the reader must be informed about the action but does not need to perform it. If you suspect a reader may in fact be a *user* (i.e., someone who uses your document to perform the process), always write instructions. Figure 6–3 provides a list of contrasting features of process explanations and instructions.

Process explanations provide information for interested readers who do not need instructional details. Although explaining a process sometimes may be the sole purpose of your document, as in Figure 6-1, often you use process explanation as a pattern of organization within a document with a larger purpose. In both cases, you are writing for a reader who wants to know what has happened or will happen but does not need to perform the process.

Instructions provide users with a road map to *do* the procedure, not just understand it—that is, someone must complete a task on the basis of words and pictures you provide. Clearly, instructions present you, the writer, with a much greater challenge and risk. The reader must be able to replicate the procedure without error and, most importantly, with full knowledge of any dangers. In each case, your instructions must explain steps so thoroughly that the reader will be able to replicate the process without having to speak in person with the writer of the instructions. The next two sections give rules for preparing both process explanations and sets of instructions.



MEMORANDUM

DATE: November 7, 2011
TO: Leonard Schwartz
FROM: Cathy Vir
SUBJECT: New E-mail System

States purpose clearly.

Yesterday I met with Jane Ansel, the installation manager at BHG Electronics, about our new e-mail system. She explained the process by which the system will be installed. As you requested, this memo summarizes what I learned about the setup process.

BHG technicians will be at our offices on November 18 to complete the following tasks:

Describes five main tasks, using parallel grammatical form.

1. Removing old cable from the building conduits
2. Laying cable to link the remaining unconnected terminals with the central processing unit in the main frame
3. Installing software in the system that gives each terminal the capacity to operate the new e-mail system
4. Testing each terminal to make sure the system can operate from that location
5. Instructing selected managers on the use of the system

Confirms the follow-up activities they have already discussed.

As you and I have agreed, when the installation is complete, I will send a memo to all office employees who will have access to e-mail. That memo will discuss setup procedures that each employee must complete before they are able to use their new e-mail accounts.

Gives reader opportunity to respond.

Please let me know if you have further suggestions about how I can help make our transition to the new e-mail system as smooth as possible.

■ Figure 6-1 ■ Process explanation: electronic mail

MEMORANDUM

DATE: November 20, 2011
 TO: All Employees with Access to New E-mail System
 FROM: Cathy Vir
 SUBJECT: Instructions for Setting Up New E-mail Account

Earlier this month, we had a new e-mail system installed that will be used beginning December 1, 2011. This memo provides instructions on how to set up your new e-mail account and how to migrate all of your archived e-mail so that it will be ready for use when the new system goes into effect.

Gives clear purpose.

Please follow the step-by-step instructions below for proper setup of your e-mail and migration of your saved e-mail to the new system.

Identifies result of steps.

1. Double-click the **E-mail** icon.
2. Use the same **Username** and **Password** that you have used most recently with the old e-mail system.
3. Select the **Accounts** menu.
4. Select the **Account Options** sub-menu.

Limits each step to one action.

RESULT: A window will open that prompts an **Account Name** and **Account Type**.

Separates results from actions.

5. Enter a name (i.e., "Mail").
6. Use the drop-down menu to select **IMAP4** as the Account Type.
7. Click *Next*.

RESULT: You will be prompted to enter an **Incoming** and **Outgoing Mail Server**.

8. Enter as follows:
Incoming: www.imap.mglobal.com
Outgoing: www.smtp.mglobal.com
9. Click *Next*.

RESULT: You will be asked for your **e-mail address**.

10. Use: *yourlastname@mglobal.com*
11. Click *Next*.
12. Click the radio button that reads: **Connect through my Local Area Network (LAN)**.
13. Click *Next*.
14. Name your "New Folder" (i.e., "Old Mail")
15. Click the **Finish** button.

Your new account access should now be available, and your old e-mails will move to the new folder that you just named.

Results if instructions have been followed correctly.

If you encounter any problems while performing the steps listed above, please contact a member of our IT staff for assistance.

Shows reader how to get more information.

■ Figure 6-2 ■ Instructions: electronic mail

■ **Figure 6–3** ■

Process
explanations
versus instructions

PROCESS EXPLANATIONS

- Purpose:** Explain a sequence of steps in such a way that the reader understands a process
- Format:** Use paragraph descriptions, listed steps, or some combination of the two
- Style:** Use *objective* point of view (“2. The operator started the engine ...”), as opposed to *command* point of view (“2. Start the engine ...”)

INSTRUCTIONS

- Purpose:** Describe a sequence of steps in such a way that the reader can *perform* the sequence of steps
- Format:** Employ numbered or bulleted lists, organized into subgroups of easily understandable units of information
- Style:** Use *command* point of view (“3. Plug the phone jack into the recorder unit”), as opposed to *objective* point of view (“3. The phone jack was plugged into the recorder unit”)

>>> Guidelines for Process Explanations

You have already learned that process explanations are aimed at persons who must understand the process, not perform it. Process explanations often have the following purposes:

- Describing an experiment
- Explaining how a machine works
- Recording steps in developing a new product
- Describing procedures to ensure compliance with regulations
- Describing what will happen during a medical procedure

>> Process Guideline 1: Know Your Purpose and Your Audience

Your intended purpose and expected audience influence every detail of your explanation. Following are some preliminary questions to answer before writing:

- Are you supposed to give just an overview, or are details needed?
- Do readers understand the technical subject, or are they laypersons?
- Do readers have mixed technical backgrounds?
- Does the process explanation supply supporting information (perhaps in an appendix), or is it the main part of the document?

When process explanations are directed to a mixed technical audience, write for your least technical readers.

>> Process Guideline 2: Follow the ABC Format

In Chapter 1, you learned about the ABC format (**A**bstract/**B**ody/**C**onclusion) that applies to all documents. The abstract gives a summary, the body supplies details, and the conclusion provides a wrap-up or leads to the next step in the communication process. Whether a process explanation forms all or part of a document, it usually subscribes to the following version of the three-part ABC plan:

- The **Abstract** component includes three background items:
 1. Purpose statement
 2. Overview or list of the main steps that follow
 3. List of equipment or materials used in the process
- The **Body** component of the process explanation moves logically through the steps of the process. By definition, all process explanations follow a *chronological*, or step-by-step, pattern of organization. These steps can be conveyed in two ways, paragraphs or a list of steps.
- The **Conclusion** component of a process explanation keeps the process from ending abruptly with the last step. Here you should help the reader put the steps together into a coherent whole. In Figure 6–1 (p. 110), the last two paragraphs identify the process outlined in the memo as part of a larger change within the organization. When the process explanation is part of a larger document, you can show how the process fits into a larger context.

>> Process Guideline 3: Use an Objective Point of View

Process explanations describe a process rather than direct how it is to be done. Thus they are written from an objective point of view—not from the personal *you* or *command* point of view common to instructions. Note the difference in these examples:

Process: The concrete is poured into the two-by-four frame.

or

The technician pours the concrete into the two-by-four frame.

Instructions: Pour the concrete into the two-by-four frame.

The process excerpts *explain* the steps, whereas the instructions excerpt *gives a command* for completing the instructions.

>> Process Guideline 4: Choose the Right Amount of Detail

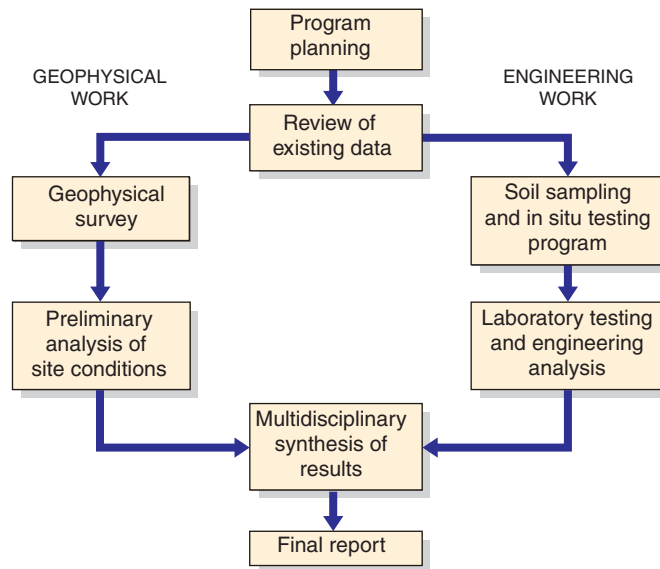
Only a thorough audience analysis will tell you how much detail to include. Figure 6–1 (p. 110), for example, could contain much more detail about the substeps for installing the e-mail system, but the writer decided that the recipient would not need more technical detail.

>> Process Guideline 5: Use Flowcharts for Complex Processes

Some process explanations contain steps that are occurring at the same time. In this case, you may want to supplement a paragraph or list explanation with a flowchart. Such charts

■ **Figure 6–4** ■

Flowchart from explanation of site investigation process



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For an example of an explanation of procedures see Procedures 2: Pest Control.

use boxes, circles, and other geometric shapes to show progression and relationships among various steps. Figure 6–4 is a flowchart that clarifies how multiple steps in a process will be occurring at the same time.

>>> Guidelines for Instructions

A survey of technical communication managers found that instructional materials, such as manuals and online help remain the most common and most important documents in the field.¹ Thus most writers would benefit from being able to create clear instructions.

Both process explanations and instructions are organized by time, but the similarity stops there. Instructions walk readers through the process so that they can do it, not just understand it. It is one thing to explain the process by which a word processing program works; it is quite another to write a set of instructions for using that word processing program. This section explores the challenge of writing instructions by giving you some basic writing and design guidelines.

These guidelines for instructions also apply to complete operating *manuals*, a document type that many technical professionals will help to write during their careers. Those manuals include the instructions themselves, as well as related information, such as (1) features, (2) physical parts, and (3) troubleshooting tips.

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See Instructions 1 for an example written for multiple audiences.

>> Instructions Guideline 1: Select the Correct Technical Level

Know exactly who will read your instructions. Are readers technicians, engineers, managers, general users, or some combination of these groups? Once you answer this question, select language that every reader can understand. If, for example, the instructions

¹Kenneth T. Rainey, Roy K. Turner, and David Dayton, “Do Curricula Correspond to Managerial Expectations? Core Competencies for Technical Communicators,” *Technical Communication* 52 (2005): 321–352.

include technical terms or names of objects that may not be understood, use the techniques of definition and description discussed in Chapter 5.

>> **Instructions Guideline 2: Provide Introductory Information**

Like process explanations, instructions follow the ABC format (Abstract/Body/Conclusion) described in Chapter 1. The introductory (or abstract) information should include (1) a purpose statement, (2) a summary of the main steps, and (3) a list or an illustration giving the equipment or materials needed (or a reference to an attachment with this information). These three items set the scene for the procedure itself.

Besides these three “musts,” you should consider whether some additional items might help set the scene for your user:

- Pointers that help with installation
- Definitions of terms
- Theory of how something works
- Notes, cautions, warnings, or dangers that apply to all steps

>> **Instructions Guideline 3: Use Numbered Lists in the Body**

A simple format is crucial to the body of the instructions—that is, the steps themselves. Most users constantly go back and forth between these steps and the project to which they apply. Thus, you should avoid paragraph format and instead use a simple numbering system.

>> **Instructions Guideline 4: Group Steps under Task Headings**

Readers prefer that you group together related steps under headings, rather than present an uninterrupted “laundry list” of steps. Figure 6–5 shows how this technique has been used in one section of a fairly long set of instructions for operating a scanner. Given the number of steps in this case, the writer has used a separate numbering system within each grouping.

>> **Instructions Guideline 5: Place One Action in a Step**

A common error is to bury several actions in a single step. This approach can confuse and irritate readers. Instead, break up complex steps into discrete units.

>> **Instructions Guideline 6: Lead Off Each Action Step with a Verb**

Instructions should include the *command* form of a verb at the start of each step. This style best conveys a sense of action to your readers. Figure 6–2 on page 111 uses command verbs consistently for all steps throughout the procedures.



■ **Figure 6–5** ■

Grouped steps for instructions from installing and operating a scanner

1. Turning on Your Scanner

- a. Locate the On/Off switch on the front of the scanner.
- b. Switch to the On position.

2. Scanning

- a. Open the scanning program by double-clicking the desktop icon.
- b. Place a piece of paper on the scanner bed, in the upper right hand corner.
- c. Select *Scan Document*.
- d. Click *Preview Document*.

NOTE: This will take 15–20 seconds.

- e. Click and drag the edges of the crop box to fit the document.
- f. Click *Scan*.

RESULT: The scanner will scan the selected area of the preview image.

3. Saving Scanned Files

- a. To save your scanned file(s), go to File > Save.
- b. Enter a name for your document.
- c. Choose to save it as either a JPG (for pictures) or PDF (for text).
- d. Find the file that you want to save the document in.
- e. Click *Save*.

>> Instructions Guideline 7: Remove Extra Information from the Step

Sometimes you may want to follow the command sentence with an explanatory sentence or two. In this case, distinguish such helpful information from actions by giving it a label, such as *Note* or *Result* as in Figure 6–2.

>> Instructions Guideline 8: Use Bullets or Letters for Emphasis

Sometimes you may need to highlight information, especially within a particular step. Avoid using numbers for this purpose, because you are already using them to signify steps. Bullets work best if there are just a few items; letters are best if there are many, especially if they are in a sequence. In particular, consider using bullets at any point at which users have an *option* as to how to respond.

>> Instructions Guideline 9: Emphasize Cautions, Warnings, and Dangers

Instructions often require alerts that draw attention to risks in using products and equipment. Your most important obligation is to highlight such information. Because professional associations and individual companies differ in the way they define terms associated with risk, make sure the alerts in your document follow the appropriate guidelines for your readers. If you have no specific guidelines, however, the following

definitions can serve as “red flags” to the reader. The level of risk increases as you move from 1 to 3:

1. **Caution:** possibility of damage to equipment or materials
2. **Warning:** possibility of injury to people
3. **Danger:** probability of injury or death to people

If you are not certain that these distinctions will be understood by your readers, define the terms *caution*, *warning*, and *danger* in a prominent place before you begin your instructions.

As for placement of the actual cautions, warnings, or danger messages, your options are as follows:

- **Option 1:** *In a separate section, right before the instructions begin.* This approach is most appropriate when you have a list of general warnings that apply to much of the procedure or when one special warning should be heeded throughout the instructions—for example: “WARNING: Keep main breaker on *off* during entire installation procedure.”
- **Option 2:** *In the text of the instructions.* This approach works best if the caution, warning, or danger message applies to the step that immediately follows it. Thus, users are warned about a problem *before* they read the step to which it applies.
- **Option 3:** *Repeatedly throughout the instructions.* This strategy is preferable with instructions that repeatedly pose risk to the user. For example, steps 4, 9, 12A, and 22—appearing on several different pages—may *all* include the hazard of fatal electrical shock. Your *danger* notice should appear in each step, as well as in the introduction to the document.

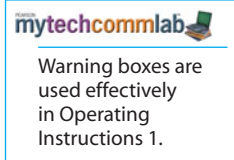
Give information about potential risks *before* the operator has the chance to make the mistake. Also, the caution, warning, or danger message can be made visually prominent by using font choices such as underlining, bold, or full caps, or graphic elements such as boxes. Color graphics can be another effective indicator of risk. You have probably seen examples, such as a red flame in a box for fire, a jagged line in a triangle for electrical shock, or an actual drawing of a risky behavior with an X through it.

The International Organization for Standardization (ISO) established international standards for safety alerts in ISO 3864, and the American National Standards Institute (ANSI) established domestic standards for safety alerts in ANSI Z535. If the organization you work for complies with ISO or ANSI, you should make sure that you are using the most recent version of the appropriate standards to reinforce the message in your text about cautions, warnings, and dangers.

>> Instructions Guideline 10: Keep a Simple Style

Perhaps more than any other type of technical communication, instructions must be easy to read. Readers expect a no-nonsense approach to writing that gives them required information without fanfare. Following are some useful techniques:

- Keep sentences short, with an average length of fewer than 10 words.
- Use informal definitions (often in parentheses, like this) to define terms not understood by all readers.



- Never use a long word when a short one will do.
- Be specific and avoid words with multiple interpretations (*frequently, seldom, occasionally, etc.*).

>> Instructions Guideline 11: Use Graphics

Illustrations are essential for instructions that involve equipment. Place an illustration next to every major step when (1) the instructions or equipment is quite complicated or (2) the audience may contain unskilled readers or people who are in a hurry. Such word–picture associations create a page design that is easy to follow.

Another useful graphic in instructions is the table. Sometimes within a step you must show correspondence between related data. For example, the instructions below include a spot table.

Step 1: Use pyrometric cones to determine when a kiln has reached the proper temperature for firing pottery. Common cone ratings are as follows:

Cone 018	1200°F
Cone 07	1814°F
Cone 06	1859°F
Cone 04	1940°F

>> Instructions Guideline 12: Test Your Instructions for Usability

Testing instructions for *usability* ensures that your users are able to follow them easily.² When you design for usability, you should be focused primarily on the user, not the product itself. This is true whether you are designing a document, software, a computer interface, or a piece of machinery. Products that are usable have the following qualities:

- Learning them is easy.
- Operating them requires the minimum number of steps.
- Remembering how to use them is easy.
- Using them satisfies the user's goals.

Usability does not happen automatically, but should be a concern from the earliest stages of the design of products and documentation.

Professional writers often test their instructions on potential users before completing the final draft. You can adapt the following user-based approach to testing assignments in this class and projects in your career. Specifically, follow these four steps:

1. Team up with another class member (or a colleague on the job). This person should be unfamiliar with the process and should approximate the technical level of your intended audience.

²Adapted from Carol M. Barnum, *Usability Testing and Research* (New York: Longman, 2002).

2. Give this person a draft of your instructions and provide any equipment or materials necessary to complete the process. For the purposes of a class assignment, this approach works only for a simple process with little equipment or few materials.
3. Observe your colleague following the instructions you provide. You should record both your observations and any verbal responses this person makes while moving through the steps.
4. Revise your instructions to solve problems your user encountered during the test.

>>> Chapter Summary

Both process explanations and instructions share the same organization principle: time. That is, both relate a step-by-step description of events. Process explanations address an audience that wants to be informed but does not need to perform the process itself. Instructions are geared specifically for persons who must complete the procedure themselves.

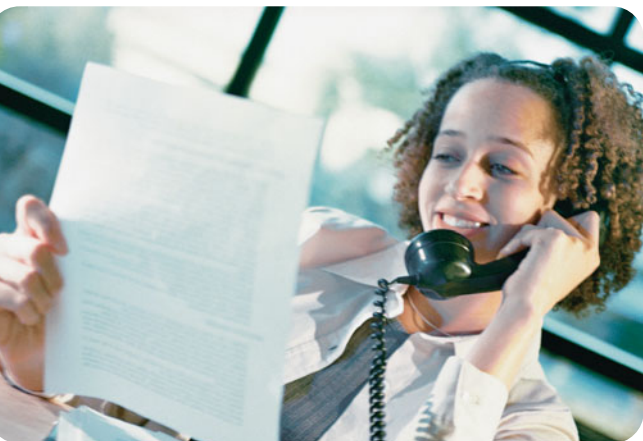
In writing good process explanations, follow these basic guidelines:

1. Know your purpose and audience.
2. Follow the ABC format.
3. Use an objective point of view.
4. Choose the right amount of detail.
5. Use flowcharts for complex processes.

For instructions, follow these 12 rules:

1. Select the correct technical level.
2. Provide introductory information.
3. Use numbered lists in the body.
4. Group similar steps under heads.
5. Place one action in a step.
6. Lead off each action step with a verb.
7. Remove extra information from the step.
8. Use bullets or letters for emphasis.
9. Emphasize cautions, warnings, and dangers.
10. Keep a simple style.
11. Use graphics.
12. Test your instructions.

Chapter | 7 | Reports



>>> Chapter Outline

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Reports communicate the results of research and activities in an organization. They may also be archived to provide a record of what an organization has accomplished. This chapter provides general guidelines for writing reports, and specific guidelines

for the two report formats—informal reports and formal reports, which differ primarily in their length and scope. This chapter also introduces you to four common types of reports.

>>> General Guidelines for Reports

You will write many reports in your career. Reports may be as simple as a single-page form, like an activity report that you submit daily or weekly so that clients can be billed for your time. Or they may be over 100-page long, like an analysis of a major equipment failure. This section provides general guidelines for effective reports.

>> Report Guideline 1: Plan Well Before You Write

As with all types of workplace writing, planning is the most important part of the writing process. For each project, complete the Planning Form at the end of this book to record specific information about these points:

- The document's purpose
- The variety of readers who will receive the document
- The needs and expectations of readers, particularly decision makers
- An outline of the main points to be covered in the body
- Strategies for writing an effective document

>> Report Guideline 2: Separate Fact from Opinion

Avoid any confusion about what constitutes fact or opinion. The safest approach in the report discussion is to move logically from findings to your conclusions and, finally, to your recommendations. Because these terms are often confused, some working definitions are as follows:

- **Findings:** Facts you uncover (e.g., you observed severe cracks in the foundations of two adjacent homes in a subdivision).
- **Conclusions:** Ideas or beliefs you develop based on your findings (e.g., you conclude that foundation cracks occurred because the two homes were built on soft fill, where original soil had been replaced by construction scraps).
- **Recommendations:** Suggestions or action items based on your conclusions (e.g., you recommend that the foundation slab be supported by adding concrete posts beneath it). Though consisting of opinions, recommendations should be grounded in facts presented in the report.

>> Report Guideline 3: Make Text Visually Appealing

Following are three visual devices that help get attention, maintain interest, and highlight important information:

- Bulleted points for short lists (like this one)
- Numbered points for lists that are longer or that include a list of ordered steps
- Frequent use of headings and subheadings

>> Report Guideline 4: Use Illustrations for Clarification and Persuasion

A simple table or figure can sometimes be just the right complement to a technical discussion in the text. Incorporate illustrations into the report body to make technical information accessible and easier to digest.

>> Report Guideline 5: Edit Carefully

The Handbook at the end of this text gives information about editing. For now, remember the following basic guidelines:

- Keep most sentences short and simple.
- Proofread several times for mechanical errors, such as misspellings (particularly personal names).
- Triple-check all cost figures for accuracy.
- Make sure all attachments are included, are mentioned in the text
- Check the format and wording of all headings and subheadings.
- Ask a colleague to check over the report.

Remember—both supervisors and clients will judge you as much on communication skills as they do on technical ability.

>>> Guidelines for Informal Reports

A working definition of informal reports follows:

Informal report: A document that contains about two to five pages of text, not including attachments. It has more substance than a simple letter or memo, but is presented in letter or memo format. It can be directed to readers either outside or inside your organization. If outside, it may be called a *letter report*; if inside, it may be called a *memo report*. In either case, its purpose can be *informative* (to clarify or explain), *persuasive* (to convince), or both.



Following are six guidelines that focus on informal report format:

>> Informal Report Guideline 1: Use Letter or Memo Format

Figure 7–1 shows that letter reports follow about the same format as typical business letters (see Chapter 4). Yet, the format of letter reports differs from that of letters in the following respects:

- The greeting is sometimes left out or replaced by an attention line, especially when your letter report will go to many readers in an organization.
- A report title often comes immediately after the inside address. It identifies the specific project covered in the report.
- Spacing between lines might be single, one-and-one-half, or double, depending on the reader's preference.

Memo reports follow the same format as typical business memos. Both memos and memo reports have a subject line that should engage interest, give readers their first quick look at your topic, and be both specific and concise—for example, “Fracture Problems with Molds 43-D and 42-G” is preferable to “Problems with Molds.” Because memo reports are usually longer than memos, they tend to contain more headings than routine memos.

>> Informal Report Guideline 2: Use the ABC Format for Organization

Most technical documents, including informal reports, follow what this book calls the *ABC format*. This approach to organization includes three parts: (1) **Abstract**, (2) **Body**, and (3) **Conclusion**. The next four guidelines give details on the ABC format as applied to informal reports.

>> Informal Report Guideline 3: Create the Abstract as an Introductory Summary

Abstracts should give readers a summary, the “big picture.” This text suggests that in informal reports, you label this overview *Introduction*, *Summary*, or *Introductory Summary*, terms that give the reader a good idea of what the section contains. Just one or two paragraphs in this first section give readers three essential pieces of information:

1. **Purpose** for the report—why are you writing it?
2. **Scope** statement—what range of information does the report contain?
3. **Summary** of essentials—what main information does the reader most want or need to know?

>> Informal Report Guideline 4: Put Important Details in the Body

The body section expands on the outline presented in the introductory summary. If your report goes to a diverse audience, managers often read the quick overview in the

ABC Format: Organization

- **ABSTRACT:** Start with a capsule version of the information most needed by decision makers.
- **BODY:** Give details in the body of the report, where technical readers are most likely to linger a while to examine supporting evidence.
- **CONCLUSION:** Reserve the end of the report for a description or list of findings, conclusions, or recommendations.

12 Post Street
Houston Texas 77000
(713) 555-9781

April 22, 2010

Big Muddy Oil Company Inc
12 Rankin St
Abilene TX 79224

ATTENTION: Mr. James Smith, Engineering Manager

**SHARK PASS STUDY
BLOCK 15, AREA 43-B
GULF OF MEXICO**

INTRODUCTORY SUMMARY

You recently asked our firm to complete a preliminary soils investigation at an offshore rig site. This report presents the tentative results of our study, including major conclusions and recommendations. A longer, formal report will follow at the end of the project.

On the basis of what we have learned so far, it is our opinion that you can safely place an oil platform at the Shark Pass site. To limit the chance of a rig leg punching into the sea floor, however, we suggest you follow the recommendations in this report.

WORK AT THE PROJECT SITE

On April 15 and 16, 2010, BoomCo's engineers and technicians worked at the Block 15 site in the Shark Pass region of the gulf. Using BoomCo's leased drill ship, *Seeker II*, as a base of operations, our crew performed these main tasks:

- Seismic survey of the project study area
- Two soil borings of 40 feet each

Both seismic data and soil samples were brought to our Houston office the next day for laboratory analysis.

LABORATORY ANALYSIS

On April 17 and 18, our lab staff examined the soil samples, completed bearing capacity tests, and evaluated seismic data. Here are the results of that analysis.

Soil Layers

Our initial evaluation of the soil samples reveals a 7–9 feet layer of weak clay starting a few feet below the seafloor. Other than that layer, the composition of the soils seems fairly typical of other sites nearby.

Includes specific title.

Uses optional heading for abstract part of ABC format.

Draws attention to main point of report.

Gives on-site details of project—dates, location, tasks.

Uses lead-in to subsections that follow.

Highlights most important point about soil layer—that is the weak clay.



M-Global Inc. | 127 Rainbow Lane | Baltimore MD 21202 | 410.555.8175

James Smith
April 22, 2010
Page 2

Bearing Capacity

We used the most reliable procedure available, the XYZ method, to determine the soil's bearing capacity (i.e., its ability to withstand the weight of a loaded oil rig). That method required that we apply the following formula:

$Q = cN_v + tY$, where

Q = ultimate bearing capacity

c = average cohesive shear strength

N_v = the dimensionless bearing capacity factor

t = footing displacement

Y = weight of the soil unit

The final bearing capacity figure will be submitted in the final report, after we repeat the tests.

Seafloor Surface

By pulling our underwater seismometer back and forth across the project site, we developed a seismic "map" of the seafloor surface. That map seems typical of the flat floor expected in that area of the gulf. The only exception is the presence of what appears to be a small sunken boat. This wreck, however, is not in the immediate area of the proposed platform site.

CONCLUSIONS AND RECOMMENDATIONS

Based on our analysis, we conclude that there is only a slight risk of instability at the site. Although unlikely, it is possible that a rig leg could punch through the sea floor, either during or after loading. We base this opinion on (1) the existence of the weak clay layer, noted earlier and (2) the marginal bearing capacity.

Nevertheless, we believe you can still place your platform if you follow careful rig-loading procedures. Specifically, take these precautions to reduce your risk:

- 1 Load the rig in 10-ton increments, waiting 1 hour between loadings.
- 2 Allow the rig to stand 24 hours after the loading and before placement of workers on board.
- 3 Have a soils specialist observe the entire loading process to assist with any emergency decisions if problems arise.

As noted at the outset, these conclusions and recommendations are based on preliminary data and analysis. We will complete our final study in 3 weeks and submit a formal report shortly thereafter.

BoomCo, Inc., enjoyed working once again for Big Muddy Oil at its Gulf of Mexico lease holdings. I will phone you this week to see if you have any questions about our study. If you need information before then, please give me a call.

Sincerely,



Bartley Hopkins, Project Manager
BoomCo, Inc.

hg

Notes why this method was chosen (i.e., reliability).

Explains both *how* the mapping procedure was done and *what* results it produced.

Leads off section with major conclusion, for emphasis.

Restates points (made in body) that support conclusion.

Uses list to emphasize recommendations to reduce risk.

Again mentions tentative nature of information, to prevent misuse of report.

Maintains contact and shows initiative by offering to call client.

introductory summary and then skip to conclusions and recommendations. Technical readers, however, may look first to the body section(s), where they expect to find supporting details.

- **Use headings generously.** Each time you change a major or minor point, consider whether a heading change would help the reader.
- **Precede subheadings with a lead-in passage.** For example, “This section covers these three phases of the field study: clearing the site, collecting samples, and classifying samples.” The preceding sentence does for the entire section exactly what the introductory summary does for the entire report—it provides a roadmap for what’s ahead.
- **Move from general to specific in paragraphs.** Start each paragraph with a topic sentence that includes your main point and then give supporting details.

>> Informal Report Guideline 5: Focus Attention in Your Conclusion

Letter and memo reports can end with a section labeled *Findings*, *Conclusions*, or *Conclusions and Recommendations*. This section gives details about major findings, conclusions, and, if called for, recommendations. People often remember best what they read last, so think hard about what you place at the end of a report.

The precise amount of detail in your conclusion depends on which of these two options you choose for your particular report:

Option 1: If your major conclusions or recommendations have already been stated in the discussion, then you only need to restate them briefly to reinforce their importance.

Option 2: If the discussion leads up to, but has not covered, these conclusions or recommendations, then you may want to give more detail in this final section (see Figure 7–1, pp. 126–127).

>> Informal Report Guideline 6: Use Attachments for Less Important Details

The trend today is to avoid lengthy text in informal reports. Replace as much report text as possible with clearly labeled attachments that include items, such as tables, figures, or costs.

>>> Guidelines for Formal Reports

You will write a number of long, formal reports during your career, often with colleagues. This text uses the following working definition for a formal report:

Formal report: A formal report covers complex projects and is directed to readers at different technical levels. Although not defined by length, a formal report usually contains at least 6 to 10 pages of text, not including appendices. It can be directed to readers either inside or outside your organization.

Because formal reports may be longer and more complex than other forms of technical communication, it is important to help your readers navigate through the report. This is accomplished with special front and end materials, clear headings, and other navigation devices like running headers and footers. This section provides guidelines for writing the main parts of a long report and includes a complete long report that follows this chapter's guidelines.

Though formats differ among organizations and disciplines, one approach to good organization applies to all formal reports. This approach is based on three main principles, discussed in detail in Chapter 1:

Principle 1: Write different parts for different readers.

Principle 2: Place important information first.

Principle 3: Repeat key points when necessary.

ABC Format: Formal Report

■ **ABSTRACT:**

- Cover/title page
- Letter or memo of transmittal
- Table of contents
- List of illustrations
- Executive summary
- Introduction

■ **BODY:**

- Discussion sections
- [Appendices—appear after text but support Body section]

■ **CONCLUSION:**

- Conclusions
 - Recommendations
-

These apply to formal reports even more than they do to short documents. Because formal reports often have a mixed technical audience, most readers focus on specific sections that interest them most, and few readers have time to wade through a lot of introductory information before reaching the main point.

You can respond to these reader needs by following the ABC format (for Abstract, Body, Conclusion). As noted in Chapter 1, the three main rules are that you should (1) start with an abstract for decision makers, (2) put supporting details in the body, and (3) use the conclusion to produce action. This simple ABC format should be evident in all formal reports, despite their complexity. The particular sections of formal reports fit within the ABC format as shown on the right:

Nine Parts of Formal Reports

The nine parts of the formal report are as follows:

1. Cover/title page
2. Letter or memo of transmittal
3. Table of contents
4. List of illustrations
5. Executive summary
6. Introduction
7. Discussion sections

8. Conclusions and recommendations
9. End material

Following are some guidelines for these parts.

Cover/Title Page

Formal reports are usually bound, often with a cover used for all reports in the writer's organization. Because the cover is the first item seen by the reader, it should be attractive and informative. It usually contains the same four pieces of information mentioned in the following list with regard to the title page.

Inside the cover is the title page, which should include the following four pieces of information:

- Project title (exactly as it appears on the letter/memo of transmittal)
- Your client's name ("Prepared for...")
- Your name and/or the name of your organization ("Prepared by...")
- Date of submission

To make your title page or cover distinctive, you might want to place a simple illustration on it; however, do not clutter the page. Use a visual only if it reinforces a main point and if it can be done simply and tastefully, as in Figure 7–2.

Letter/Memo of Transmittal

Letters or memos of transmittal give the readers a taste of what is ahead. If your formal report is to readers outside your own organization, write a letter of transmittal. If it is to readers inside your organization, write a memo of transmittal.

>> Transmittal Guideline 1: Place the Letter/Memo Immediately after the Title Page

This placement means that the letter/memo is bound with the document, to keep it from becoming separated.

>> Transmittal Guideline 2: Include a Major Point from Report

Readers are heavily influenced by what they read first in reports. Therefore, take advantage of the position of this section by including a major finding, conclusion, or recommendation from the report—besides supplying necessary transmittal information.

>> Transmittal Guideline 3: Acknowledge Those Who Helped You

Recognizing those who have been particularly helpful with your project gives them recognition and identifies you as a team player. It reflects well on you and on your organization.

>> Transmittal Guideline 4: Follow Letter and Memo Conventions

Like other letters and memos, letters and memos of transmittal should be easy to read, inviting readers into the rest of the report. Keep introductory and concluding paragraphs relatively short—no more than three to five lines each. Also, write in a conversational style, free of technical jargon.

Table of Contents

Your contents page acts as an outline. Many readers go there right away to grasp the structure of the report, and then return repeatedly to locate report sections of most interest to them. Guidelines follow for assembling this important component of your report; see p. 137 for an example in Figure 7–2.

>> Table of Contents Guideline 1: Make It Very Readable

The table of contents must be pleasing to the eye so that readers can find sections quickly and see their relationship to each other. Be sure to

- Space items well on the page
- Use indenting to draw attention to subheadings
- Include page numbers for every heading and subheading, unless there are many headings in a relatively short report, in which case you can delete page numbers for all of the lowest level headings listed in the table of contents

>> Table of Contents Guideline 2: Use the Contents Page to Reveal Report Emphases

Be specific yet concise so that each heading listed in the table of contents gives the reader a good indication of what the section contains. Readers associate the importance of report sections with the number of headings and subheadings listed in the table of contents.

>> Table of Contents Guideline 3: Consider Leaving Out Low-Level Headings

In very long reports, you may want to declutter the table of contents by removing lower-level headings. As always, the needs of the readers are the most important criterion to use in making this decision.

>> Table of Contents Guideline 4: List Appendices

Appendices include items, such as tables of data or descriptions of procedures that are inserted at the end of the report. Typically, they are listed at the end of the table of contents. Tabs on the edges of pages help the reader locate these sections.

>> Table of Contents Guideline 5: Use Parallel Form in All Entries

All headings in one section, and sometimes even all headings and subheadings in the report, have parallel grammatical form.

>> Table of Contents Guideline 6: Proofread Carefully

Wrong page numbers and incorrect headings are two common mistakes. Proofread this section carefully.

List of Illustrations

Illustrations within the body of the report are usually listed on a separate page right after the table of contents. Another option is to list them at the bottom of the table of contents page rather than on a separate page. In either case, this list should include the number, title, and page number of every table and figure within the body of the report.

Executive Summary

No formal report would be complete without an executive summary. Consider it a stand-alone section that provides a capsule version of the report and is free of technical jargon. In some cases, a copy of the executive summary may be circulated and filed separate from the report.

>> Executive Summary Guideline 1: Put It on One Page

It is comforting to most readers to know that somewhere within a long report there is one page to which they can turn for an overview. Moreover, a one-page length permits easy distribution at meetings.

Some extremely long formal reports may require that you write an executive summary of several pages or longer. In this case, you can still provide the reader with a section that summarizes the report in less than a page by including a brief *abstract*. The abstract is a condensed version of the executive summary directed to the highest-level decision makers, that is placed right before the executive summary.

>> Executive Summary Guideline 2: Avoid Technical Jargon

Include only that level of technical language the decision makers understand. Do not talk over the heads of the most important readers.

>> Executive Summary Guideline 3: Include Only the Important Conclusions and Recommendations

The executive summary mentions only the major points. An exhaustive list of findings, conclusions, and recommendations can come later at the end of the report.

>> **Executive Summary Guideline 4: Avoid References to the Report Body**

Avoid the tendency to say that the report provides additional information. It is understood that the executive summary is only a generalized account of the report's contents.

An exception is those instances when you are discussing issues that involve danger or liability in which it may be necessary to add qualifiers in your summary—for example, “As noted in this report, further study will be necessary.” Such statements protect you and the client in the event the executive summary is removed from the report and used as a separate stand-alone document.

>> **Executive Summary Guideline 5: Use Paragraph Format**

Whereas lists are often appropriate for body sections of a report, they can give executive summaries a fragmented effect. Instead, the best summaries create unity with a series of relatively short paragraphs that flow well together. Within a paragraph, there can be a short listing of a few points for emphasis, but the listing should not be the main structural element of the summary.

>> **Executive Summary Guideline 6: Write the Executive Summary Last**

Only after finishing the report do you have the perspective to write a summary. Sit back and review the report from beginning to end, asking yourself, “What would my readers really need to know if they had only a minute or two to read?” The answer to that question becomes the core of your executive summary.

Introduction

View this section as your chance to prepare both technical and nontechnical readers for the discussion ahead. Give information on the report's purpose, scope, and format, as well as a project description.

>> **Introduction Guideline 1: State Your Purpose and Lead into Subsections**

The purpose statement for the document should appear immediately after the main introduction heading. Follow it with a sentence that mentions the introduction subdivisions to follow.

>> **Introduction Guideline 2: Include a Project Description**

Here you must be precise about the project. Depending on the type of project, you may be describing a physical setting, a set of problems that prompted the report study, or some other data. When the project description is too long for the introduction, sometimes it is placed in the body of the report.

>> Introduction Guideline 3: Include Scope Information

This section outlines the precise objectives of the study. Your listing or description should parallel the order of the information presented in the body of the report. Like the project description, this subsection must be accurate in every detail.

>> Introduction Guideline 4: Consider Including Information on Report Format

Often, the scope section lists information as it is presented in the report. If this is not the case, end the introduction with a short subsection on the report format where you can give readers a brief preview of the main sections that follow. In effect, the section acts as a condensed table of contents and may list the report's major sections and appendices.

Discussion Sections

Discussion sections make up the longest part of formal reports and are written for the most technical members of your audience. You can focus on facts and opinions, demonstrating the technical expertise that the reader expects from you.

>> Discussion Guideline 1: Move from Facts to Opinions

As you have learned, the ABC format requires that you start your formal report with a summary of the most important information—that is, you skip to essential conclusions and recommendations. Once into the discussion section, however, you back up and adopt a strategy that parallels the stages of the technical project itself. You begin with data and move toward conclusions and recommendations.

One way to view the discussion is that it should follow the order of a typical technical project, which usually involves the following stages:

First, collect data (e.g., samples, interviews, records).

Second, subject these data to verification or testing (e.g., lab tests, computer analyses).

Third, analyze all the information, using your professional experience and skills to form conclusions (or convictions based on the data).

Fourth, develop recommendations that flow directly from the conclusions you have formed.

Thus, the body of your report gives technical readers the same movement from fact toward opinion that you experience during the project itself.

>> Discussion Guideline 2: Use Frequent Headings and Subheadings

Headings give readers handles by which to grasp the content of your report. Your readers view headings, collectively, as a sort of outline by which they can make their way easily through the report.

>> Discussion Guideline 3: Use Listings to Break Up Long Paragraphs

Long paragraphs full of technical details irritate readers. Use paragraphs for brief explanations, not for descriptions of processes or other details that could be listed.

>> Discussion Guideline 4: Place Excessive Detail in Appendices

Appendices give readers access to supporting information without cluttering the text of the report. Of course, you must refer to appendices in the body of the report and label appendices clearly so that readers can locate them easily.

Conclusions and Recommendations

This section gives readers a description—sometimes in the form of a listing—of all conclusions and recommendations. The points may or may not have been mentioned in the body of the report, depending on the length and complexity of the document. *Conclusions*, on the one hand, are convictions or beliefs based on the findings of your study. *Recommendations*, on the other hand, are actions you are suggesting based on your conclusions. For example, your conclusion may be that there is a dangerous level of toxic chemicals in a town's water supply, and your recommendation may be that the toxic site near the reservoir should be cleaned immediately.

The “Conclusions and Recommendations” section provides a complete list of conclusions and recommendations for technical and management readers, whereas the “Executive Summary” provides a selected list or description of the most important conclusions and recommendations for decision makers. In other words, view the “Conclusions and Recommendations” section as an expanded version of the “Executive Summary.” It usually assumes one of these three headings, depending, of course, on the content:

1. Conclusions
2. Recommendations
3. Conclusions and Recommendations

Another option for reports that contain many conclusions and recommendations is to separate this last section into two sections: (1) “Conclusions” and (2) “Recommendations.”

End Material

One kind of end material—appendices—is mentioned in the context of the discussion section. Note that formal reports may also contain works-cited pages or bibliographies, which should be included in the end materials. Finally, very long reports may include indices.

>>> Formal Report Example

Figure 7–2 provides a long and formal technical report. The report results from a study completed for the city of Winslow, Georgia. Members of the audience come from both technical and nontechnical backgrounds. Some are full-time professionals hired by the city, whereas others are part-time, unpaid citizens appointed by the mayor to explore

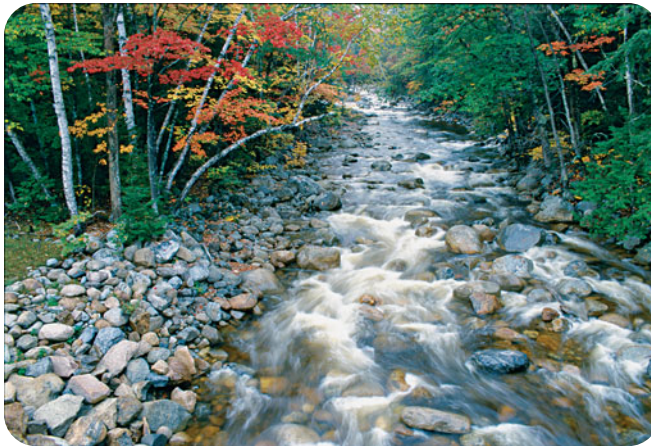
STUDY OF WILDWOOD CREEK

WINSLOW, GEORGIA

Prepared for:
The City of Winslow

Prepared by:
Christopher S. Rice, Hydro/Environmental Engineer
D-Lynn, Inc.

November 28, 2010



Uses graphic on title page to reinforce theme of environmental protection.

12 Peachtree Street
Atlanta GA 30056
(404) 555-7524

D-Lynn Project #99-119
November 28, 2010

Adopt-a-Stream Program
City of Winslow
300 Lawrence Street
Winslow
Georgia 30000

Attention: Ms. Elaine Sykes, Director

**STUDY OF WILDWOOD CREEK
WINSLOW, GEORGIA**

Lists project title as it
appears on title page.

Gives brief statement of
project information.

Provides major point
from report.

We have completed our seven-month project on the pollution study of Wildwood Creek. This project was authorized on May 16, 2010. We performed the study in accordance with our original proposal No. 14-P72, dated April 24, 2010.

This report mentions all completed tests and discusses the test results. Wildwood Creek scored well on many of the tests, but we are concerned about several problems—such as the level of phosphates in the stream. The few problems we observed during our study have led us to recommend that several additional tests should be completed.

Thank you for the opportunity to complete this project. We look forward to working with you on further tests for Wildwood Creek and other waterways in Winslow.

Sincerely,

Christopher S. Rice

Christopher S. Rice, P.E.
Hydro/Environmental Engineer



M-Global Inc | 127 Rainbow Lane | Baltimore MD 21202 | 410.555.8175

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Uses white space,
indenting, and bold to
accent organization of
report.

Includes illustration
titles as they appear in
text.



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EXECUTIVE SUMMARY

The City of Winslow hired D-Lynn, Inc., to perform a pollution study of Wildwood Creek. The section of the creek that was studied is a one-mile-long area in Burns Nature Park, from Newell College to U.S. Highway 42. The study lasted seven months.

Summarizes purpose and scope of report.

D-Lynn completed 13 tests on four different test dates. Wildwood scored fairly well on many of the tests, but there were some problem areas—for example, high levels of phosphates were uncovered in the water. The phosphates were derived either from fertilizer or from animal and plant matter and waste. Also uncovered were small amounts of undesirable water organisms that are tolerant to pollutants and can survive in harsh environments.

Describes major findings and conclusions.

D-Lynn recommends that (1) the tests done in this study be conducted two more times, through spring 2011; (2) other environmental tests be conducted, as listed in the conclusions and recommendations section; and (3) a voluntary cleanup of the creek be scheduled. With these steps, we can better analyze the environmental integrity of Wildwood Creek.

Includes main recommendation from report text.

3

INTRODUCTION

Gives lead-in to Introduction.

D-Lynn, Inc., has completed a follow-up to a study completed in 2002 by Ware County on the health of Wildwood Creek. This introduction describes the project site, scope of our study, and format for this report.

PROJECT DESCRIPTION

Briefly describes project.

By law, all states must clean up their waterways. The State of Georgia shares this responsibility with its counties. Ware County has certain waterways that are threatened and must be cleaned. Wildwood Creek is one of the more endangered waterways. The portion of the creek that was studied for this report is a one-mile stretch in the Burns Nature Park between Newell College and U.S. Highway 42.

SCOPE OF STUDY

The purpose of this project was to determine whether the health of the creek has changed since the previous study in 2002. Both physical and chemical tests were completed. The nine physical tests were as follows:

Uses bulleted list to emphasize scope of activities.

- Air temperature
- Water temperature
- Water flow
- Water appearance
- Habitat description
- Algae appearance
- Algae location
- Visible litter
- Bug count

The four chemical tests were as follows:

- pH
- Dissolved oxygen (DO)
- Turbidity
- Phosphate

REPORT FORMAT

Provides “map” of main sections in report.

This report includes three main sections:

1. Field Investigation: a complete discussion of all the tests that were performed for the project
2. Test Comparison: charts of the test results and comparisons
3. Conclusions and Recommendations

■ Figure 7-2 ■ continued

FIELD INVESTIGATION

Wildwood Creek has been cited repeatedly for environmental violations in the pollution of its water. Many factors can generate pollution and affect the overall health of the creek. In 2002, the creek was studied in the context of a study of all water systems in Ware County. Wildwood Creek was determined to be one of the more threatened creeks in the county.

The city needed to learn if much has changed in the past nine years, so D-Lynn was hired to perform a variety of tests on the creek. Our effort involved a more in-depth study than that done in 2002. Tests were conducted four times over a seven-month period. The 2002 study lasted only one day.

The field investigation included two categories of tests: physical tests and chemical tests.

PHYSICAL TESTS

The physical tests covered a broad range of environmental features. This section discusses the importance of the tests and some major findings. The Test Comparison section on page 9 includes a table that lists results of the tests and the completion dates. The test types were as follows: air temperature, water temperature, water flow, water appearance, habitat description, algae appearance, algae location, visible litter, and bug count.

Amplifies information
presented later in report.

Air Temperature

The temperature of the air surrounding the creek will affect life in the water. Unusual air temperature for the seasons will determine if life can grow in or out of the water.

Three of the four tests were performed in the warmer months. Only one was completed on a cool day. The difference in temperature from the warmest to coolest day was 10.5°C, an acceptable range.

Water Temperature

The temperature of the water determines which species will be present. Also affected are the feeding, reproduction, and metabolism of these species. If there are one or two weeks of high temperature, the stream is unsuitable for most species. If water temperature changes more than 1° to 2°C in 24 hours, thermal stress and shock can occur, killing much of the life in the creek.

During our study, the temperature of the water averaged 1°C cooler than the temperature of the air. The water temperature did not get above 23°C or below 13°C. These ranges are acceptable by law.

5

Water Flow

The flow of the water influences the type of life in the stream. Periods of high flow can cause erosion to occur on the banks and sediment to cover the streambed. Low water flow can decrease the living space and deplete the oxygen supply.

The flow of water was at the correct level for the times of year the tests were done—except for June, which had a high rainfall. With continual rain and sudden flash floods, the creek was almost too dangerous for the study to be performed that month.

In fact, in June we witnessed the aftermath of one flash flood. Figure 1 shows the creek with an average flow of water, and Figure 2 shows the creek during the flood. The water's average depth is 10 inches. During the flash flood, the water level rose and fell 10 feet in about one hour. Much dirt and debris were washing into the creek, while some small fish were left on dry land as the water receded.

Incorporates graphic into
page of text.

**KEY**

- water
- streambed
- running track

■ **Figure 1** ■ Wildwood Creek—Normal Water Level



- KEY
- water
 - streambed
 - running track

■ **Figure 2** ■ Wildwood Creek—Flash Flood Water Level

Water Appearance

The color of the water gives a quick but fairly accurate view of the health of the creek. If the water is brown or dirty, then silt or human waste may be present. Black areas of water may contain oil or other chemical products.

On each of the four test days, the water was always clear. Thus the appearance of the creek water was considered excellent.

Habitat Description

The habitat description concerns the appearance of the stream and its surroundings. An important criterion is the number of pools and the number of ripples—that is, points where water flows quickly over a rocky area. Both pools and ripples provide good locations for fish and other stream creatures to live and breed.

In describing habitat, D-Lynn also evaluates the amount of sediment at the bottom of the stream. Too much sediment tends to cover up areas where aquatic life lays eggs and hides them from predators. We also evaluate the stability of the stream banks; a stable bank indicates that erosion has not damaged the habitat. Finally, we observe the amount of stream cover. Such vegetation helps keep soil in place on the banks.

Elaborates on importance of information shown in Table 1. Description parallels five items in table.

7

Wildwood Creek tested fairly well for habitat. The number of pools and ripples was about average for such creeks. Stream deposits and stream bank stability were average to good, and stream cover was good to excellent. For more detail about test results, see the chart in the Test Comparison section on page 9.

Algae Appearance and Location

Algae is naturally present in any creek. The amount of algae can be a warning of pollution in the water. If algae is growing out of control, disproportionate amounts of nutrients such as nitrogen or phosphate could be present. These chemicals could come from fertilizer washed into the creek. Excessive amounts of algae cause the oxygen level to drop when they die and decompose.

During the four studies, algae was everywhere, but it was especially heavy on the rocks in the ripples of the creek. The algae was always brown and sometimes hairy.

Visible Litter

Litter can affect the habitat of a creek. Although some litter has chemicals that can pollute the water, other litter can cover nesting areas and suffocate small animals. Whether the litter is harmful or not, it is always an eyesore.

On all four test dates, the litter we saw was heavy and ranged from tires to plastic bags. Some of the same trash that was at the site on the first visit was still there seven months later.

Bug Count

The bug count is a procedure that begins by washing dirt and water onto a screen. As water drains, the dirt with organisms is left on the screen. The bugs are removed and classified. Generally, the lower the bug count, the higher the pollution levels. Bug counts were considered low to average.

Two types of aquatic worms were discovered every time during our count, but in relatively small amounts. In addition, the worms we observed are very tolerant of pollution and can live in most conditions. Finally, we observed only two crayfish, animals that are somewhat sensitive to pollution.

CHEMICAL TESTS

Although physical tests cover areas seen with the naked eye, chemical tests can uncover pollutants that are not so recognizable. Certain chemicals can wipe out all life in a creek. Other chemicals can cause an overabundance of one life-form, which in turn could kill more sensitive animals.

A chart of results of chemical tests is included in the Test Comparison section on page 9. The chemical tests that D-Lynn performed were pH, dissolved oxygen (DO), turbidity, and phosphate.

Gives specific details that support the report's conclusions and recommendations, which come later.

pH

The pH test is a measure of active hydrogen ions in a sample. The range of the pH test is 0–14. If the sample is in the range of 0–7.0, it is acidic; but if the sample is in the range of 7.0–14, it is basic. By law, the pH of a water sample must be within the range of 6.0–8.5.

For the tests we completed, the water sample was always 7.0, which is very good for a creek.

Dissolved Oxygen (DO)

Normally, oxygen dissolves readily into water from surface air. Once dissolved, it diffuses slowly in the water and is distributed throughout the creek. The amount of DO depends on different circumstances. Oxygen is always highest in choppy water, just after noon, and in cooler temperatures.

In many streams, the level of DO can become critically low during the summer months. When the temperature is warm, organisms are highly active and consume the oxygen supply. If the amount of DO drops below 3.0 ppm (parts per million), the area can become stressful for the organisms. An amount of oxygen that is 2.0 ppm or below will not support fish. DO that is 5.0 ppm to 6.0 ppm is usually required for growth and activity of organisms in the water.

According to the Water Quality Criteria for Georgia, average daily amounts of DO should be 5.0 ppm with a minimum of 4.0 ppm. Wildwood Creek scored well on this test. The average amount of DO in the water was 6.9 ppm, with the highest amount being 9.0 ppm on November 11, 2010.

Turbidity

Turbidity is the discoloration of water due to sediment, microscopic organisms, and other matter. One major factor of turbidity is the level of rainfall before a test.

Three of our tests were performed on clear days with little rainfall. On these dates, the turbidity of Wildwood Creek was always 1.0, the best that creek water can score on the test. The fourth test, which scored worse, occurred during a rainy period.

Phosphate

Phosphorus occurs naturally as phosphates—for example, orthophosphates and organically bound phosphates. Orthophosphates are phosphates that are formed in fertilizer, whereas organically bound phosphates can form in plant and animal matter and waste.

Phosphate levels higher than 0.03 ppm contribute to an increase in plant growth. If phosphate levels are above 0.1 ppm, plants may be stimulated to grow out of control. The phosphate level of Wildwood was always 0.5 ppm, considerably higher than is desirable.

TEST COMPARISON

There was little change from each of the four test dates. The only tests that varied greatly from one test to another were air temperature, water temperature, water flow, and DO. On the basis of these results, it would appear that Wildwood Creek is a relatively stable environment.

Brings together test results for easy reference.

Table 1 Physical Tests

TEST DATES	5/26/10	6/25/10	9/24/10	11/19/10
Air Temperature in °C	21.5	23.0	24.0	13.5
Water Temperature in °C	20.0	22.0	23.0	13.0
Water Flow	Normal	High	Normal	Normal
Water Appearance	Clear	Clear	Clear	Clear
Habitat Description				
Number of Pools	2.0	3.0	2.0	5.0
Number of Ripples	1.0	2.0	2.0	2.0
Amount of Sediment Deposit	Average	Average	Good	Average
Stream Bank Stability	Average	Good	Good	Good
Stream Cover	Excellent	Good	Excellent	Good
Algae Appearance	Brown	Brown/hairy	Brown	Brown
Algae Location	Everywhere	Everywhere	Attached	Everywhere
Visible Litter	Heavy	Heavy	Heavy	Heavy
Bug Count	Low	Average	Low	Average

Table 2 Chemical Tests

Test	5/26/10	6/25/10	9/24/10	11/19/10
PH 7.0	7.0	7.0	7.0	
Dissolved Oxygen (DO)	6.8	6.0	5.6	9.0
Turbidity	1.0	3.0	1.0	1.0
Phosphate	0.50	0.50	0.50	0.50

CONCLUSIONS AND RECOMMENDATIONS

This section includes the major conclusions and recommendations from our study of Wildwood Creek.

CONCLUSIONS

Generally, we were pleased with the health of the stream bank and its floodplain. The area studied has large amounts of vegetation along the stream, and the banks seem to be sturdy. The floodplain has been turned into a park, which handles floods in a natural way. Floodwater in this area comes in contact with vegetation and some dirt. Floodwater also drains quickly, which keeps sediment from building up in the creek.

However, we are concerned with the number and types of animals uncovered in our bug counts. Only two bug types were discovered, and these were types quite tolerant to pollutants. The time of year these tests were performed could affect the discovery of some animals. However, the low count still should be considered a possible warning sign about water quality. Phosphate levels were also high and probably are the cause of the large amount of algae.

We believe something in the water is keeping sensitive animals from developing. One factor that affects the number of animals discovered is the pollutant problems in the past (see Appendix A). The creek may still be in a redevelopment stage, thus explaining the small numbers of animals.

RECOMMENDATIONS

On the basis of these conclusions, we recommend the following actions for Wildwood Creek:

1. Conduct the current tests two more times, through Spring 2011. Spring is the time of year that most aquatic insects are hatched. If sensitive organisms are found then, the health of the creek could be considered to have improved.
2. Add testing for nitrogen. With the phosphate level being so high, nitrogen might also be present. If it is, then fertilizer could be in the water.
3. Add testing for human waste. Some contamination may still be occurring.
4. Add testing for metals, such as mercury, that can pollute the water.
5. Add testing for runoff water from drainage pipes that flow into the creek.
6. Schedule a volunteer cleanup of the creek.

With a full year of study and additional tests, the problems of Wildwood Creek can be better understood.

← Draws conclusions that flow from data in body of report.

← Uses paragraph format instead of lists because of lengthy explanations needed.

← Gives numbered list of recommendations for easy reference.

APPENDIX A**Background on Wildwood Creek**

Wildwood Creek begins from tributaries on the northeast side of the city of Winslow. From this point, the creek flows southwest to the Chattahoochee River. Winslow Wastewater Treatment Plant has severely polluted the creek in the past with discharge of wastewater directly into the creek. Wildwood became so contaminated that signs warning of excessive pollution were posted along the creek to alert the public.

Today, all known wastewater discharge has been removed. The stream's condition has dramatically improved, but nonpoint contamination sources continue to lower the creek's water quality. Nonpoint contamination includes sewer breaks, chemical dumping, and storm sewers.

Another problem for Wildwood Creek is siltration. Rainfall combines with bank erosion and habitat destruction to wash excess dirt into the creek. This harsh action destroys most of the macroinvertebrates. At the present time, Wildwood Creek may be one of the more threatened creeks in Ware County.

APPENDIX B

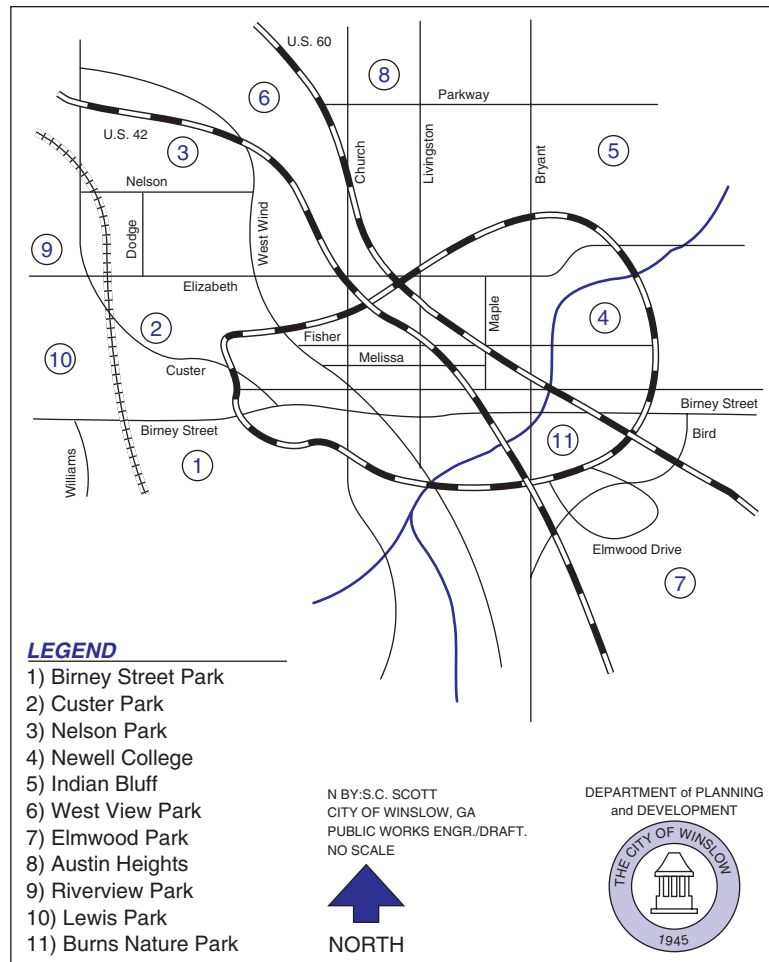
Water Quality Criteria for Georgia

All waterways in Georgia are classified in one of the following categories: fishing, recreation, drinking, and wild and scenic. Different protection levels apply to the different uses. For example, the protection level for dissolved oxygen is stricter in drinking water than fishing water. All water is supposed to be free from all types of waste and sewage that can settle and form sludge deposits.

In Ware County, all waterways are classified as “fishing,” according to Chapter 391-3-6.03 of “Water Use Classifications and Water Quality Standards” in the Georgia Department of Natural Resources *Rules and Regulations for Water Quality Control*. The only exception is the Chattahoochee River, which is classified as “drinking water supply” and “recreational.”

APPENDIX C

Map 6
Location of City of Winslow
Parks and Recreation Facilities



environmental problems. The paid professionals include engineers, environmental specialists, accountants, city planners, managers, lawyers, real estate experts, and public relations specialists. The part-time appointees include citizens who work in a variety of blue-collar and white-collar professions or who are homemakers.

>>> Four Common Reports

Report types vary from company to company and can be produced in either informal or formal report format. The four types described here are only a sampling of what you will be asked to write on the job. If you master these, you can probably handle other types that come your way.

Equipment Evaluations

All organizations use some kind of equipment that someone has to buy, maintain, or replace. Because companies put so much money into this part of their business, evaluating equipment is an important activity. Equipment evaluations provide objective data about how machinery, tools, computer software, or other equipment has functioned. An equipment evaluation may focus only on problems; or it may go on to suggest a change in equipment. Whatever its focus, an equipment evaluation must provide a well-documented review of the exact manner in which equipment performed.



Report 14: Research Report is an example of a formal equipment evaluation report.

Progress/Periodic Reports

Some reports are intended to cover activities that occurred during a specific period of time. They provide managers or clients with details about work on a specific project. A special type of progress report is the *project completion report*, which may be presented as a formal report when the project is completed. Progress and periodic reports contain mostly objective data. Yet both of them, especially progress reports, sometimes may be written in a persuasive manner. After all, you are trying to put forth the best case for the work you have completed. Figure 7–1 is an example of a progress report.



Report 12: Final Report is an example of a formal project completion report.

Problem Analyses

Every organization faces both routine and complex problems. Routine problems are often handled without much paperwork; they are discussed and then solved. However, other problems must often be described in reports, particularly if they involve many people, are difficult to solve, or have been brewing for a long time. Problem analysis reports present readers with a detailed description of problems in areas, such as personnel, equipment, products, and services.



Report 5: Research Report is an example of a problem analysis.



Most problem analyses contain both facts and opinions. As the writer, you must make special efforts to separate the two because most readers want the opportunity to draw their own conclusions about the problem.

Recommendation Reports

Recommendation reports use objective information to support suggestions that affect personnel, equipment, procedures, products, and services.

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Report 6: Report of Findings is an example of a recommendation report.

>>> Chapter Summary

This chapter deals with the common types of reports you will write in your career. When writing reports, follow these guidelines:

1. Plan well before you write.
2. Separate fact from opinion.
3. Make text visually appealing.
4. Use illustrations for clarification and persuasion.
5. Edit carefully.

On the job, you write informal reports for readers inside your organization (as *memo reports*) and outside your organization (as *letter reports*). In both cases, follow these basic guidelines:

1. Use letter or memo format.
2. Use the ABC format for organizing information.
3. Start with an introductory summary.
4. Put detailed support in the body.
5. Focus attention in your conclusion.
6. Use attachments for details.

You also will write formal reports for large and complex projects, either inside or outside your organization, for readers with mixed technical backgrounds. Although long report formats vary according to company and profession, most have these nine basic parts: (1) cover/title page, (2) letter/memo of transmittal, (3) table of contents, (4) list of illustrations, (5) executive summary, (6) introduction, (7) discussion sections, (8) conclusions and recommendations, and (9) end material. Follow the specific guidelines in this chapter for these sections. The annotated example can serve as your reference.

Although formal and informal reports come in many varieties, this chapter describes only four common types: equipment evaluations, problem analyses, progress/periodic reports, and recommendations.

Chapter | 8 | Proposals



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Everyone of us relies on persuasion. Usually we apply our persuasive strategies in less formal situations—for example, in asking for an extension on a paper deadline, gaining approval to use a sales coupon after its void date, convincing a friend to choose one vacation destination over another, or selling retail products on commission. When persuasion is applied to business situations, it often results in a written document called a *proposal*, and you may write many of them in your career. Proposals are defined as follows:

Proposal: A document written to convince your readers to adopt an idea, a product, or a service. They can be directed to colleagues inside your own organization (*in-house proposals*), to clients outside your organization (*sales proposals*), or to organizations that fund research and other activities (*grant proposals*).

In all three cases, proposals can be presented in either a short, simple format (*informal proposal*) or a longer, more complicated format (*formal proposal*). Also, proposals can be either requested by the reader (*solicited*) or submitted without a request (*unsolicited*).

Like informal reports, informal proposals are short documents that cover projects with a limited scope. Following are some guidelines to help you decide when to use informal and formal proposals:

Use Informal Proposals When

- The text of the proposal (excluding attachments) is no more than five pages
- The size of the proposed project is such that a long formal proposal appears to be inappropriate
- The client has expressed a preference for a leaner and less-formal document

Use Formal Proposals When

- The text of the proposal (excluding attachments) is more than five pages

- The size and importance of the project is such that a formal proposal is appropriate
- The client has expressed a preference for a more-formal document

These two formats can be used for proposals that are either *in-house* (to readers within your own organization) or *external* (to readers outside your organization).

The flowchart in Figure 8–1 shows one possible communication cycle that would involve both a proposal and an RFP, which stands for *request for proposal*.

Request for proposal (RFP): A document sometimes sent out by organizations that want to receive proposals for a product or service. The RFP gives guidelines on (1) what the proposal should cover, (2) when it should be submitted, and (3) to whom it should be sent. As writer, you should follow the RFP religiously in planning and drafting your proposal.

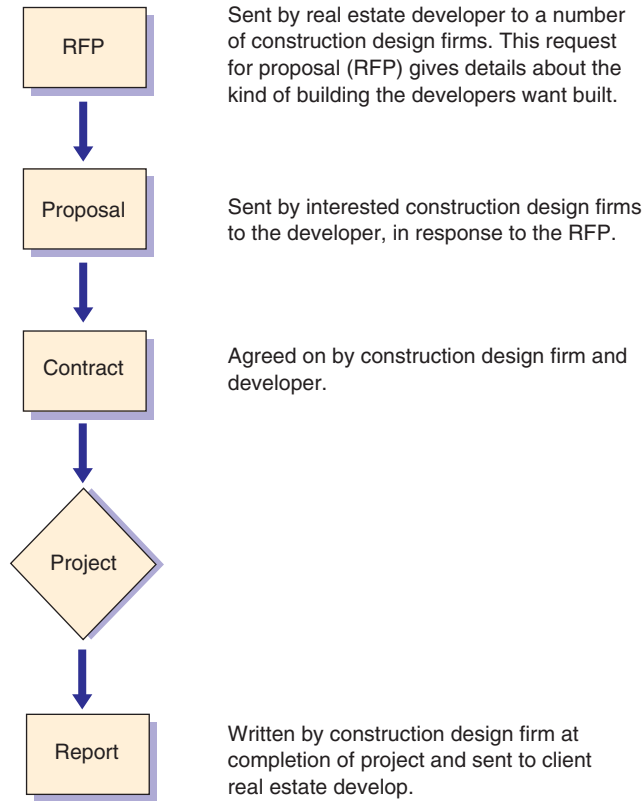
RFPs generally are used when an organization:

- Wants to receive multiple approaches to address a problem.
- Wants the best price for a well-defined scope of work.
- Is required to solicit bids for projects.
- Provides grants or fellowships for research or community projects.

If you are responding to an RFP, it is key to your understanding the needs of the readers to whom you will send your proposal. You must read RFPs very carefully to understand the main concerns and interests of your readers, as well as the requirements for proposals. You should precisely follow the content and formatting guidelines outlined in the RFP, as proposals that do not follow the guidelines are usually rejected immediately.

■ **Figure 8-1** ■

Flowchart showing the main documents involved in an external, solicited proposal process.



>>> Guidelines for Informal Proposals

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The Model Sales Proposal for Home Repair is an example of a short, informal proposal.

This section provides writing guidelines and an annotated model for informal proposals, which have two formats: (1) memos (for in-house proposals) and (2) letters (for external proposals). With some variations, these guidelines are similar to those suggested in Chapter 7 for informal reports. The formats are much the same, although the content and tone are different. Reports *explain*, whereas proposals *persuade*.

>> Informal Proposal Guideline 1: Plan Well Before You Write

Complete the Planning Form at the end of the book for all proposal assignments. Carefully consider your purpose, audience, and organization. Two factors make this task especially difficult in sales-proposal writing:

1. You may know nothing more about the client than what is written on the RFP.
2. Proposals often are on a tight schedule, which limits your planning time.

Nevertheless, try to find out exactly who will be making the decision about your proposal. Many clients will tell you if you give them a call. In fact, they may be pleased that you care enough about the project to target the audience. Once you identify the decision makers, spend time brainstorming about their needs before you begin writing.

>> **Informal Proposal Guideline 2: Use Letter or Memo Format**

Letter proposals usually follow the format of routine business letters (see Chapter 4). This casual style gives readers the immediate impression that your document will be *approachable*—that is, easy to get through and limited in scope. Memo proposals, such as the example shown in Figure 8–2, follow the format of an internal memorandum (see Chapter 4). Following are a few highlights:

- Line spacing is usually single, but it may be one-and-a-half or double, depending on the reader's or company's preference.
- The recipient's name, date, and page number appear on sheets after the first.
- Most readers prefer an uneven or ragged-edge right margin, as opposed to an even or full-justified margin.

Your subject line in a memo proposal gives readers the first impression of the proposal's purpose. Choose concise yet accurate wording. See Figure 8–2 for wording that gives the appropriate information and tries to engage the reader's interest.

>> **Informal Proposal Guideline 3: Make Text Visually Appealing**

The page design of informal proposals must draw readers into the document. Remember—you are trying to sell a product, a service, or an idea. Also, remember that your proposal may be competing with others. Put yourself in the place of the reader who is wondering which one to pick up first. How the text looks on the page can make a big difference. Following are a few techniques to follow to help make your proposal visually appealing:

- Use lists (with bullets or numbered points) to highlight main ideas.
- Follow your readers' preferences as to font size, type, line spacing, and so forth. Proposals written in the preferred format of the reader gain an edge.
- Use headings and subheadings to break up blocks of text.

These and other techniques help guide readers through the informal proposal. Given that there is no table of contents, you must take advantage of such strategies.



DATE: October 3, 2010
TO: Gary Lane
FROM: Jeff Bilstrom **JB**
SUBJECT: Creation of Logo for Montrose Service Center

Gives concise view of problem—and his proposed solution.

Part of my job as director of public relations is to get the Montrose name firmly entrenched in the minds of metro Atlanta residents. Having recently reviewed the contacts we have with the public, I believe we are sending a confusing message about the many services we offer retired citizens in this area.

To remedy the problem, I propose we adopt a logo to serve as an umbrella for all services and agencies supported by the Montrose Service Center. This proposal gives details about the problem and the proposed solution, including costs.

Includes effective lead-in.

The Problem

The lack of a logo presents a number of problems related to marketing the center's services and informing the public. Here are a few:

Uses bulleted list to highlight main difficulties posed by current situation.

- The letterhead mentions the organization's name in small type, with none of the impact that an accompanying logo would have.
- The current brochure needs the flair that could be provided by a logo on the cover page, rather than just the page of text and headings that we now have.
- Our 14 vehicles are difficult to identify because there is only the lettered organization name on the sides without any readily identifiable graphic.
- The sign in front of our campus, a main piece of free advertising, could better spread the word about Montrose if it contained a catchy logo.
- Other signs around campus could display the logo, as a way of reinforcing our identity and labeling buildings.

Ends section with good transition to next section.

It is clear that without a logo, the Montrose Service Center misses an excellent opportunity to educate the public about its services.

Starts with main point—need for logo.

The Solution

I believe a professionally designed logo could give the Montrose Service Center a more distinct identity. Helping to tie together all branches of our operation, it would give the public an easy-to-recognize symbol. As a result, there would be a stronger awareness of the center on the part of potential users and financial contributors.

■ Figure 8-2 ■ Memo proposal

Gary Lane
October 3, 2010
Page 2

The new logo could be used immediately to do the following:

- Design and print letterhead, envelopes, business cards, and a new brochure.
- Develop a decal for all company vehicles that would identify them as belonging to Montrose.
- Develop new signs for the entire campus, to include a new sign for the entrance to the campus, one sign at the entrance to the Blane Workshop, and one sign at the entrance to the Administration Building.

Focuses on *benefits* of proposed change.

Cost

Developing a new logo can be quite expensive. However, I have been able to get the name of a well-respected graphic artist in Atlanta who is willing to donate his services in the creation of a new logo. All that we must do is give him some general guidelines to follow and then choose from among eight to ten rough sketches. Once a decision is made, the artist will provide a camera-ready copy of the new logo.

Emphasizes *benefit* of possible price break.

• Design charge	\$0.00
• Charge for new letterhead, envelopes, business cards, and brochures (min. order)	545.65
• Decal for vehicles 14 @ \$50.00 + 4%	728.00
• Signs for campus	415.28
Total Cost	<u>\$1,688.93</u>

Uses listing to clarify costs.

Conclusion

As the retirement population of Atlanta increases in the next few years, there will be a much greater need for the services of the Montrose Service Center. Because of that need, it is in our best interests to keep this growing market informed about the organization.

Closes with major benefit to reader and urge to action.

I'll stop by later this week to discuss any questions you might have about this proposal.

Keeps control of next step.

ABC Format:

Informal Proposal

- **ABSTRACT:** Gives “big picture” for those who make decisions. Usually includes a statement of the problem or other information that will entice the audience to read further.
- **BODY:** Gives details about exactly what you are proposing to do.
- **CONCLUSION:** Drives home the main benefit and makes clear the next step.

>> Informal Proposal Guideline 4: Use the ABC Format for Organization

The ABC Format used throughout this book also applies to informal proposals.

Note: Beginning and end sections should be easy to read and stress just a few points. They provide a short buffer on both ends of the longer and more technical body section in the middle. The next four guidelines give more specific advice for writing the main parts of an informal proposal.

>> Informal Proposal Guideline 5: Create the Abstract as an Introductory Summary

In an abstract, you capture the client’s attention with a capsule summary of the entire proposal. This is especially important in unsolicited proposals. This one- or two-paragraph starting section permits space only for what the reader really must know at the outset, such as the following:

- Purpose of proposal
- Reader’s main need
- Main features you offer, as well as related benefits
- Overview of proposal sections to follow

Keep this overview very brief. Answer the one question readers are thinking: “Why should I accept this proposal?”

>> Informal Proposal Guideline 6: Put Important Details in the Body

The discussion of your proposal should address these basic questions:

1. What problems are you trying to solve, and why?
2. What are the technical details of your approach?
3. Who will do the work, and with what?
4. When will it be done?
5. How much will it cost?

Discussion formats vary from proposal to proposal, but here are some sections commonly used to respond to these questions:

1. **Description of problem or project and its significance.** Give a precise technical description, along with any assumptions you have made on the basis of previous

contact with the reader. Explain the importance or significance of the problem to the reader of the proposal.

2. **Proposed solution or approach.** Describe specific tasks you propose in a manner that is clear and well organized. If you are presenting several options, discuss each one separately—making it easy for the reader to compare and contrast information.

3. **Personnel.** If the proposal involves people performing tasks, it may be appropriate to explain qualifications of participants.

4. **Schedule.** Even the simplest proposals require some information about the schedule for delivering goods, performing tasks, and so forth. Be both clear and realistic in this portion of the proposal. Use graphics when appropriate.

5. **Costs.** Place complete cost information in the body of the proposal unless you have a table that would be more appropriately placed in an attachment. Above all, do not bury dollar figures in paragraph format. Instead, highlight figures with indented or bulleted lists. Because your reader will be looking for cost data make that information easy to find. Finally, be certain to include all costs—materials, equipment, personnel, salaries, and so forth.

>> Informal Proposal Guideline 7: Use a Problem/Solution Organization

As any good salesperson knows, customers must feel that they need your product, service, or idea before they can be convinced to purchase or support it. Lay the groundwork for acceptance by first showing the readers that a strong need exists.

Establishing need is most crucial in unsolicited proposals, of course, when readers may not be psychologically prepared to accept a change that costs money. Even in proposals that have been solicited, however, you should give some attention to restating the basic needs of the readers. You should show your readers your understanding of the problem, and you should emphasize the connections between the problem and your proposed solution.



>> Informal Proposal Guideline 8: Focus Attention in Your Conclusion

Called *conclusion* or *closing*, this section gives you the opportunity to control the readers' last impression. It also helps avoid the awkwardness of ending proposals with the costs. In this closing section, you can

- Emphasize a main benefit or feature of your proposal
- Restate your interest in doing the work
- Indicate what should happen next

Regarding the last point, sometimes you may ask readers to call if they have questions. In other situations, however, it is appropriate to say that you will follow up the proposal with a phone call. This approach leaves you in control of the next step.

>> Informal Proposal Guideline 9: Use Attachments for Less Important Details

The text of informal proposals is usually less than five pages, so you may have to place supporting data or illustrations in attachments that follow the conclusion. Cost and schedule information, in particular, is best placed at the end in well-labeled sections.

Make sure the proposal text includes clear references to all visuals. If you have more than one attachment, give each one a letter and a title (for example, “Attachment A: Project Costs”). If you have only one attachment, include the title but no letter (for example, “Attachment: Resumes”).

>> Informal Proposal Guideline 10: Edit Carefully

Build in enough time for a series of editing passes, preferably by different readers. There are two reasons why proposals of all kinds deserve this special attention.

1. They can be considered contracts in a court of law. If you make editing mistakes that alter meaning (such as an incorrect price figure), you could be bound to the error.
2. Proposals often present readers with their first impression of you. If the document is sloppy, they can make assumptions about your professional abilities as well.

>>> Guidelines for Formal Proposals

Sometimes the complexity of the proposal may be such that a formal response is best. Ask yourself questions such as the following in deciding whether to write an informal or a formal proposal:

- Is there too much detail for a letter or memo?
- Is a table of contents needed so that sections can be found quickly?
- Will the professional look of a formal document lend support to the cause?
- Are there so many attachments that a series of lengthy appendices would be useful?
- Are there many different readers with varying needs, such that there should be different sections for different people?

If you answer “yes” to one or more of these questions, you probably should write a formal proposal. Although this long format is most common in external sales proposals, some in-house proposals may require the same approach—especially in large organizations in which you may be writing to unknown persons in distant departments.

Formal proposals can be long and complex, so this part of the chapter treats each proposal section separately—from title page through conclusion. Two points will become evident as you use these guidelines. First, formal reports and formal proposals are very much alike. A quick look at Chapter 7 shows you the similarities in format. Second, a formal proposal follows the basic ABC format described in Chapter 1.

Specifically, the parts of the formal proposal fit the pattern as shown in the following box. Notice that formal proposals, like informal proposals, are organized with a problem/solution pattern.

As you read through and apply these guidelines, refer to Figure 8–3 on pp. 174–182 for an annotated example of the formal proposal.

Cover/Title Page

Like formal reports, formal proposals are usually bound documents with a cover, which includes one or more of the items listed on the right for inclusion on the title page. The cover should be designed to attract the reader's interest—with good page layout and perhaps even a graphic.

Inside the cover is the title page, which contains these four pieces of information:

- **Project title**, sometimes preceded by *Proposal for* or similar wording
- **Your reader's name** (sometimes preceded by *Prepared for...*)
- **Your name or the name of your organization** spelled out in full (sometimes preceded by *Prepared by...*)
- **Date of submission**

The title page gives clients their first impression of you. For that reason, consider using some tasteful graphics to make the proposal stand out from those of your competitors.

Letter/Memo of Transmittal

Internal proposals have *memos of transmittal*; external sales proposals have *letters of transmittal*. The guidelines for format and organization presented here help you write attention-getting prose. In particular, note that the letter or memo should be in single-spaced, ragged-right-edge format, even if the rest of the proposal is double-spaced copy with right-justified margins.

For details of letter and memo format, see Chapter 4. The guidelines for the letter/memo of transmittal for formal reports also apply to formal proposals (see “Transmittal Guideline 4” in Chapter 7). For now, here are some highlights of format and content that apply especially to letters and memos of transmittal:

1. Use short beginning and end paragraphs.
2. Use a conversational style, with little or no technical jargon.

ABC Format: Formal Proposal

■ **ABSTRACT:**

- Cover/Title Page
- Letter of Transmittal
- Table of Contents
- List of Illustrations
- Executive Summary
- Introduction

■ **BODY:**

- Technical information
- Management information
- Cost information
- (Appendices—appear after text, but support Body section)

■ **CONCLUSION:**

- Conclusion
-

3. Use the first paragraph for introductory information, mentioning what your proposal responds to (e.g., a formal RFP, a conversation with the client, your perception of a need).
4. Use the middle of the letter to emphasize one main benefit of your proposal. Stress what you can do to solve a problem, using the words *you* and *your* as much as possible (rather than *I* and *we*).
5. Use the last paragraph to retain control by orchestrating the next step in the proposal process. When appropriate, indicate that you will call the client soon to follow up on the proposal.
6. Follow one of the letter formats described in Chapter 4. Following are some exceptions, additions, or restrictions:
 - Use single-spaced, ragged-right-edged copy, which makes your letter stand out from the proposal proper.
 - Keep the letter on one page—a two-page letter loses that crisp and concise impact you want to make.
 - Place the company proposal number (if there is one) at the top, above the date. Exact placement of both number and date depends on your organization's letter style.
 - Include the client's company name or personal name on the first line of the inside address, followed by the mailing address used on the envelope. Include the full name (and title, if appropriate) of your contact person at the client firm. If you use a personal name, follow the last line of the inside address with a conventional greeting ("Dear Mr. Adams:").
 - (Optional) Include the project title beneath the attention line, using the exact wording that appears on the title page.
 - Close with "Sincerely" and your name at the bottom of the page. Also include your company affiliation.

Table of Contents

Create a very readable table of contents by spacing items well on the page. List all proposal sections, subsections, and their page references. At the end, list any appendices that may accompany the proposal.

Given the tight schedule on which most proposals are produced, errors can be introduced at the last minute because of additions or revisions. Therefore, take time to proofread the table of contents carefully. In particular, make sure to follow these guidelines:

- Wording of headings should match within the proposal text.
- Page references should be correct.
- All headings of the same order should be parallel in grammatical form.

List of Illustrations

When there are many illustrations, the list of illustrations appears on a separate page after the table of contents. When there are few entries, however, the illustrations may be listed at the end of the table of contents page. In either case, the list should include the number, title, and page number of every illustration appearing in the body of the text. (If there is only one illustration, a number need not be included.) You may divide the list into tables and figures.

Executive Summary

Executive summaries are the most frequently read parts of proposals. Often read by decision makers in an organization, the summary should present a concise overview (usually one page) of the proposal's most important points. It should also accomplish the following objectives:

- Avoid technical language.
- Be as self-contained as possible.
- Make brief mention of the problem, proposed solution, and cost.
- Emphasize the main benefits of your proposal.

Start the summary with one or two sentences that command readers' attention and engage their interest, and then focus on just a few main selling points (three to five is best). You might even want to highlight these benefits with indented lead-ins, such as *Benefit 1* and *Benefit 2*. When possible, use the statement of benefits to emphasize what is unique about your company or your approach so that your proposal attracts special attention. Finally, remember to write the summary after you have completed the rest of the proposal. Only at this point do you have the perspective to sit back and develop a reader-oriented overview.

Introduction

The introduction provides background information for both nontechnical and technical readers. Although the content varies from proposal to proposal, some general guidelines apply. You should include information on the (1) purpose, (2) description of the problem to which you are responding, (3) scope of the proposed study, and (4) format of the proposal.

- Use subheadings if the introduction goes over a page. In this case, begin the section with a lead-in sentence or two that mention the sections to follow.
- Start with a purpose statement that concisely states the reason you are writing the proposal.

- Include a description of the problem or need to which your proposal is responding. Use language directly from the request for proposal or other document the reader may have given you so that there is no misunderstanding. For longer problem or need descriptions, adopt the alternative approach of including a separate needs section or problem description after the introduction.
- Include a scope section in which you briefly describe the range of proposed activities covered in the proposal, along with any research or preproposal tasks that have already been completed.
- Include a proposal format section if you believe the reader would benefit from a listing of the major proposal sections that follow.

Some formal sales proposals include information about the history, background, and expertise of an organization. This is especially appropriate in proposals that are responding to an RFP, where a number of organizations are competing to win the contract. This material is often *boilerplate*, or text that can be reused in all similar documents.



The Model Sales Proposal opens with boilerplate text.

Discussion Sections

Aim the discussion or body toward readers who need supporting information. Traditionally, the discussion of a formal sales proposal contains three basic types of information: (1) technical, (2) management, and (3) cost. Following are some general guidelines for presenting each type. Remember that the exact wording of headings and subheadings varies depending on proposal content.

1. Technical Sections

- Respond thoroughly to the client's concerns, as expressed in writing or meetings.
- Follow whatever organization plan that can be inferred from the request for proposal.
- Use frequent subheadings with specific wording.
- Back up all claims with facts.

2. Management Sections

- Describe who will do the work.
- Explain when the work will be done.
- Display schedule information graphically.
- Highlight personnel qualifications (but put resumes in appendices).

3. Cost Section

- Make costs extremely easy to find.
- Use formal or informal tables when possible.
- Emphasize value received for costs.

- Be clear about add-on costs or options.
- Always total your costs.

Conclusion

Formal proposals should always end with a section labeled “Conclusion” or “Closing.” This final section of the text gives you the chance to restate a main benefit, summarize the work to be done, and assure clients that you plan to work with them closely to satisfy their needs. Just as important, this brief section helps you end on a positive note. You come back full circle to what you stressed at the beginning of the document—benefits to the client and the importance of a strong personal relationship. (Without the conclusion, the client’s last impression would be made by the cost section in the discussion.)

Appendices

Because formal proposals are so long, readers sometimes have trouble locating information they need. Headings help, but they are not the whole answer. Another way you can help readers is by transferring technical details from the proposal text into appendices. The proposal still contains detail—for technical readers who want it—but details do not intrude into the text.

Proposals often include boilerplate, such as the resumes of all major personnel who will be working on a project or project sheets that show expertise and experience with projects like those on which the company is bidding. Creating boilerplate for this information can save you or your employer’s considerable time by eliminating the need to recreate this material for each new proposal.

Any supporting information can be placed in appendices, but following are some common items included there as well:

- Resumes
- Organization charts
- Company histories
- Detailed schedule charts
- Contracts
- Cost tables
- Detailed options for technical work
- Summaries of related projects already completed
- Questionnaire samples

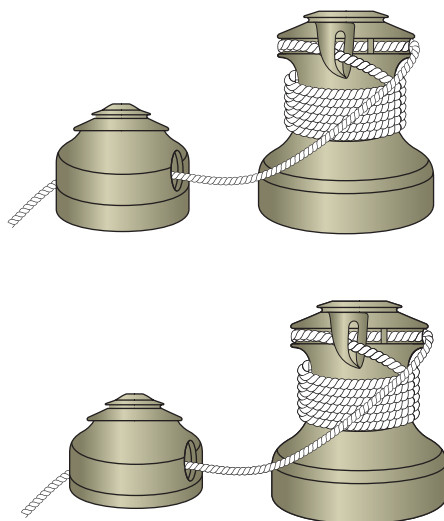
This boilerplate is often printed from separate files, and thus is not paged in sequence with your text. Instead, it is best to use individual paging within each appendix. For example, pages in an Appendix B are numbered as B-1, B-2, B-3, and so forth.

**PROPOSAL FOR SUPPLYING
TEAK CAM CLEAT SPACERS**

Prepared by
Totally Teak, Inc.

Prepared for
John L. Riggini
Bosun's Locker Marine Supply

August 22, 2011



■ Figure 8-3 ■ Formal proposal



August 22, 2011

John L Riggini President
Bosun's Locker Marine Supply
38 Oakdale Parkway
Norcross OH 43293

Dear Mr. Riggini:

I enjoyed talking with you last week about inventory needs at the 10 Bosun's stores. In response to your interest in our products, I'm submitting this proposal to supply your store with our Teak Cam Cleat Spacers.

This proposal outlines the benefits of adding Teak Cam Cleat Spacers to your line of sailing accessories. The potential for high sales volume stems from the fact that the product satisfies two main criteria for any boat owner:

1. It enhances the appearance of the boat.
2. It makes the boat easier to handle.

Your store managers will share my enthusiasm for this product when they see the response of their customers.

I'll give you a call next week to answer any questions you have about this proposal.

Sincerely,

William G. Rugg
President
Totally Teak, Inc.

WR/rr

In this example, the letter of transmittal appears immediately following the title page. It can also appear before the title page (see Model 8-4).

Establishes link with previous client contact.

Stresses two main benefits.

Says he will call (rather than asking client to call).

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Organizes entire proposal around *benefits*.

EXECUTIVE SUMMARY

This proposal outlines features of a custom-made accessory designed for today's sailors—whether they be racers, cruisers, or single-handed skippers. The product, Teak Cam Cleat Spacers, has been developed for use primarily on the Catalina 22, a boat owned by many customers of the 10 Bosun's stores. However, it can also be used on other sailboats in the same class.

The predictable success of Teak Cam Cleat Spacers is based on two important questions asked by today's sailboat owners:

- Will the accessory enhance the boat's appearance?
- Will it make the boat easier to handle and, therefore, more enjoyable to sail?

This proposal answers both questions with a resounding affirmative by describing the benefits of teak spacers to thousands of people in your territory who own boats for which the product is designed. This potential market, along with the product's high profit margin, will make Teak Cam Cleat Spacers a good addition to your line of sailing accessories.

Briefly mentions main need to which proposal responds.

Reinforces main points mentioned in letter (selective repetition of crucial information is acceptable).

INTRODUCTION

Makes clear the proposal's purpose.



The purpose of this proposal is to show that Teak Cam Cleat Spacers will be a practical addition to the product line at the Bosun's Locker Marine Supply stores. This introduction highlights the need for the product, as well as the scope and format of the proposal.

Gives lead-in about section to follow.



Background

Sailing has gained much popularity in recent years. The high number of inland impoundment lakes, as well as the vitality of boating on the Great Lakes, has spread the popularity of the sport. With this increased interest, more and more sailors have become customers for a variety of boating accessories.

Establishes need for product.



What kinds of accessories will these sailors be looking for? Accessories that (1) enhance the appearance of their sailboats and (2) make their sailboats easier to handle and, consequently, more enjoyable to sail. With these customer criteria in mind, it is easy to understand the running joke among boat owners (and a profitable joke among marine supply dealers): "A boat is just a hole in the water that you pour your money into."

Shows his understanding of need (personal experience of designing owner survey).



The development of this particular product originated from our designers' first-hand sailing experiences on the Catalina 22 and knowledge obtained during manufacture (and testing) of the first prototype. In addition, we conducted a survey of owners of boats in this general class. The results showed that winch and cam cleat designs are major concerns.

Proposal Scope and Format

The proposal focuses on the main advantages that Teak Cam Cleat Spacers will provide your customers. These six sections follow:

Gives list of sections to follow, to reinforce organization of proposal.



1. Practicality
2. Suitability for a Variety of Sailors
3. High-Quality Construction and Appearance
4. Dealer Benefits
5. Sizable Potential Market
6. Affordable Price

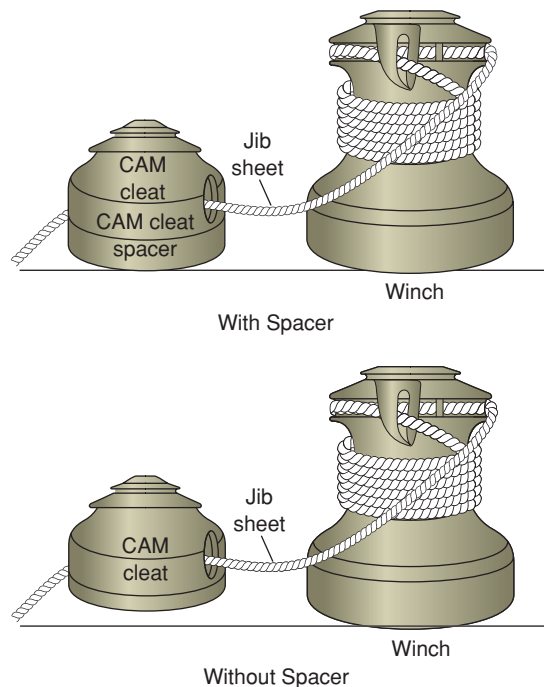
FEATURES AND BENEFITS

Teak Cam Cleat Spacers offer Bosun's Locker Marine Supply the best of both worlds. On the one hand, the product solves a nagging problem for sailors. On the other hand, it offers your store managers a good opportunity for profitability. Described here are six main benefits for you to consider.

Practicality

This product is both functional and practical. When installed in the typical arrangement shown in the figure below, the Teak Cam Cleat Spacer raises the height of the cam cleat, thereby reducing the angle between the deck and the sheet as it feeds downward from the winch. As a result of this increased height, a crewmember is able to cleat a sheet with one hand instead of two.

SIDE VIEWS OF CLEATING ARRANGEMENT



Uses main heading that engages reader's interest.

Phrases each side heading in "benefit-centered" language.

Explains exactly how product will work.

Moves logically from racing to cruising to single-handed sailing—all to show usefulness of product.

Stresses quality.

Appeals to self-interest of individual Bosun's dealers.

Such an arrangement allows a skipper to maintain steerage of the boat, keeping one hand on the helm while cleating the sheet with the other. Securing a sheet in this manner can be done more quickly and securely. Also, this installation reduces the likelihood of a sheet "popping out" of the cam cleat during a sudden gust of wind.

Suitability for a Variety of Sailors

For the racer, cruiser, and single-handed sailor, sailing enjoyment is increased as sheets and lines become easier to handle and more secure. In a tight racing situation, these benefits can be a deciding factor. The sudden loss of sail tension at the wrong moment as a result of a sheet popping out of the cam cleat could make the difference in a close race.

A cruising sailor is primarily concerned with relaxation and pleasure. A skipper in this situation wants to reduce his or her workload as much as possible. In the instance of a sheet popping loose, the sudden chaos of a sail flapping wildly interrupts an otherwise tranquil atmosphere. Teak Cam Cleat Spacers reduce the chance of this happening.

A cruising sailor often has guests aboard. In this situation, as well as in a race, the skipper wants to maintain a high level of seamanship, especially where the control of the boat and the trim of its sails are concerned.

The single-handed sailor derives the greatest benefit from installing Teak Cam Cleat Spacers. Without crew nearby to assist with handling lines or sheets, anything that makes work easier for the skipper is welcome.

High-Quality Construction and Appearance

The teakwood frame from which this product is manufactured is well suited for use around water, since teak will not rot. It also looks nice when oiled or varnished.

The deck of most sailboats is made primarily of fiberglass. The appearance of such a boat can be significantly enhanced by the addition of some teak brightwork.

Each spacer is individually handcrafted by Totally Teak, Inc., to guarantee a consistent level of high quality.

Dealer Benefits

Teak Cam Cleat Spacers make a valuable addition to the dealer's product line. They complement existing sailing accessories as well as provide the customer with the convenience of a readily available prefabricated product.

A customer who comes in to buy a cam cleat is a ready prospect for the companion spacer. Such a customer will likely want to buy mounting hardware as well.

With this unique teak product readily available, a dealer can save the customer the time and trouble of fabricating makeshift spacers.

Sizable Potential Market

These Teak Cam Cleat Spacers are designed with a large and growing potential market in mind. They are custom-made for the Catalina 22, one of the most popular sailboats in use today. More than 13,000 of these sailboats have been manufactured to date. These spacers are also well suited for other similar-class sailboats.

Includes number of owners to emphasize potential sales.

Affordable Price

The Teak Cam Cleat Spacers made by Totally Teak, Inc., wholesale for \$3.95/pair. Suggested retail is \$6.95/pair. This low price is easy on the skipper's wallet and should help this product move well. And, of course, the obviously high profit margin should provide an incentive to your store managers.

Keeps price information short and clear.

Links list of *benefits* with their order as presented in the body of the proposal—drives home advantages of product to user and dealer.

CONCLUSION

Why should a marine supply dealer consider carrying Teak Cam Cleat Spacers? This product satisfies two common criteria of sailboat owners today: it enhances the appearance of any sailboat, and it makes the boat easier to handle. The potential success of this product is based on its ability to meet these criteria and the following features and benefits:

1. It is practical, allowing quick, one-handed cleating.
2. It is ideally suited for a variety of sailors, whether they are racing, cruising, or sailing single-handedly.
3. It is a high-quality, handcrafted product that enhances the appearance of any sailboat.
4. It is a product that benefits the dealer by making a valuable addition to her or his product. It complements existing sail accessories and satisfies a customer need.
5. It is geared toward a sizable potential market. Today there are thousands of sailboats in the class for which this accessory is designed.
6. It is affordably priced and provides a good profit margin.

>>> Chapter Summary

Proposals stand out as documents that aim to convince readers. You are writing to convince someone inside or outside your organization to adopt an idea, a product, or a service.

This chapter includes lists of writing guidelines for informal proposals and formal proposals. For informal proposals, follow these basic guidelines:

1. Plan well before you write.
2. Use letter or memo format.
3. Make text visually appealing.
4. Use the ABC format for organization.
5. Use the heading “Introductory Summary” for the generic abstract section.
6. Put important details in the body.
7. Use a problem/solution pattern of organization.
8. Focus attention in your conclusion.
9. Use attachments for less important details.
10. Edit carefully.

In formal proposals, abide by the same general format presented in Chapter 7 for formal reports. To be sure, formal proposals have a different tone and substance because of their more persuasive purpose. Yet they do have the same basic parts, with minor variations: cover/title page, letter/memo of transmittal, table of contents, list of illustrations, executive summary, introduction, discussion sections, conclusion, and appendices.

Chapter | 9 | Presentations



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Your career will present you with many opportunities for oral presentations, both formal and informal. At the time they arise, however, you may not consider them to be “opportunities.” They may seem to loom on the horizon as stressful obstacles. That response is normal. The purpose of this chapter is to provide the tools that help oral presentations contribute to your career success. The entire chapter is based on one simple principle: *Almost anyone can become an excellent speaker.* Certainly some people have more natural talent at thinking on their feet or have a more

resonant voice, but success at speaking can come to all speakers, whatever their talent, if they follow the 3 Ps:

Step 1: Prepare carefully

Step 2: Practice often

Step 3: Perform with enthusiasm

These steps form the foundation for all specific guidelines that follow. Before presenting these guidelines, this chapter examines specific ways that formal and informal presentations become part of your professional life.

>>> Presentations and Your Career

The following examples present some realistic situations in which the ability to speak well can lead to success for you and your organization:

- **Getting hired:** You may be asked to present information about your accomplishments to a hiring committee or small group of managers.
- **Getting customers:** If you are on a proposal team, presentations to potential clients may be a part of your regular responsibilities.
- **Keeping customers:** You may be asked to present findings or even periodic reports to clients, so that they can ask questions.
- **Contributing to your profession:** You may present new procedures that your organization has developed at professional conferences, such as the Society for Technical Communication’s annual Summit.
- **Contributing to your community:** You may be asked to speak to a community group or local governing body, as a representative of your company or organization.
- **Getting promoted:** As an employee about to be considered for promotion, you may be evaluated on your ability to present information orally.

As you can see from this list, oral presentations are defined quite broadly.

Throughout your career, you will speak to different-size groups, on diverse topics, and in varied formats. The next two sections provide some common guidelines on preparation, delivery, and graphics.

>>> Guidelines for Preparation and Delivery

The goal of most presentations is quite simple: You must present a few basic points, in a fairly brief time, to an interested but usually impatient audience. If you deliver what you expect when *you* hear a speech, then you will give good presentations yourself.

Although the guidelines here apply to any presentation, they relate best to those that precede or follow a written report, proposal, memo, or letter. With this connection in mind, note that there are many similarities between the guidelines for good speaking and those for good writing covered in earlier chapters.

>> **Presentation Guideline 1: Know Your Listeners**

The following features are common to most listeners:

- They cannot “rewind the tape” of your presentation, as opposed to the way they can skip back and forth through the text of a report.
- They are impatient after the first few minutes, particularly if they do not know where a speech is going.
- They will daydream and often must have their attention brought back to the matter at hand (expect a 30-second attention span).

To respond to these realities, learn as much as possible about your listeners. For example, you can (1) consider what you already know about your audience, (2) talk with colleagues who have spoken to the same group, and (3) find out which listeners make the decisions.

Most important, make sure not to talk over anyone’s head. If there are several levels of technical expertise represented by the group, decrease the technical level of your presentation accordingly. Remember—decision makers are often the ones without current technical experience. They may want only highlights; later, they can review written documents for details or solicit more technical information during the question-and-answer session after you speak.

>> **Presentation Guideline 2: Use the Preacher’s Maxim**

The well-known preacher’s maxim goes like this:

First you tell ’em what you’re gonna tell ’em, then you tell ’em, and then you tell ’em what you told ’em.

This plan gives the speech a simple three-part structure that most listeners can grasp easily.

1. **Abstract (beginning of presentation):** At the outset, you should (1) get the listeners’ interest (with an anecdote, a statistic, or other technique), (2) state the exact purpose of the speech, and (3) list the main points you will cover. Do not try the patience of your audience with an extended introduction—use no more than a minute.
2. **Body (middle of presentation):** Here you discuss the points mentioned briefly in the introduction, in the same order that they were mentioned. Provide the kinds of obvious transitions that help your listeners stay on track.
3. **Conclusion (end of presentation):** In the conclusion, review the main ideas covered in the body of the speech and specify actions you want to occur as a result of your presentation.

This simple three-part plan for all presentations gives listeners the handle they need to understand your speech. First, there is a clear *road map* in the introduction so that they know what lies ahead. Second, there is an organized pattern in the body, with clear transitions between points. And third, there is a strong finish that brings the audience back full circle to the main thrust of the presentation.

>> **Presentation Guideline 3: Stick to a Few Main Points**

Our short-term memory holds limited items. It follows that listeners are most attentive to speeches organized around a few major points. In fact, a good argument can be made for organizing information in groups of threes whenever possible. Listeners seem to remember groups of three items more than they do any other size groupings.

>> **Presentation Guideline 4: Put Your Outline on Cards or Paper**

The best presentations are *extemporaneous*, meaning the speaker shows great familiarity with the material but uses notes for occasional reference. Avoid extremes of reading a speech verbatim, which many listeners consider the ultimate insult, or memorizing a speech, which can make your presentation seem somewhat wooden and artificial.

Ironically, you appear more natural if you refer to notes during a presentation. Such extemporaneous speaking allows you to make last-minute changes in phrasing and emphasis that may improve delivery, rather than locking you into specific phrasing that is memorized or written out word for word.

>> **Presentation Guideline 5: Practice, Practice, Practice**

Many speakers prepare a well-organized speech but then fail to add the essential ingredient: practice. Constant practice distinguishes superior presentations from mediocre ones. It also helps eliminate the nervousness that most speakers feel at one time or another.

In practicing your presentation, make use of four main techniques, listed here from least to most effective:

- **Practice before a mirror:** This old-fashioned approach allows you to hear and see yourself in action. The drawback, of course, is that it is difficult to evaluate your own performance while you are speaking.
- **Use of an audio recording:** The portability of electronics allows you to practice almost anywhere. Although recording a presentation does not improve gestures, it helps you discover and eliminate verbal distractions such as *filler words* (e.g., *uhhhh*, *um*, *ya know*).
- **Use of live audience:** Groups of your colleagues, friends, or family—simulating a real audience—can provide the kinds of responses that approximate those of a real audience. Make certain that observers understand the criteria for a good presentation and are prepared to give an honest and forthright critique.
- **Use of video recording:** This practice technique allows you to see and hear yourself as others do. Your careful review of the recording can help you identify and eliminate problems with posture, eye contact, vocal patterns, and gestures.

>> Presentation Guideline 6: Speak Vigorously and Deliberately

Vigorously means with enthusiasm; *deliberately* means with care, attention, and appropriate emphasis on words and phrases. The importance of this guideline becomes clear when you think back to how you felt during the last speech you heard. At the very least, you expected the speaker to show interest in the subject and demonstrate enthusiasm.

>> Presentation Guideline 7: Avoid Filler Words

Avoiding filler words presents a tremendous challenge to most speakers. When they think about what comes next or encounter a break in the speech, they may tend to fill the gap with words and phrases, such as “um,” “okay,” or “you know.” To eliminate such distractions, follow these three steps:

- Step 1: Use pauses to your advantage.** Short gaps or pauses inform the listener that you are shifting from one point to another. In signaling a transition, a pause serves to draw attention to the point you make right after the pause. Do *not* fill these strategic pauses with filler words.
- Step 2: Practice with a recorder.** When you play it back, you become instantly aware of fillers that occur more than once or twice. Keep a tally sheet of the fillers you use and their frequency. Your goal is to reduce this frequency with every practice session.
- Step 3: Ask for help from others.** After working with audio recorders in step 2, give your speech to an individual who has been instructed to stop you after each filler. This technique gives immediate reinforcement.

>> Presentation Guideline 8: Use Rhetorical Questions

Enthusiasm, of course, is your best delivery technique for capturing the attention of the audience. Another technique is the use of rhetorical questions at pivotal points in your presentation.

Rhetorical questions are those you ask to get listeners thinking about a topic, not those that you would expect them to answer out loud. They prod listeners to think about your point and set up an expectation that important information follows. Also, they break the monotony of standard declarative sentence patterns. You must make a conscious effort to insert them at points when it is most important to gain or redirect the attention of the audience. Three effective uses follow:

1. **As a grabber at the beginning of a speech:** “Have you ever wondered how you might improve the productivity of your clerical staff?”
2. **As a transition between major points:** “We’ve seen that a centralized copy center can improve the efficiency of report production, but will it simplify report production for your staff?”
3. **As an attention-getter right before your conclusion:** “Now that we’ve examined the features of a centralized copy center, what’s the next step you should make at Dark Star Publishing?”

>> Presentation Guideline 9: Maintain Eye Contact

Your main goal is to keep listeners interested in what you are saying. This goal requires that you maintain control, and frequent eye contact is one good strategy.

The simple truth is that listeners pay closer attention to what you are saying when you look at them. Think how you react when a speaker makes constant eye contact with you. If you are like most people, you feel as if the speaker is speaking to you personally—even if there are 100 people in the audience. Also, you tend to feel more obligated to listen when you know that the speaker's eyes will be meeting yours throughout the presentation.

>> Presentation Guideline 10: Use Appropriate Gestures and Posture

Speaking is only one part of giving a speech; another is adopting appropriate posture and using gestures that reinforce what you are saying. Good presenters do the following:

1. Use their hands and fingers to emphasize major points.
2. Stand straight, without leaning on or gripping the lectern.
3. Step out from behind the lectern on occasion.
4. Point toward visuals on screens or charts, without losing eye contact.

With work on this facet of your presentation, you can avoid problems like keeping your hands constantly in your pockets, rustling change (remove pocket change and keys beforehand), tapping a pencil, scratching nervously, slouching over a lectern, and shifting from foot to foot.



>>> Guidelines for Presentation Graphics

Listeners expect good graphics during oral presentations. Much like gestures, graphics transform the words of your presentation into true communication with the audience. When you display graphics and text during a presentation, they should illustrate and clarify your speech. Therefore, we include displayed text in our discussion of graphics in this section.

>> Graphics Guideline 1: Discover Listener Preferences

Some professionals prefer simple speech graphics, such as flip charts or transparencies. Others prefer more sophisticated presentations, such as animations, audio, or video. Your listeners are usually willing to indicate their preferences when you call on them. Contact the audience ahead of time and make some inquiries. Also ask for information about the room in which you will be speaking. If possible, request a setting that allows you to make best use of your graphics choice.

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See these guidelines at work in Model Presentation 4: Presentation to Persuade.

>> Graphics Guideline 2: Match the Graphics to the Content

Plan graphics while you prepare the text so that the final presentation seems fluid. Remember that everything you project on a screen or present on a flip chart should support and enhance your presentation.

>> Graphics Guideline 3: Keep the Message Simple

When Edward Tufte critiqued PowerPoint slides in *The Cognitive Style of PowerPoint*, one of the problems he noted was the use of too many graphic elements on each slide, the equivalent of *chartjunk* that he had argued against in his earlier studies of graphics¹ (see Chapter 3). Some basic design guidelines apply, whether you are using posters, overhead transparencies, or computer-aided graphics such as PowerPoint. Figure 9–1 is an example of an introductory slide that uses these principles.

- Use few words, emphasizing just one idea on each frame.

Note: A common PowerPoint mistake is the use of too much text, which the speaker then reads to the audience.

- Use more white space, perhaps as much as 60%–70% per frame.
- Use landscape format more often than portrait, especially because it is the preferred default setting for most presentation software.
- Use sans-serif large print, from 14 pt. to 18 pt. minimum for text to 48 pt. for titles.

Your goal is to create graphics that are seen easily from anywhere in the room and that complement—but do not overpower—your presentation.

By avoiding asbestos contamination, you can

1. Prevent health problems
2. Satisfy regulatory requirements
3. Give yourself peace of mind



Three main reasons why building owners should be concerned about asbestos:

1. To prevent future health problems of your tenants.
2. To satisfy regulatory requirements of the government.
3. To give yourself peace of mind for the future.

■ **Figure 9–1** ■
Notes view of
presentation slide

¹Edward R. Tufte, *The Cognitive Style of PowerPoint* (Cheshire, CT: Graphics Press, 2003).

You should also use audio and video elements sparingly. Most presentation software programs include sound effects to accompany slide changes or the appearance of text or images. These are distracting and annoying, and should be avoided. You should also use video carefully.

>> Graphics Guideline 4: Consider Alternatives to Bulleted Lists

Recently, there has been a move away from the default slide layouts in most presentation software. One recommendation is to use full-sentence headings on slides to help the audience understand and remember the information being presented; another is to combine text with graphics on slides when appropriate.² Figure 9–2 uses the sentence headline and image format.

>> Graphics Guideline 5: Use Colors Carefully

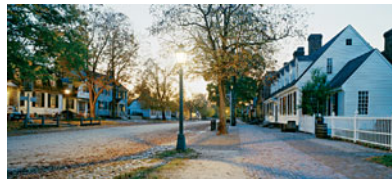
Colors can add flair to visuals. Use the following simple guidelines to make colors work for you:

- Have a good reason for using color (such as the need to highlight three different bars on a graph with three distinct colors).
- Be sure that a color contrasts with its background (e.g., yellow on white does not work well).
- Use no more than three or four colors in each graphic (to avoid a confused effect).

■ Figure 9–2 ■

Notes view of presentation slide using sentence headline-image format

Prevent future health problems of your tenants.



Most important reason be concerned?

Long term health of your tenants.

Asbestos linked to:

Lung cancer

Colon cancer

Asbestosis (debilitating lung disease)

Connection first documented in 1920s, but only taken seriously in last few decades.

By this time, asbestos common in buildings.

²Michael Alley and Kathryn A. Neeley, “Rethinking the Design of Presentations Slides: A Case for Sentences Headlines and Visual Evidence.” *Technical Communication* 52, no. 4 (2005): 417–426.

>> Graphics Guideline 6: Leave Graphics Up Long Enough

Because graphics reinforce text, they should be shown only while you address the particular point at hand. For example, reveal a graph just as you are saying, “As you can see from the graph, the projected revenue reaches a peak in 2007.” Then pause and leave the graph up a bit longer for the audience to absorb your point.

>> Graphics Guideline 7: Avoid Handouts

Because timing is so important in your use of speech graphics, handouts are often a bad idea. Readers move through a handout at their own pace, rather than at the pace the speaker might prefer. Use them only if (1) no other visual will do, (2) your listener has requested them, or (3) you distribute them as reference material after you have finished talking.

>> Graphics Guideline 8: Maintain Eye Contact While Using Graphics

Do not stare at your visuals while you speak. Maintain control of listeners’ responses by looking back and forth from the visual to faces in the audience. To point to the graphic aid, use the hand closest to the visual. Using the opposite hand causes you to cross over your torso, forcing you to turn your neck and head away from the audience.

>> Graphics Guideline 9: Include All Graphics in Your Practice Sessions

Dry runs before the actual presentation should include every graphic you plan to use, in its final form. This is a good reason to prepare graphics as you prepare text, rather than as an afterthought. If you are going to be projecting images from a transparency or computer program, the projected image may appear different than the original image. By previewing your graphics, you are able to fix them before your presentation.

You should also practice timing your graphics with your speech. Running through a final practice without graphics would be much like doing a dress rehearsal for a play without costumes and props—you would be leaving out parts that require the greatest degree of timing and orchestration.

>> Graphics Guideline 10: Plan for Technology to Fail

Murphy’s Law always seems to apply when you use another person’s audiovisual equipment: Whatever can go wrong, will, and at the worst possible moment. For example, a new bulb burns out and there is no extra bulb in the equipment drawer, an extension cord is too short, the screen does not stay down, the client’s computer does not read your file—many speakers have experienced these problems and more. Even if the equipment works, it often operates differently from what you are used to. The only sure way to put the odds in your favor is to carry your own equipment and set it up in advance.

If you must rely on someone else’s equipment, following are a few ways to prevent problems:

- Find out exactly who is responsible for providing the equipment and contact that person in advance.

- Have some easy-to-carry backup supplies in your car—an extension cord, a large easel pad, felt-tip markers, and dry erase markers, for example.
- Bring handout versions of your visuals to use as a last resort.

In short, you want to avoid putting yourself in the position of having to apologize. Plan well.

>>> Chapter Summary

Anyone can become a good speaker by preparing well, practicing often, and giving an energetic performance. Your effort pays off richly by helping you deal effectively with employers, customers, and professional colleagues.

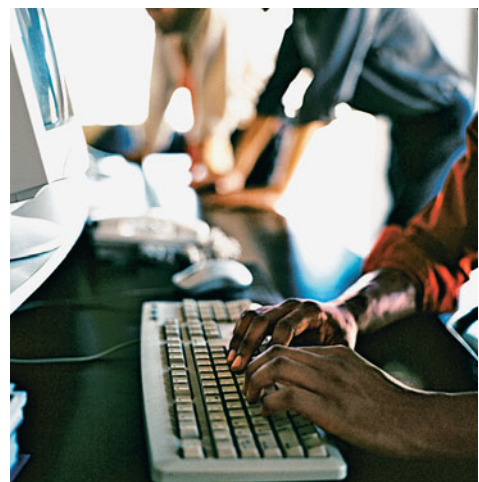
Ten guidelines for preparation and delivery lead to first-class presentations:

1. Know your listeners.
2. Use the preacher's maxim.
3. Stick to a few main points.
4. Put your outline on cards or paper.
5. Practice, practice, practice.
6. Speak vigorously and deliberately.
7. Avoid filler words.
8. Use rhetorical questions.
9. Maintain eye contact.
10. Use appropriate gestures and posture.

You should also strive to incorporate illustrations into your speeches by following these 10 guidelines:

1. Discover listener preferences.
2. Match the graphics to the content.
3. Keep the message simple.
4. Consider alternatives to bulleted lists.
5. Use colors carefully.
6. Leave graphics up long enough.
7. Avoid handouts.
8. Maintain eye contact while using graphics.
9. Include all graphics in your practice sessions.
10. Plan for technology to fail.

Chapter | 11 | Style in Technical Writing



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This chapter, as well as the Handbook (Appendix A), focuses on the final stage of the writing process—revising. As you may already have discovered, revision sometimes gets short shrift during the rush to finish documents on time. That is a big mistake. Your writing must be clear, concise, and correct if you expect the reader to pay attention to your message. Toward that end, this chapter offers a few basic guidelines on style. The Handbook contains alphabetized entries on grammar, mechanics, and usage.

After defining style and its importance, this chapter gives suggestions for achieving five main stylistic goals:

- Writing clear sentences
- Being concise
- Being accurate in wording
- Using the active voice
- Using nonsexist language

>>> Overview of Style

This section (1) provides an overview of *style* as it applies to technical writing and (2) defines one particularly important aspect of style—called *tone*—that relates to every guideline in this chapter.

Definition of Style

Just as all writers have distinct personalities, they also display distinct features in their writing. Writing style can be defined as follows:

Style: The features of one's writing that show its individuality, separating it from the writing of others and shaping it to fit the needs of particular situations. Style results from the conscious and subconscious decisions each writer makes in matters like word choice, word order, sentence length, and active and passive voice. These decisions are different from the “right and wrong” matters of grammar and mechanics (see the Handbook). Instead, they are composed of choices writers make in deciding how to transmit ideas.

Style is usually a series of personal decisions you make when you write. As noted in Chapter 2, however, much writing is being done these days by teams of writers. Collaborative writing requires individual writers to combine their efforts to produce a consensus style, usually a compromise of stylistic preferences of the individuals involved. Thus, personal style becomes absorbed into a jointly produced product. Similarly, many companies tend to develop a company style in documents like reports and proposals.

Importance of Tone

Tone is a major component of style and thus deserves special mention here. Through tone, you express an attitude in your writing—for example, neutral objectivity on the one hand, or unbridled enthusiasm on the other. The attitude evident in your tone exerts great influence over the reader. Indeed, it can determine whether your document achieves its

objectives. Much like the broader term *style*, *tone* refers to the way you say something rather than what you say.

The following adjectives show a few examples of the types of tone or attitude that can be reflected in your writing. Here they are correlated with specific examples of documents:

1. **Casual tone:** E-mail to three colleagues working with you on a project.
2. **Objective tone:** Formal report to a client in which you present data comparing cost information for replacing the company's computer infrastructure.
3. **Persuasive tone:** Formal proposal to a client in hopes of winning a contract for goods or services.
4. **Enthusiastic tone:** Recommendation letter to a university to accept one of your employees in a master's program.
5. **Serious tone:** Memorandum to employees about the need to reduce the workforce and close an office.
6. **Authoritative tone:** Memo to an employee in which you reprimand him or her for violations of a policy about documenting absences.
7. **Friendly tone:** Letter to long-term clients inviting them to an open house at your new plant location.

Although there are almost as many variations in tone as there are occasions to write documents, one guideline always applies: Be as positive as you can possibly be, considering the context. Negative writing has little place in technical communication. In particular, a condescending or sarcastic tone should be avoided at all costs. It is the kind of writing you will regret. When you stress the positive, you stand the best chance of accomplishing your purpose and gaining the reaction you want from the reader.

Despite the need to make style conform to team or company guidelines, each individual remains the final arbiter of her or his own style in technical writing. Most of us will be our own stylists, even in firms in which in-house editors help clean up writing errors. This chapter helps such writers deal with everyday decisions of sentence arrangement, word choice, and the like. However, although style is a personal statement, you should not presume that anything goes. Certain fundamentals are part of all good technical style in the professional world. Let us take a look at these basics.

>>> Writing Clear Sentences

Each writer has his or her own approach to sentence style; yet, everyone has the same tools with which to work: words, phrases, and clauses. This section defines some basic terminology in sentence structure, and then it provides simple stylistic guidelines for writing clear sentences.

Sentence Terms

The most important sentence parts are the subject and verb. The *subject* names the person doing the action or the thing being discussed (e.g., *He* completed the study/*The*

figure shows that); the *verb* conveys action or state of being (e.g., She *visited* the site/He *was* the manager).

Whether they are subjects, verbs, or other parts of speech, words are used in two main units: phrases and clauses. A *phrase* lacks a subject or verb or both and it thus must always relate to or modify another part of the sentence (She relaxed *after finishing her presentation*./As *project manager*, he had to write the report). A *clause*, however, has both a subject and a verb. Either it stands by itself as a *main clause* (He *talked to the team*) or it relies on another part of the sentence for its meaning and is thus a *dependent clause* (After she left the site, she went home).

Beyond these basic terms for sentence parts, you also should know the four main types of sentences:

- A *simple sentence* contains one main clause (He *completed his work*).
- A *compound sentence* contains two or more main clauses connected by conjunctions (He *completed his work*, but she *stayed at the office to begin another job*).
- A *complex sentence* includes one main clause and at least one dependent clause (After he *finished the project*, he *headed for home*).
- A *compound–complex sentence* contains at least two main clauses and at least one dependent clause (After they *studied the maps*, they *left the fault line*, but they *were unable to travel much farther that night*).

Guidelines for Sentence Style

Knowing the basic terms of sentence structure makes it easier to apply stylistic guidelines. Following are a few fundamental guidelines that form the underpinnings for good technical writing. As you review and edit your own writing or that of others, put these principles into practice.

>> Guideline 1: Place the Main Point Near the Beginning

One way to satisfy this criterion for good style is to avoid excessive use of the passive voice (see “Using the Active Voice” on pp. 231–232); another way is to avoid lengthy phrases or clauses at the beginnings of sentences. Remember that the reader usually wants the most important information first.

Original: “After reviewing the growth of the Cleveland office, it was decided by the corporate staff that an additional lab should be constructed at the Cleveland location.”

Revision: “The corporate staff decided to build a new lab in Cleveland after reviewing the growth of the office there.”

>> Guideline 2: Focus on One Main Clause in Each Sentence

When you string together too many clauses with *and* or *but*, you dilute the meaning of your text. However, an occasional compound or compound–complex sentence is acceptable, just for variety.

Original: “The hiring committee planned to interview Jim Steinway today, but bad weather delayed his plane departure, and the committee had to reschedule the interview for tomorrow.”

Revision: “The hiring committee had to change Jim Steinway’s interview from today to tomorrow because bad weather delayed his flight.”

>> **Guideline 3: Vary Sentence Length, but Seek an Average Length of 15–20 Words**

Of course, do not inhibit your writing process by counting words while you write. Instead, analyze one of your previous reports to see how you fare. If your sentences are too long, make an effort to shorten them, such as by making two sentences out of one compound sentence connected by *and* or *but*.

You should also vary the length of sentences. Such variety keeps your reader’s attention engaged. Make an effort to place important points in short but emphatic sentences. Reserve longer sentences for supporting main points.

Original: “Our field trip for the project required that we conduct research on Cumberland Island, a national wilderness area off the Georgia Coast, where we observed a number of species that we had not seen on previous field trips. Armadillos were common in the campgrounds, along with raccoons that were so aggressive that they would come out toward the campfire for a handout while we were still eating. We saw the wild horses that are fairly common on the island and were introduced there by explorers centuries ago, as well as a few bobcats that were introduced fairly recently in hopes of checking the expanding population of armadillos.”

Revision: “Our field trip required that we complete research on Cumberland Island, a wilderness area off the Georgia Coast. There we observed many species we had not seen on previous field trips. Both armadillos and raccoons were common in the campgrounds. Whereas the armadillos were docile, the raccoons were quite aggressive. They approached the campfire for a handout while we were still eating. We also encountered Cumberland’s famous wild horses, introduced centuries ago by explorers. Another interesting sighting was a pair of bobcats. They were brought to the island recently to check the expanding armadillo population.”

>>> **Being Concise**

Some experts believe that careful attention to conciseness could shorten technical documents by 10% to 15%. As a result, reports and proposals would take less time to read and cost less to produce. This section on conciseness offers several techniques for reducing verbiage without changing meaning.

>> **Guideline 1: Put Actions in Verbs**

Concise writing depends more on verbs than it does on nouns. Sentences that contain abstract nouns that hide actions can be shortened by putting the action in strong verbs

instead. By converting abstract nouns to action verbs, you can eliminate wordiness, as the following sentences illustrate:

Wordy: “The *acquisition* of the property was accomplished through long and hard negotiations.”

Concise: “The property was *acquired* through long and hard negotiations.”

Wordy: “*Confirmation* of the contract occurred yesterday.”

Concise: “The contract was *confirmed* yesterday.”

Wordy: “*Exploration* of the region had to be effected before the end of the year.”

Concise: “The region had to be *explored* before the end of the year.”

As the examples show, abstract nouns often end with *-tion* or *-ment* and are often followed by the preposition *of*. These words are not always “bad” words; they cause problems only when they replace action verbs from which they are derived. The following examples show some noun phrases along with the preferred verb substitutes:

assessment of	assess
classification of	classify
computation of	compute
delegation of	delegate
development of	develop
disbursement of	disburse
documentation of	document
elimination of	eliminate
establishment of	establish
negotiation of	negotiate
observation of	observe
requirement of	require
verification of	verify

>> Guideline 2: Shorten Wordy Phrases

Many wordy phrases have become common in business and technical writing. Weighty expressions add unnecessary words and rob prose of clarity. Following are some of the culprits, along with their concise substitutes:

afford an opportunity to	permit
along the lines of	like
an additional	another
at a later date	later
at this point in time	now
by means of	by

come to an end	end
due to the fact that	because
during the course of	during
for the purpose of	for
give consideration to	consider
in advance of	before
in the amount of	of
in the event that	if
in the final analysis	finally
in the proximity of	near
prior to	before
subsequent to	after
with regard to	about

>> Guideline 3: Replace Long Words with Short Ones

In grade school, most students are taught to experiment with long words. Although this effort helps build vocabularies, it also can lead to a lifelong tendency to use long words when short ones will do. Of course, sometimes you want to use longer words just for variety—for example, using an occasional *approximately* for the preferred *about*. As a rule, however, the following long words (in the left column) should routinely be replaced by the short words (in the right column):

advantageous	helpful
alleviate	lessen, lighten
approximately	about
cognizant	aware
commence	start, begin
demonstrate	show
discontinue	end, stop
endeavor	try
finalize	end, complete
implement	carry out
initiate	start, begin
inquire	ask
modification	change
prioritize	rank, rate
procure	buy
terminate	end, fire

transport	move
undertake	try, attempt
utilize	use

>> **Guideline 4: Leave Out Clichés**

Clichés are worn-out expressions that add words to your writing. Although they once were fresh phrases, they became clichés when they no longer conveyed their original meaning. You can make writing more concise by replacing clichés with a good adjective or two. Following are some clichés to avoid:

ballpark figure
efficient and effective
last but not least
needless to say
reinvent the wheel
skyrocketing costs

>> **Guideline 5: Make Writing More Direct by Reading It Aloud**

Much wordiness results from talking around a topic. Sometimes called *circumlocution*, this stylistic flaw arises from a tendency to write indirectly. It can be avoided by reading passages aloud. Hearing the sound of the words makes problems of wordiness quite apparent. It helps condense all kinds of inflated language, including the wordy expressions mentioned earlier. Remember, however, that direct writing must also retain a tactful and diplomatic tone when it conveys negative or sensitive information.

Indirect: “At the close of the last phase of the project, a bill for your services should be expedited to our central office for payment.”

Direct: “After the project ends, please send your bill immediately to our central office.”

Indirect: “It is possible that the well-water samples collected during our investigation of the well on the site of the subdivision could possibly contain some chemicals in concentrations higher than is allowable according to the state laws now in effect.”

Direct: “Our samples from the subdivision’s well might contain chemical concentrations beyond those permitted by the state.”

>> **Guideline 6: Avoid *There Are*, *It Is*, and Similar Constructions**

There are and *it is* should not be substituted for concrete subjects and action verbs, which are preferable in good writing. Such constructions delay the delivery of information about who or what is doing something. They tend to make your writing lifeless and abstract. Avoid them by creating (1) main subjects that are concrete nouns and (2) main verbs that are action words. Note that the following revised passages give readers a clear idea of who is doing what in the subject and verb positions.

Original: “It is clear to the hiring committee that writing skills are an important criterion for every technical position.”

Revision: “The hiring committee believes that writing skills are an important criterion for every technical position.”

Original: “There were 15 people who attended the meeting at the client’s office in Charlotte.”

Revision: “Fifteen people attended the meeting at the client’s office in Charlotte.”

>> Guideline 7: Cut Out Extra Words

This guideline covers all wordiness errors not mentioned earlier. You must keep a vigilant eye for any extra words or redundant phrasing. Sometimes the problem comes in the form of needless connecting words, like *to be* or *that*. Other times it appears as redundant points—that is, those that have been made earlier in a sentence, paragraph, or section and do not need repeating.

Delete extra words when their use (1) does not add a necessary transition between ideas or (2) does not provide new information to the reader. (One important exception is the intentional repetition of main points for emphasis, as in repeating important conclusions in different parts of a report.)

>>> Being Accurate in Wording

Good technical writing also demands accuracy in phrasing. Technical professionals place their reputations and financial futures on the line with every document that goes out the door. That fact shows the importance of taking your time on editing that deals with the accuracy of phrasing. Accuracy often demands more words, not fewer. The main rule is:

Never sacrifice clarity for conciseness.

Careful writing helps to limit liability that your organization may incur. Your goal is very simple: Make sure words convey the meaning you intend—no more, no less. Some basic guidelines to follow include:

>> Guideline 1: Distinguish Facts from Opinions

In practice, this guideline means you must identify opinions and judgments as such by using phrases like *we recommend*, *we believe*, *we suggest*, or *in our opinion*.

Example: “In our opinion, spread footings would be an acceptable foundation for the building you plan at the site.”

If you want to avoid repetitious use of such phrases, group your opinions into listings or report sections. Thus, a single lead-in can show the reader that opinions, not facts, are forthcoming.

Example: “On the basis of our site visit and our experience at similar sites, we believe that (1) _____, (2) _____, and (3) _____.”

>> Guideline 2: Include Obvious Qualifying Statements When Needed

This guideline does not mean you must be overly defensive in every part of the report; it means that you must be wary of possible misinterpretations.

Example: “Our summary of soil conditions is based only on information obtained during a brief visit to the site. We did not drill any soil borings.”

>> Guideline 3: Use Absolute Words Carefully

Avoid words that convey an absolute meaning or that convey a stronger meaning than you intend. One notable example is *minimize*, which means to reduce to the lowest possible level or amount. If a report claims that a piece of equipment will *minimize* breakdowns on the assembly line, the passage could be interpreted as an absolute commitment. The reader could consider any breakdown at all to be a violation of the report’s implications. If instead the writer had used the verb *limit* or *reduce*, the wording would have been more accurate and less open to misunderstanding.

>>> Using the Active Voice

Striving to use the active voice can greatly improve your technical writing style. This section defines the active and passive voices and then gives examples of each. It also lists some practical guidelines for using both voices.

What Do Active and Passive Mean?

Active-voice sentences emphasize the person (or thing) performing the action—that is, somebody (or something) does something (“Matt completed the field study yesterday”). Passive-voice sentences emphasize the recipient of the action itself—that is, something is being done to something by somebody (“The field study was completed [by Matt] yesterday”). Following are some other examples of the same thoughts being expressed in first the active and then the passive voice:

■ Examples: Active-Voice Sentences:

1. “We *reviewed* aerial photographs in our initial assessment of possible fault activity at the site.”
2. “The study *revealed* that three underground storage tanks had leaked unleaded gasoline into the soil.”
3. “We *recommend* that you use a minimum concrete thickness of 6 in. for residential subdivision streets.”

■ Examples: Passive-Voice Sentences:

1. “Aerial photographs *were reviewed* [by us] in our initial assessment of possible fault activity at the site.”

2. “The fact that three underground storage tanks had been leaking unleaded gasoline into the soil *was revealed* in the study.”
3. “*It is recommended* that you use a minimum concrete thickness of 6 in. for residential subdivision streets.”

Just reading through these examples gives the sense that passive constructions are wordier than active ones. Also, passive voices tend to leave out the person or thing doing the action. Although occasionally this impersonal approach is appropriate, the reader can become frustrated by writing that fails to say who or what is doing something.

When Should Active and Passive Voices Be Used?

Both the active and passive voices have a place in your writing. Knowing when to use each is the key. Following are a few guidelines that will help:

■ *Use the active voice when you want to:*

1. Emphasize who is responsible for an action (“*We recommend* that you consider...”)
2. Stress the name of a company, whether yours or the reader’s (“*PineBluff Contracting* expressed interest in receiving bids to perform work at...”)
3. Rewrite a top-heavy sentence so that the person or thing doing the action is up front (“*Figure 1 shows* the approximate locations of...”)
4. Pare down the verbiage in your writing, because the active voice is usually a shorter construction.

■ *Use the passive voice when you want to:*

1. Emphasize the object of the action or the action itself rather than the person performing the action (“*Samples will be sent* directly from the site to our laboratory in Sacramento”).
2. Avoid the kind of egocentric tone that results from repetitious use of *I*, *we*, and the name of your company (“*The project will be directed* by two programmers from our Boston office”).
3. Break the monotony of writing that relies too heavily on active-voice sentences.

Although the passive voice has its place, it is far too common in business and technical writing. This stylistic error results from the common misperception that passive writing is more objective. In fact, excessive use of the passive voice only makes writing more tedious to read. In modern business and technical writing, strive to use the active voice.

>>> Using Nonsexist Language

Language usually follows changes in culture rather than anticipating such changes. An example is today’s shift away from sexist language in business and technical writing—indeed, in all writing and speaking. The change reflects the increasing number of women entering previously male-dominated professions, such as engineering, management, medicine,

and law. It also reflects the fact that many men have taken previously female-dominated positions as nurses and flight attendants.

This section on style defines sexist and nonsexist language. Then it suggests ways to avoid using gender-offensive language in your writing.

Sexism and Language

Sexist language is the use of wording, especially masculine pronouns like *he* or *him*, to represent positions or individuals who could be either men or women. For many years, it was perfectly appropriate to use *he*, *his*, *him*, or other masculine words in sentences such as the following:

- “The operations specialist should check page 5 of his manual before flipping the switch.”
- “Every physician was asked to renew his membership in the medical association before next month.”
- “Each new student at the military academy was asked to leave most of his personal possessions in the front hallway of the administration building.”

The masculine pronoun was understood to represent any person—male or female. Such usage came under attack for several reasons:

1. As previously mentioned, the entry of many more women into male-dominated professions has called attention to the inappropriate generic use of masculine pronouns.
2. Many people believe the use of masculine pronouns in a context that could include both genders constrains women from achieving equal status in the professions and in the culture—that is, the use of masculine pronouns encourages sexism in society as a whole.

Either point supplies a good enough reason to avoid sexist language. Many women in positions of responsibility may read your on-the-job writing. If you fail to rid your writing of sexism, you risk drawing attention toward sexist language and away from your ideas. Common sense argues for following some basic style techniques to avoid this problem.

Techniques for Nonsexist Language

This section offers techniques for shifting from sexist to nonsexist language. When shifting to nonsexist language, many writers have problems with subject–verb agreement. (The *engineer* recorded *their* data.) The strategies that follow help you avoid this problem. Not all these strategies will suit your taste in writing style; use the ones that work for you.

>> Technique 1: Avoid Personal Pronouns Altogether

One easy way to avoid sexist language is to delete or replace unnecessary pronouns:

Example:

Sexist Language: “During *his* first day on the job, any new employee in the toxic-waste laboratory must report to the company doctor for *his* employment physical.”

Nonsexist Language: “During *the* first day on the job, each new employee in the toxic-waste laboratory must report to the company doctor for *a* physical.”

>> Technique 2: Use Plural Pronouns Instead of Singular

In most contexts you can shift from singular to plural pronouns without altering meaning. The plural usage avoids the problem of using masculine pronouns.

Example:

Sexist Language: “Each geologist should submit *his* time sheet by noon on the Thursday before checks are issued.”

Nonsexist Language: “All geologists should submit *their* time sheets on the Thursday before checks are issued.”

Interestingly, you may encounter sexist language that uses generic female pronouns inappropriately. For example, “Each nurse should make every effort to complete *her* rounds each hour.” As in the preceding case, a shift to plural pronouns is appropriate: “Nurses should make every effort to complete *their* rounds each hour.”

>> Technique 3: Alternate Masculine and Feminine Pronouns

Writers who prefer singular pronouns can avoid sexist use by alternating *he* and *him* with *she* and *her*. When using this technique, writers should avoid the unsettling practice of switching pronoun use within too brief a passage, such as a paragraph or page. Instead, writers may switch every few pages, or every section or chapter.

Although this technique is not yet in common use, its appeal is growing. It gives writers the linguistic flexibility to continue to use masculine and feminine pronouns in a generic fashion. However, one problem is that the alternating use of masculine and feminine pronouns tends to draw attention to itself. Also, the writer must work to balance the use of masculine and feminine pronouns, in a sense to give equal treatment.

>> Technique 4: Use Forms Like *He or She*, *Hers or His*, and *Him or Her*

This solution requires the writer to include pronouns for both genders.

Example:

Sexist Language: “The president made it clear that each branch manager will be responsible for the balance sheet of *his* respective office.”

Nonsexist Language: “The president made it clear that each branch manager will be responsible for the balance sheet of *his or her* respective office.”

This stylistic correction of sexist language may bother some readers. They believe that the doublet structure of *her or his*, is wordy and awkward. Many readers are bothered even more by the slash formations of *he/she*, *his/her*, and *her/him*. Avoid using these.

>> Technique 5: Shift to Second-Person Pronouns

Consider shifting to the use of *you* and *your*, words without any sexual bias. This technique is effective only with documents in which it is appropriate to use an instructions-related command tone associated with the use of *you*.

Example:

Sexist Language: “After selecting *her* insurance option in the benefit plan, each new nurse should submit *her* paperwork to the Human Resources Department.”

Nonsexist Language: “Submit *your* paperwork to the Human Resources Department after selecting *your* insurance option in the benefit plan.”

>> Technique 6: Be Especially Careful of Titles and Letter Salutations

Today, most women in business and industry are comfortable being addressed as *Ms.* If you know that the recipient prefers *Miss* or *Mrs.*, use that in your salutation. If a person's gender is not obvious from the name, call the person's employer and ask how the person prefers to be addressed. (When calling, also check on the correct spelling of the person's name and the person's current job title.) Receptionists and secretaries expect to receive such inquiries.

When you do not know who will read your letter, never use *Dear Sir* or *Gentlemen* as a generic greeting. Such a mistake may offend women reading the letter and may even cost you some business. *Dear Sir* or *Madam* is also inappropriate. It shows you do not know your audience, and it includes the archaic form *Madam*. Instead, call the organization for the name of a particular person to whom you can direct your letter. If you must write to a group of people, replace the generic greeting with an *Attention* line that denotes the name of the group.

Examples:

Sexist Language: Dear Miss Finnegan: [to a single woman for whom you can determine no title preference]

Nonsexist Language: Dear Ms. Finnegan:

Sexist Language: Dear Sir: or Gentlemen:

Nonsexist Language: Attention: Admissions Committee

No doubt the coming years will bring additional suggestions for solving the problem of sexist language. Whatever the culture finally settles on, it is clear that good technical writing style no longer tolerates the use of such language.

>>> Plain English and Simplified English

When you are writing technical or business documents, you may be asked to use one of two important styles of workplace writing: Plain English or Simplified English. Both of these styles include specific recommendations about sentence structure and word choice, but they are designed for particular audiences and purposes.

Plain English

Plain English is a specific style recommended for the U.S. government documents and for documents, such as proposals and reports that are submitted to federal agencies. Although people had been discussing clearer government documents for years, the Plain Language movement gained strong support during the mid-1990s. In 1995, a group of people began creating standards for Plain English in government writing. This group became the Plain Language Action and Information Network (PLAIN).

Plain English guidelines include many of the elements of clear technical communication: audience awareness, good document design, effective use of headings, and clear organization. However, Plain English is most clearly defined by its style recommendations, which include the following:

- Use active voice
- Put actions in strong verbs
- Use *you* to speak directly to the reader
- Use short sentences (no longer than about 20 words)
- Use concrete words
- Use simple and compound sentences with a subject–verb structure
- Make sure that modifiers are clear
- Use parallel structure for parallel ideas
- Avoid wordiness

The Plain Language Web site at <http://www.plainlanguage.gov> includes a complete discussion of Plain English with examples and links to other resources.

Simplified English

Simplified English includes many of the same recommendations as Plain English, and it is sometimes confused with Plain English. However, it serves a different purpose and is designed for a different audience. Simplified English, sometimes called *Controlled English* or *Internationalized English*, is designed for the global economy. It is designed for an audience for whom English is a second language, to be easily translatable from English into other languages. A leading organization for the development of Simplified English is the European Association of Aerospace Manufacturers (AECSMA), which created the original standard in the 1980s.

Simplified English is designed to be clear and unambiguous, so it recommends specific sentence structures and limited vocabulary. Simplified English includes the following:

- Use only approved words
- Use one word for each meaning (avoid synonyms)
- Use only one meaning for each word (e.g., *close* is used only as a verb)
- Use active voice

- Use strong verbs
- Use articles (*a, an, the*) or demonstrative adjectives (*this, that, these, those*) for clarity
- Avoid strings of more than three nouns
- Use short sentences (less than 20 words)

More information about Simplified English standards is available at <http://www.asd-ste100.org>, and an overview of Simplified English is available at <http://www.userlab.com/SE.html>, which also includes a sample list of approved words at <http://www.userlab.com/Downloads/SE.pdf>. Because the standards were developed for the aerospace industry, the word lists are specialized for that industry. Other industries are developing their own word lists. A more general word list can be downloaded from the Publications/Documents section of Intecom's Web site at <http://www.intecom.org>.

>>> Chapter Summary

Style is an important part of technical writing. During the editing process, writers make the kinds of changes that place their personal stamp on a document. Style can also be shaped (1) by a team, in that writing done collaboratively can acquire features of its diverse contributors or (2) by an organization, in that an organization may require writers to adopt a particular writing style. Yet the decision-making process of individual writers remains the most important influence on the style of technical documents.

This chapter offers five basic suggestions for achieving good technical writing style. First, sentences should be clear, with main ideas at the beginning and with one main clause in most sentences. Although sentences should average only 15 to 20 words, you should vary sentence patterns in every document. Second, technical writing should be concise. You can achieve this goal by reading prose aloud as you rewrite and edit. Third, wording should be accurate. Fourth, the active voice should be dominant, although the passive voice also has a place in good technical writing. And fifth, the language of technical documents should be free of sexual bias.

Although clarity and conciseness are important to all workplace writing, you may be asked to follow a specific style sheet. Many organizations create their own in-house style guides, and some styles, such as Plain English and Simplified English are used for industries or to clarify cross-cultural communication.

IEEE Citation



What Is IEEE?

The IEEE stands for Institute of Electrical and Electronics Engineers. Its reference format is a commonly accepted style for writing, formatting, and citing research papers. The format was created based on the Chicago Manual of Style and is widely used by students and researchers in Computer Science, Engineering and relevant technical fields. It is a professional association of electrical engineering and associated disciplines. It was formed in 1963 based on two American establishments – the Institute of Radio Engineers and the American Institute of Electrical Engineers.

The IEEE style consists of 2 critical elements:

- In-text citations (direct & indirect)
- References

In-text citations are short quotations taken from primary or secondary sources. In IEEE in-text citation, they are numbered in square brackets. Every number next to their corresponding citation identifies the complete citation listed in the References section in numerical order.

Why Citing is Important for Your Academic Performance?

Collecting sources for an academic paper and then citing them can be time-consuming and tedious. Many students find the citation process too complicated and often overlook its importance. So, why is citing sources vital for your academic performance? First, citing external sources helps students show their professors that they have invested time and effort into researching the topic. Also, this process has a few other purposes:

1. Providing Credibility:

The main reasons for using information from external sources are either to complement the contents of your work with relevant data or to provide a different look/opinion on the issue stated. In both cases, it means adding credibility to your own paper, making it look more trustworthy and complete, and establishing yourself as an authoritative author.

2. Ensuring Academic Integrity and Uniqueness:

By not giving credit to other people's works when relevant, you lose academic integrity and misrepresent your own academic achievements. At the same time, you risk the uniqueness of your paper; as simply using external resources without citing them is most often considered plagiarism. Both issues can lead to a low grade, task failure, or even expulsion.

3. Acknowledging the Works of Other Specialists:

The main purpose of citing sources is to give credit to the works of other people, wherever it is pertinent. When creating each citation, you acknowledge the hard work, time, and effort another person has put into his or her research.

4. Contributing to Further Research:

Finally, the last reason that proves the importance of citations is the fact that by giving accurate credit to others' works, you are contributing to your own further research and also helping other researchers locate the relevant sources you've cited with ease.

Citing Sources in the Text

Every reference has a corresponding IEEE citation within the main body of the essay or research paper. Each time a writer adds a citation, they should insert a number within square brackets. This is known as an **in-text citation**. Later, each citation should be assigned a complete reference on the 'References' page. Relevant citations within the text and on the 'References' page should correspond to each other by numbers to make it possible to find the complete reference. The **references page** should be on its own separate page of the paper.

The in-text citations numbers start at [1], and continue in ascending order throughout the paper – unless you are referring to a source you have already cited in your text, in which case you can use the previously assigned number. Each in-text citation number should be enclosed by square brackets and appear on the text line, inside sentence punctuation, with a space before the bracket.

For example:

"Thevenin's and Norton's theorems are widely used for circuit analysis simplification [13]."

Citations of references may be given simply as "in [1]...", rather than as "in reference [1] ...".

Furthermore, citations may be grammatically treated either as if they were:

Footnote Numbers:	Nouns:
As shown by Jones [4] ...	As seen in [2] ...
For more details, see [1], [3], [7].	According to [4] and [6]–[8] ...
as mentioned earlier [3], [4]–[6], [8] ...	In contrast to [5, p. 7], it is evident that...
Taylor et al. [5] have noted ...	As demonstrated in [4] ...

When authors are mentioned, they may be treated in the following way:

Rickard [5] has shown ...

Jones [6], and Zheng and Rogers [7] have stated ...

Azzarello et al. [3] stated that they were unable to determine why ...

If there are more than three authors, provide **et al.** (meaning 'and others') after the first author in the text of the paper. Note that **et al.** is not italicised in the in-text citations. In the reference list, however,

list all the authors for up to six authors– use **et al.** only if the names are not given. Also use **et al.** in the reference list for more than 6 authors, e.g.:

[8] J. D. Bellamy *et al.*, *Computer Telephony Integration*, New York: Wiley, 2010.

Do not mention authors of a source or provide date of publication within the text (e.g. “in Jones [1]” should be changed to “in [1]”) except in such cases where the author’s name is integral to the understanding of the sentence (e.g. “Jones [1] proposed a new approach for sensor and actuator selection problems). Editing the in-text citation numbers may require renumbering the whole reference list. Please check that the in-text citation numbers match the reference list numbers.

Paraphrasing in IEEE:

When you are paraphrasing, that is, expressing an idea or a fact found in a source using different words, a reference citing the source should always be given. Provide your citation number directly after the reference - this is not necessarily at the end of the sentence, unless it is where the reference occurs. Also, page numbers are generally not given for paraphrases:

In contrast to ‘data partitioning’ structures [13], the ‘space partitioning’ structures show better performance for dynamic memory resident data [14]–[15].

Quotes in IEEE:

Direct quotes are used to support an argument showing the exact words and phrases of an author according to the original source. Enclose quotes in single quotation marks and provide the citation in square brackets after the quotation or after the author’s name along with the page number(s).

An example of a short direct quote:

Baez et al. have noted that ‘full 3D stacking can potentially offer additional advantages for memory and processor applications’ [7, p. 14].

If you use longer quotations (of three lines and more), use a block quotation by setting the block of quoted text as a paragraph. Use smaller font size for block quotations, and also indent them from both margins, for example:

An example of a longer direct quote:

As Abad notes:

It is also desirable to minimize the parasitic capacitance of electronic packages because it is another source of signal delay. Consequently, a very low relative dielectric constant insulating material should be used whenever possible, since the relative dielectric constant is a measure of a material's total polarizability and determines its charge storage capacity with respect to a vacuum [7, p. 63].

the source where the quoted words or phrases are taken from.

Reference list

In the Reference list, provide page numbers if you are referencing a section or chapter of the source:

[7] W. Brown, "Electrical Design Considerations," in *Advanced Electronic Packaging: With Emphasis on Multichip Modules*: Wiley-IEEE Press, 2013, pp. 51-74.

References in IEEE Style

Each full IEEE reference must include all necessary bibliographic information to help the reader find more details about its topic. **The elements of an IEEE format reference are:**

- A corresponding number to the in-text citation
- Author's surname and initials
- The full title of the work
- Place of publication
- Date of publication
- Some smaller details like page number, issue, or volume (if available)

Now, that you know the basic rules of citations, let's look at some IEEE format citation examples from our political science essay service to help you understand how to make a reference page:

1. Book: Single Author

[corresponding number] Author. Book title. Location: Publishing company, year, pp.

Example: [1] W.-K. Chen. Linear Networks and Systems. Belmont, CA: Wadsworth, 1993, pp. 123-135.

2. Book: Two or More Authors

[corresponding number] Author, Author, and Author. Book title. Location: Publishing company, year, pp.

Example: [2] U. J. Gelinis, Jr., S. G. Sutton, and J. Fedorowicz. Business Processes and Information Technology. Cincinnati: South-Western/Thomson Learning, 2004, pp. 98-100.

3. Article in a Journal

[corresponding number] Author. "Article title". Journal title, vol., pp, date.

Example:[4] G. Pever. "Infrared Nation." The International Journal of Infrared Design, vol. 33, pp. 56-99, Jan. 1979.

Sample of In-Text Citation

Some vibration and noise sources give rise to annoyance to people and every so often damage to property. Vibration in either digital or analogue panel meters can cause their malfunction or difficulty in reading the meters [1]. Hence one of the significant goals of vibration study is to monitor, analyse and reduce vibrations through proper design of machines and their mountings [2], [3].

According to [4], the complexity of mechanical and electrical systems and applications of vibratory equipment in them have increased markedly in recent years. For example, vibration is put to work in vibrating power tools like jackhammers and chainsaws, vibratory feeders and conveyors, hoppers, sieves, electric toothbrushes, dentist's drills, mobile phones, including the fitness machines in recent times [5]. It has been emphasised in [6] that an important consideration in designing of electronic equipment is that it must work with a high degree of reliability in severe vibration and shock environments. Ozansoy [7] has further argued that the process of designing the electronic equipment also needs to account for the severe thermal environments in which the equipment will be operating.

Sample of a Reference Page

Reference List

- [1] R. E. Ziemer and W. H. Tranter, *Principles of Communications: Systems, Modulation, and Noise*, 7th ed. Hoboken, NJ: Wiley, 2015, pp. 13–17.
- [2] J. D. Bellamy et al., *Computer Telephony Integration*, New York: Wiley, 2010.
- [3] C. Jacks, *High Rupturing Capacity (HRC) Fuses*, New York: Penguin Random House, 2013, pp. 175–225.
- [4] N. B. Vargaftik, J. A. Wiebelt, and J. F. Malloy, "Radiative transfer," in *Convective Heat*. Melbourne: Engineering Education Australia, 2011, ch. 9, pp. 379–398.
- [5] H. C. Hottel and R. Siegel, "Film condensation," in *Handbook of Heat Transfer*, 2nd ed. W. C. McAdams, Ed. New York: McGraw-Hill, 2011, ch. 9, pp. 78–99.
- [6] W. M. Rohsenow, "Heat transmission," in *Thermal Radiation Properties*, vol. 3, M. W. Catton and J. P. Hartnett, Eds. New York: Macmillan, 2012, ch. 9, pp. 37–62.
- [7] H. Schmidt-Walter and R. Kories, *Electrical Engineering. A Pocket Reference*. Boston: Artech House, 2007. Accessed on: Oct. 16, 2016. [Online]. Available: <http://ebruary.com>
- [8] H. H. Gaynor, *Leading and Managing Engineering and Technology, Book 2: Developing Managers and Leaders*. IEEE-USA, 2011. Accessed on: Oct. 15, 2016. [Online]. Available: <http://www.ieeeusa.org/communications/ebooks/files/sep14/n2n802/Leading-and-Managing-Engineering-and-Technology-Book-2.pdf>
- [9] G. H. Gaynor, "Dealing with the manager leader dichotomy," in *Leading and Managing Engineering and Technology, Book 2, Developing Leaders and Managers*. IEEE-USA, 2011, pp. 27–28. Accessed on Jan. 23, 2017. [Online]. Available: <http://www.ieeeusa.org/communications/ebooks/files/sep14/n2n802/Leading-and-Managing-Engineering-and-Technology-Book-2.pdf>
- [10] M. Cvijetic, "Optical transport system engineering," in *Wiley Encyclopedia of Telecommunications*, vol. 4, J. G. Proakis, Ed. New York: John Wiley & Sons, 2003, pp. 1840–1849. Accessed on: Feb. 5, 2017. [Online]. Available: <http://ebscohost.com>
- [11] T. Kaczorek, "Minimum energy control of fractional positive electrical circuits", *Archives of Electrical Engineering*, vol. 65, no. 2, pp.191–201, 2016.

Guide on Writing a Research Paper in IEEE Format

Every student who studies Computer Science and Engineering should know how to write a paper in the IEEE format to set themselves up for success. The IEEE paper format is widely used for styling academic papers written in these subjects, which is why it is vital to master it.

Basic Format of a Research Paper

The standard IEEE template contains the following sections in the same order:

1. Title Page (including the article's title, byline, membership, and first footnote)
2. Abstract – should be one paragraph long (preferably between 150 to 250 words)
3. Index Terms
4. Nomenclature (optional)
5. Introduction
6. Body of Article
7. Conclusion
8. Appendix(es)
9. Acknowledgment(s)
10. References
11. Photos and Biographies

To write an IEEE-style paper, students should follow these basic rules:

- **Paper title** – placed on the first page, centred at the top, and 24pt font size.
- **By-line** – placed below the paper title after a line break, centred, 10pt font size. The by-line has to include the following information (every point needs to be on a separate line):
 - Author's name(s)
 - Author's affiliation(s)
 - City & country location(s)
 - E-mail address(es).
- **Main body** – all written in 10pt font size, the text should appear in two columns on the page. Columns on the last page must be the same length, which means the author may need to add a column break.
- The paper should start with the abstract and index terms.
- Depending on the subject and context, papers may include additional sections like:
 - Acknowledgments
 - Appendices
 - Note to Practitioners
 - Nomenclature
- The main body of the paper can be divided into relevant sections and subsections. Each section and subsection should have a heading formatted according to the rules specified later in this guide.
- All tables, equations, and figures must be numbered in consecutive order and centred in the column.

- IEEE papers should start with a drop cap two lines deep, followed by the next 8-12 characters or 1-2 words (depending on which is more appropriate) in all caps.

Parts of a Research Paper Based on IEEE Style

Headings

The IEEE format suggests authors of papers use four levels of hierarchy for the text's section headings:

Type of Heading	Suggested Format	Example
Primary	Centred, enumerated by Roman numerals, followed by a period. The first letter is set to 10pt. font size, the rest should be 8pt., all caps	I. INTRODUCTION
Secondary	Flush left, enumerated by capital letters, followed by a period, upper and lower case, italic, 10pt. font size	<i>A. Formal Frameworks</i>
Tertiary	Indented one em, enumerated by Arabic numerals followed by a parenthesis, it runs into the text in its sections, upper and lower case, italic, and followed by a colon. All set to 10pt. font size, as is the whole text in the main body	<i>1) Sophisticated Local Control:</i> The term sophisticated local control refers to...
Quaternary	Same as tertiary, with the following exceptions: indented two ems, enumerated by lower case letters, and only requires capitalizing the first letter of the heading	<i>a) Communication policies:</i> Communication policies are developed to...

Enumeration of section headings is preferable, but not mandatory – the author can use his or her preference. However, the chosen format should be consistently followed throughout the entire paper. Apart from section headings, there are also Appendix, Reference, and Acknowledgement headings, which are formatted differently.

Body of the Paper

Abstract

The IEEE format requires authors to include an Abstract in every paper. An abstract should follow the Title Page of the document and appear in the text in boldface type. It has to be concise and should not include any numbered references or mathematical equations.

Index Terms

Authors should also provide Index Terms in their IEEE papers. Index Terms should also appear in boldface type, in alphabetic order, and located within the Abstract as its final paragraphs. You should only define acronyms in this section if they have already been defined in the paper itself. The authors can request a list of keywords by sending a blank email to [keywords\[at\]ieee.org](mailto:keywords[at]ieee.org).

Text Equations

When placing equations within the main body of the paper, one should number them in inconsequential order from the beginning of the work to its end. In some transactions, it is permitted to use the author's own numbering system, for example, numbering by section, e.g., (1.1), (1.2.1), (A1).

Acknowledgement

The Acknowledgement section should always follow the final part of the text, after any Appendix(es) and preceding the References section. It should be written in the third person.

If you need to cite names in the Acknowledgement section, do not use full names and honorables like Mr., Mrs., Miss, and Ms. Instead, only list first initials followed by a family name. You can still use titles like Prof. and Dr., but only in the singular form, placing it separately next to each name.

Do NOT include information about any financial support in the Acknowledgement section, place it in the first paragraph of the first footnote instead.

References

In IEEE papers, all references should be numbered and there should be a separate entry for each number. It is not allowed to use the same reference number of a group of IEEE citations.

Text Citation of Figures and Tables

IEEE citation format for figures and tables requires following numerical order. Citations of figures located within the paper should always be labeled by the abbreviation "Fig." followed by the relevant figure number. Authors of papers should use this abbreviation even if it has to be put at the beginning of a sentence.

Biographies

IEEE Transactions require including author biographies. **As a rule, biographies have to be divided into three paragraphs:**

1. The first paragraph should mention the full name of the author and the history of his or her IEEE membership.
2. The second paragraph should contain information about the author's work and military experience (if any); all job titles should be capitalized.

3. The third paragraph should list information about the author's memberships in professional societies (except IEEE).

Footnotes

A footnote is any additional information, explanation, or comment related to the main text that is located at the bottom of the page. All footnotes within the IEEE format paper should be numbered and followed in consecutive order within the body of the paper. The numbers, both within the text and in the actual footnotes, should be marked with superscripts. Within the body of the paper, the superscript footnote numbers should always be placed after the punctuation marks, such as parentheses, commas, and periods. However, if the sentence is a compound one, the numbers should be placed before the punctuation marks, such as semicolons, dashes, colons, and quotation marks. The actual footnotes should be placed at the bottom of the page where the cited information is located.

List in Text

All lists within the IEEE style papers should be formatted according to the following order of labeling: 1), 2), 3) followed by a), b), c), and then i), ii), iii).

Example: According to the IEEE Computer Society, the most crucial areas of computer science are: 1) algorithms and data structures; 2) theory of computation; 3) computer elements and architecture; and 4) programming methodology and languages.

Taken from: <https://essaypro.com/blog/ieee-format>

<https://researchguides.njit.edu/ieee-citation/>



English Language Centre

Faculty of Languages & Literature

University of Central Punjab, Lahore

Assessment Rubrics for English 3

**Comprehensive Guidelines for
Effective English Language Teaching & Assessment**

Tailored by

Dr. Muhammad Habib Qazi

Director, ELC

&

Tahira Munir

Lecturer, ELC

Rubric for Assessment of Writing Scale for English 3

Marks	Content	Communicative Achievement	Organisation	Language
5	- All content is relevant to the task. - Target reader is fully informed.	Uses the conventions of the communicative task effectively to hold the target reader's attention and communicate straightforward and complex ideas, as appropriate.	Text is well organised and coherent, using a variety of cohesive devices and organisational patterns to generally good effect.	- Uses a range of vocabulary, including less common lexis, appropriately. - Uses a range of simple and complex grammatical forms with control and flexibility. - Occasional errors may be present but do not impede communication.
4	<i>Performance shares features of 3 and 5 marks.</i>			
3	- Minor irrelevances and/or omissions may be present. - Target reader is on the whole informed.	Uses the conventions of the communicative task to hold the target reader's attention and communicate straightforward ideas.	Text is generally well organised and coherent, using a variety of linking words and cohesive devices.	- Uses a range of everyday vocabulary appropriately, with occasional inappropriate use of less common lexis. - Uses a range of simple and some complex grammatical forms with a good degree of control. - Errors do not impede communication.
2	<i>Performance shares features of 1 and 3 marks.</i>			
1	Irrelevances and misinterpretation of task may be present. Target reader is minimally informed.	Uses the conventions of the communicative task in generally appropriate ways to communicate straightforward ideas.	Text is connected and coherent, using basic linking words and a limited number of cohesive devices.	- Uses everyday vocabulary generally appropriately, while occasionally overusing certain lexis. - Uses simple grammatical forms with a good degree of control. - While errors are noticeable, meaning can still be determined.
0	- Content is totally irrelevant. - Target reader is not informed.	<i>Performance below 1 mark.</i>		

A Sample Question with a Candidate's Answer

Question: You have received this email from your English-speaking friend David.

From: David

Subject: Touring Holiday

Some college friends of mine are visiting your area soon for a week's touring holiday. They would like to travel around and learn about your local area and its history. Can you tell me about some of the places they could visit? What's the best way to travel around – car, bike or coach?

Thanks,

David

Write your email.

Answer by a Candidate:

Dear David,

I'm glad your friends are visiting my area soon for a week's touring holiday. I have many ideas what I can show them and tell about.

In my opinion the best way to travel around will be by bike because of small distances between the places and views are amazing.

My area includes also beautiful Baltic Sea which many tourist visit especially in summer. Your friends could sunbath or swim if they would like but the water is quite cold in this season. Beautiful sightseeing of sunrise is the best memorise!

You wrote that they are intrested in history of my local area. That's great! We have museum of our local history where I can go with them. Tickets are not so expensive and I can think about some discount.

What do you think about it? Would you mind send me some your ideas?

I look forward to hearing from you soon.

Best wishes,

Sam

Marking on the Scale and the Examiner's Comments for the Candidate

Scale	Marks	Commentary
Content	5	- All content is relevant and the target reader is fully informed. - The candidate responds to the questions that have been asked and gives details about the places David's friends can visit (Baltic Sea; museum of our local history). - The email explains that the best mode of transport is bike, because of small distances between the places and views are amazing.
Communicative Achievement	3	- The conventions of the task are used to hold the reader's attention. The response is written in an informal, friendly but polite tone, which is appropriate for the task. The conventions for opening and closing an email of this type are used appropriately, as well as standard phrases for communication to a friend (You wrote that; What do you think about it; I look forward to hearing from you soon; Best wishes). - Straightforward ideas are communicated. Information and advice are given and the overall tone is enthusiastic and positive.
Organisation	4	- The text is well organised and coherent, and uses the input email well to structure the response (I'm glad your friends are visiting; I have many ideas; In my opinion; My area includes also; which many; Your friends could; You wrote that; That's great). - A variety of linking words (because of) and cohesive devices, such as substitution and referencing, are used across some sentences and to connect simple ideas.
Language	3	- There is a range of everyday vocabulary used which is relevant to the topic (beautiful Baltic Sea; many tourist visit; in this season; Tickets are not so expensive). - There is a range of simple and some complex grammatical forms used with a good degree of control, e.g. modals, present and past forms of verbs. - There are some errors (My area includes also; send me some your ideas) and some spelling mistakes (sunbath; memorise; intrested) but these do not impede communication.

Rubric for Assessment of Speaking Scale for English 3

Marks	Grammar and Vocabulary	Discourse Management	Pronunciation	Interactive Communication
5	- Shows a good degree of control of a range of simple and some complex grammatical forms. - Uses a range of appropriate vocabulary to give and exchange views on a wide range of familiar topics.	- Produces extended stretches of language with very little hesitation. - Contributions are relevant and there is a clear organisation of ideas. - Uses a range of cohesive devices and discourse markers.	- Is intelligible. - Intonation is appropriate. - Sentence and word stress is accurately placed. - Individual sounds are articulated clearly.	- Initiates and responds appropriately, linking contributions to those of other speakers. - Maintains and develops the interaction and negotiates towards an outcome.
4	<i>Performance shares features of 3 and 5 marks.</i>			
3	- Shows a good degree of control of simple grammatical forms, and attempts some complex grammatical forms. - Uses a range of appropriate vocabulary to give and exchange views on a range of familiar topics.	- Produces extended stretches of language despite some hesitation. - Contributions are relevant, and there is very little repetition. - Uses a range of cohesive devices.	- Is intelligible. - Intonation is generally appropriate. - Sentence and word stress is generally accurately placed. - Individual sounds are generally articulated clearly.	- Initiates and responds appropriately. - Maintains and develops the interaction and negotiates towards an outcome with very little support.
2	<i>Performance shares features of 1 and 3 marks.</i>			
1	- Shows a good degree of control of simple grammatical forms. - Uses a range of appropriate vocabulary when talking about everyday situations.	- Produces responses which are extended beyond short phrases, despite hesitation. - Contributions are mostly relevant, despite some repetition. - Uses basic cohesive devices.	Is mostly intelligible, and has some control of phonological features at both utterance and word levels.	- Initiates and responds appropriately. - Keeps the interaction going with very little prompting and support.
0	<i>Performance below 1 mark.</i>			

Resource:

Cambridge Assessment English